

Jour 350 The Pharmaceutical Industry

Numbered Transcript

JOUR 350 • Gerald Posner

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Lecture 1: Snake Oil Origins

At a glance — Posner traces the snake-oil origins of the drug industry.

- 1.1 Welcome to the Secrets of Big Pharma: How to Be a Smarter Patient. I think you will find some tips. We're going to explore what I call the hidden history of the drug industry. Not that the drug industry is trying to keep it all secret today, but there are certain parts of their history that they don't talk about that much. So we're going to examine the origins of this industry that affects so many lives.
- 1.2 And there's no doubt, you're talking about the pharmaceutical industry, talking about an industry that most of us interact with, in one way or another. We've taken a prescription at some point, even if it's for an infection, for antibiotics. Many, many people, millions, tens of millions of Americans, over half of all Americans, are on a daily pill for some condition. There's always a new miracle pill, as I call it. It depends on what the miracle pill is. And today, of course, it's GLP-1s and weight loss drugs.
- 1.3 And we do the same thing in the drug industry, and you'll see this over the course of these lectures as we talk about different drugs. Something becomes popular. It's pushed heavily and marketed heavily by the drug industry in different ways to doctors and through advertisements and by sponsoring studies. And then it catches with the public. And once it does, all of the different manufacturers try to come up with their own version, so they can get a patent on that particular drug and charge as much as they can for it.
- 1.4 The public comes in, for the most part, people using the drug, full-force. They're enthusiastic about it. At that point, the industry is selling you not only just a drug for a particular condition, but it's also telling you, "Gee, there may be side effects here that are good." GLP-1s, weight-loss drugs, might help put off dementia. They may help people improve their heart. They may help men with more vigor. It's fantastic. It's only as we start to get in the inevitable list of side effects that the reports come in, that eventually you find out, "Oh, I wish we had known 5 or 10 years ago what we know now." It's a process that plays out time and time again, and I think you're going to see that.
- 1.5 What I want to talk to you in this first lecture about are what I call the roots of the industry. You plant it, you put the seeds in, and then it sort of comes around. And those roots are important because, in my view, having studied the pharmaceutical industry for a number of years now, it sets the foundation for what the industry becomes. The start is the DNA. It's the template for what you're dealing with. I think that, snake oil to science, there's no doubt it starts with snake oil, and then it does move to science, but the origins of the industry are addictive.
- 1.6 It's the selling of these potent narcotics we're going to talk about as miracle cures in the late 19th and early 20th century. No national standards when the industry gets

going in the 1800s. No prescription laws—don't exist—don't need prescriptions, so nobody's filling in a prescription at this time. No safety requirements, which is pretty amazing. And that era sets the tone for what comes up in the future, which I call profit-driven products that aren't necessarily focused, they're unregulated, and they're often dangerous.

- 1.7 So before "Big Pharma," as I like to say, there was "big hype." It's an unregulated marketplace, 19th and 20th centuries, where anyone could call themselves a doctor. No AMA, nothing else. It's remarkable. You could set up the—not many people were doing it because they didn't think about it, but they thought they could sell a concoction of chemicals and call it a miracle cure. We'll get into so-called patent medicines, as they were called.
- 1.8 The most successful ones of these particular products, they weren't selling anything based upon evidence, but they were selling what I call dreams. They grew by selling exaggerated promises of false claims about the efficacy of their products. No scientific evidence, no clinical studies, nothing like that required. And in the beginning, lack of oversight, nonexistence of oversight and accountability. No government regulations or standards. You could market the products. You didn't have to disclose the ingredients, and you didn't have to, of course, list potential harms, because nobody was talking about that at all. So that unregulated marketplace laid the foundation for the industry's model of prioritizing profits over public health. That's the era where it gets set.
- 1.9 Let's dig a little bit deeper into the unveiling of what I call the pharmaceutical industry itself. Marketing strategy shaped the development of public image of the pharmaceutical industry over time. What they were worried about is how they were selling it to you as a miracle cure, not necessarily what it was doing, and you'll see this time and time again, when we get into some of the early examples.
- 1.10 The commercial interest in medicine, the decision-making is affected by what I call the bottom line, time and time again. And a public perception, the public views on medicine were influenced entirely by what the industry was telling it was doing, by these so-called miracle cures. So this is a pattern that continues to influence health care today as we go along, so a marketplace of ignorance, as I call it.
- 1.11 In the 1800s, as I said a moment ago, no prescription laws, nothing else. I get it. No medical licensing. Any peddler could sell themselves a medical cure. This is the unregulated marketplace. The background of this that is so interesting for the pharmaceutical industry, is that doctors, and there were some real doctors, but society as a whole knew very little about what caused disease and illness. So they didn't know much about evidence.
- 1.12 Think of this. We're talking about, it had only been, when the first drugs start to come out in the mid-1800s, it's only a few decades since Louis Pasteur, this French bacteriologist, had discovered, on rotten meat and milk that had gone bad, what he

thought were germs, microscopic small little germs, that were there on the spoiled meat and the bad milk. That was the beginning of germ theory, the idea that maybe there are microbes that could exist on the planet.

1.13 People thought that was just wrong. They dismissed it time and time again. They didn't believe in the idea of the germ theory. So what happens is, they didn't know there were bacterial pathogens. They didn't know what caused diseases things like tuberculosis, until a German bacteriologist in 1882 discovered that you can actually get tuberculosis from microorganisms. They think that you inherit it. That's the idea. Things like the Immigration Act of 1882 really came in because of cholera. They thought that cholera came from only certain immigrants, and most likely Irish.

1.14 So, somehow, that had been the way that we looked at it. We didn't think there was something that was infectious. To show you where the thinking was in this country, not just this country, but in Europe as well, in 1891, we passed the Immigration Act, and I'm quoting from it here, so I'll read you this precisely. They think that cholera will be ended by passing the Immigration Act. The Immigration Act excludes, quote, "all idiots, insane persons, paupers, or persons likely to become public charges, and persons suffering from a loathsome or dangerous contagious disease."

1.15 It sounds like we think we can keep everybody out who might be actually bringing a disease, not understanding that it is passed by microbes as well. Until an American army general, Walter Reed, in 1900, discovered that mosquitoes and others are actually the cause of something like yellow fever, that they spread yellow fever. We think that you can only catch it if you run into contact with somebody else. That ignorance about medicine helps the people who are selling what I call snake oil cures. Because you don't know what's making you sick, they play to your fear, and they tell you that they have an absolute cure for it, so it's easier to believe it.

1.16 Now, the one drug that was working, that they didn't know quite how it worked in terms of the body, but they knew that the effect worked, was morphine. That had been discovered only in 1804 by a German pharmaceutical apprentice, working, he was 21 years old, and he was able to extract from the opium poppy an actual morphine alkaloid, the first time you got a pure alkaloid from the poppy. He is the one, this 21-year-old apprentice, who names it morphine, after the Greek god of dreams, because morphine will sort of put you into this out, dream state, after the Greek god of dreams, Morpheus. So that's how we get the name morphine.

1.17 Pretty quickly in the early 1800s, people realized it also helps take away pain. Two German companies ended up becoming the morphine manufacturers in the early 1800s. It's a guy called Friedrich Bayer, who founded his company that you all know as Bayer. And Friedrich, the Schering, who founded Schering, later Schering & Plough. Those two German companies, Schering & Bayer, are manufacturing what I call quality morphine. No American company is manufacturing.

- 1.18** And then what happens is you get into the Mexican-American War, 1846 to 1848, a two-year war in which it really picks up '46 and '48. Eighty-seven percent— this is an astonishing figure to me; I keep looking at this, it's hard to imagine—87 % of the fatalities for American troops in the Mexican-American War were a failure to treat yellow fever, dysentery infections, and cholera. So not from being killed by the enemy, but because of diseases that we couldn't treat.
- 1.19** And in that war, the morphine that arrived at the front lines for soldiers was often adulterated. So it was being made by mom-and-pops in the United States. Sometimes it was being imported from Europe. By the time it was being imported from Europe, it was coming through a distributor that was taking the original morphine of high quality and then diluting it a little bit to try to have more morphine to sell at a different time. When it arrived at the front lines, the soldiers weren't getting the pain relief from it.
- 1.20** Interesting history, but why does it matter for the pharmaceutical industry? Because there were a number of people who said, "There's a market there. We can fill that market. There's no American company, no American manufacturer of morphine. This war just proved that there should be." So, you get, one year after the Mexican-American War, a couple of German cousins in New York City founded Charles Pfizer & Company. They had \$2,500 in savings of \$1,000. They ended up borrowing, and they set up a factory on Bartlett Street, and they become the first real sincere, what I call, company dedicated to making morphine of high quality.
- 1.21** Their competition, soon afterwards, is Edward Robinson Squibb. We know it later as E. R. Squibb & Sons. Squibb was a colonel inside the Mexican-American War. He saw the morphine arrive that was of bad quality, that was often thrown over the ship because it was so bad. He goes into also in Brooklyn as competition, essentially creating morphine there.
- 1.22** Something happens, as you well know, unfortunately, in this country goes into a civil war in 1861. So bad for the country, one of the worst parts of history, but good for the morphine business because there are more injuries on that battlefield than at any other time in American history. And as a result, there's a boom. Charles Pfizer and E. R. Squibb are absolutely set for it, and they are booming. But now, others come into the business, seeing the opportunity for what they think is this pharmaceutical boom.
- 1.23** One year into the Civil War, two pharmacist brothers in Philadelphia, John and Frank Wyeth, set up a pharmacy. They have a small drug distributorship. They land such a lucrative contract with the Union Army. They get rid of the pharmacy, and they make just a manufacturing facility for creating morphine to be able to provide that contract to the Union Army.
- 1.24** A part of Detroit's healthcare system, Samuel Duffield, has his own concoction. Not everybody's just doing morphine. He has an ether and alcohol solution that alleviates some of the pain. He also lands a union contract that's good. And when he sells that

business, he sells it to a couple of guys called Hervey Parke, who is a mining investor, and a 26-year-old salesman, George Davis. Parke-Davis, that's what they become, and they are involved in that.

- 1.25** Eli Lilly missed the Civil War opportunity, but he served, and he knew about the demand for morphine, and he formed his company afterwards as Eli Lilly. The main products are all opium-based and morphine-based products when Eli Lilly gets going, just after the Civil War. Two American pharmacists also missed the Civil War by a couple of years: Silas Burroughs and Henry Wellcome. Wellcome and Burroughs went over to the UK, and their main products were a whole bunch of laxatives and opium-based creams as well.
- 1.26** That's what I call the start of the pharmaceutical industry. You go over it, we're talking about Pfizer, we're talking about Edward Robinson Squibb, we're talking about the Wyeth brothers who become Wyeth, the pharmaceutical company, Parke-Davis, Eli Lilly, Burroughs and Wellcome, not just companies, by the way, that came and went, but companies that survived and became some of the biggest pharmaceutical companies on the planet. This is their origin. This is how they get going on an addictive product that has no restrictions on how it's dispensed, and they understand the power of that.
- 1.27** The last thing on this, in the middle of all of this, in this marketplace of what I call ignorance, in 1859, a German doctoral student was the first person to be able to pull a pure alkaloid, which is the basis for a drug. You can make it, you can synthesize it from that, from the coca plant. So now we're just at the point around the Civil War period, afterwards, you have the first bit of cocaine. It's actually this doctoral student who writes his dissertation on how he was able to go ahead and get the alkaloid from the coca plant. He's the one who calls it cocaine, for those of you who wonder how it ever got that name. Because in Latin, I-N-A, from Latin, means "from," and all he was saying is "from the coca plant," cocaine, as we call it.
- 1.28** That becomes a stimulant that people understand and becomes the second part of drugs offered in America. He, by the way, is a footnote, smart student, he goes on to develop mustard gas, not the greatest thing in the world, which becomes the most deadly chemical agent in all of World War I, and is responsible for a lot of deaths. So the inventor of cocaine is also the inventor of nerve gas.
- 1.29** Now, cocaine and morphine, you won't be surprised, were the first big blockbusters. They were sold openly by Merck, Bayer, Pfizer, Parke-Davis, Squibb. Dentists sold cocaine-infused lozenges for toothaches, asthma sufferers bought pure cocaine inhalants, used as needed. They were sold for just about every ailment that you could imagine. Safety, not a concern. Cocaine boom. I'm telling you that cocaine is booming as a product. I'll give you a figure that will convince you that that's the case. Merck's annual cocaine production: in a two-year span, after cocaine was first discovered, they were selling products, they were using up less than a pound of cocaine in their

products, and two years later, they were using 180,000 pounds. So the products are expanding. The use for cocaine is expanding.

- 1.30** Parke-Davis—you have to love the ingenuity of the American drug industry. They are trying to patent a process to increase the purity and extend the shelf life of cocaine products. So we'll take natural cocaine, and maybe we can make it a little bit stronger, and we can say we have the best and purest version of this, and we can make it last longer on the shelf because we have to send it all around the country. So there was this real type of what I call hardcore competition. Parke-Davis also introduced cocaine-infused cigars—they called them these coca cheroots and coca-leaf cigarettes—and cocaine alcohol syrup marketed as true medicine. I don't think Parke-Davis or Merck talk much about that in their annual reports anymore, but that's all right.
- 1.31** The rapid growth of cocaine, the development of processes to refine the drug is just the highlight of this early 19th and 20th century period. Now, opiate competition, don't think that they've all turned to just cocaine products because they're competing broadly with as much as they can still with opiates. They all got their start on this. I think Merck had a powder of morphine. They boasted and advertised at the time, if you look back at some of their early material newspapers, that their purity of their morphine powder was absolutely the best, and they did have one of the most popular opiate-based products. One of Merck's most popular drugs was opium-laced cough lozenges. The opiates will put a cough at bay without any doubt. Squibb and Pfizer: opium tinctures, nearly a dozen variations of opium tinctures.
- 1.32** All of this was not just a handful of these companies that we know today competing with each other. Their real competition was coming from patent medicines, over 50,000 patent medicines. These were really the big sellers of the day. Merck, Pfizer, Parke-Davis were making money, but the patent medicine makers—we called them patent medicines, but they weren't actually patented. It's unusual. It just became the way—they were nostrums, they were called. They were put out by anybody with a clever idea. They claimed all types of cures. These people, these are the snake oils, the real snake oil salesmen and saleswomen. They were the ones who were cashing in a way that was unknown.
- 1.33** So, Lydia Pinkham had an item; you knew nothing about the ingredients. You just knew what the claim was. So Kopp's Baby Friend, one of the most successful patent drugs ever, was marketed toward new parents so that you would be able to stop the cough of your infant child, and they could sleep through the night. That company was so clever that they went through the newspapers looking for birth announcements, and then they would send a mailing to the new family saying, "By the way, don't forget Kopp's Baby Friend is the greatest thing to help you through all the sleepless nights." It was [one]-third pure opium—that's what it turned out to be—and morphine. There were deaths later of children because too much was given. But this was the market into which this first started to come, the patent drugs, as I call them.

- 1.34** From a book that I did, *Pharma*, I say, "Despite their aggressive narcotics expansion, the major drug manufacturer still had a powerful competitor: the patent medicine industry, nearly fifty thousand branded remedies, sold without prescription, disclosure, or regulation." The tens of thousands of medicines on the market, they marketed as miracle cures. I'll tell you, they didn't underplay it. If you look at the advertisements, it's really rather remarkable.
- 1.35** Secret formulas, everybody competing with a secret formula—boy, Coca-Cola wasn't the only one that said, "We had a secret formula, and we're not going to tell anyone." Or Heinz ketchup, 57 ingredients, and "we can't disclose them to everybody here." Because they didn't have to tell you what was in it. They just sold you what the product was. It did play back to what I call the fears and ignorance of the marketplace because people didn't know what caused the illness. You might think that you could cure everything from a tumor to somebody that suddenly had a stroke. You weren't sure what the cause of the stroke was.
- 1.36** Lack of regulations around ingredients, purity, and dosage—no controls over those, what went in. Brand names, bottle shapes, and label designs, they fought over that all the time. It's fantastic. I'm a lawyer, a non-practicing lawyer, but I love legal cases, and there were fights between the patent medicine makers, the most successful ones. One would say, "You copied the shape of my bottle that I'm selling that elixir in, so people are getting confused when they go out to buy it because they think it's my product." "Oh, no, you have the same type of label. You have my print." They were fighting over the way it looked, not over what the ingredients were or what the cure was.
- 1.37** Rapid growth of these products, combined with the proliferation of unregulated [ones], highlights the need at this point for stricter regulations in the industry to protect public health, but nobody's talking about that in the late 1800s, leading into the beginning of the 20th century. It's that business model that we've talked about that says: if you sell a product, you can make it work. That business model, that DNA, is what exists today.
- 1.38** Here's one of my favorite case studies. There are so many great case studies in this period, but is heroin a safe medicine? From 1887 to 1903, Bayer, the German pharmaceutical company, they're doing research inside their labs, and they have research team, and they make four discoveries over a few-year period, pretty big discoveries. In 1887, this team develops phenacetin, which is like Tylenol. Never had it existed before. They come up with it in the lab, so they're doing it. A year and a half later, they come up with aspirin, also pretty amazing. Aspirin, I never say a wonder drug because I know it's overused, but almost a wonder drug. It alleviates pain, is anti-inflammatory, and everything else. We know the downsides of aspirin because they're bleeding in the stomach, but none of that's known, but they come up with aspirin. So two analgesics over a short amount of time.
- 1.39** In 1900, they developed heroin. Now, you'd think heroin is just another form of morphine. It's 10-15 times more powerful, much stronger. It acts faster, and they

decide to market it as heroin. By the way, they also have, in 1903, they developed the first barbiturates. It comes under the brand name Luminal; it's phenobarbital. So, no barbiturates had existed before, also very powerful drugs. Acetaminophen, aspirin, heroin—which, by the way, is named after the German word heroisch, "heroic;" they give it the name heroin—you wonder how these drugs get named. And then, in 1903, we get barbiturates.

- 1.40** They didn't put out one of those drugs. They market it. By the way, these pharma companies are marketing everything. But Bayer doesn't market one of them because inside their lab, in the testing they're doing on animals, they think one of them is causing all types of effects at high doses, and so we'll keep it back. You know what that was? Acetaminophen. They didn't put Tylenol out, but they put out barbiturates, no prescription, over-the-counter. They put out heroin over-the-counter, and they put out aspirin over-the-counter.
- 1.41** Heroin goes on sale in the United States in 1900. And let me tell you, it's offered, sold, as a cough remedy, even to children, hard to believe, for colds. It's more effective than codeine, with only a 10th, they said, of codeine side effects, which is fantastic. It's a miracle solution. My favorite marketing technique for heroin was that they marketed it as a cure for morphine addiction. True—hard to imagine—you will give up your morphine addiction, it's true, but you will have a heroin addiction. It's the starkest example to me of marketing triumphant science. I don't think you see much about Bayer in their annual reports talking about how they were pioneers in the industry, and heroin was one of their items.
- 1.42** When I said to you before—by the way, the secret history of the pharmaceutical industry, it's not as though these are secrets that I just went down and found in an archive in Washington, and by the way, I have a whistleblower, and they're going to tell you something today. Well, some of that will come, maybe in later lectures, but it's things that the industry will omit about itself. You get to tell your own story at a certain point. They spend millions and millions of dollars on PR and on creating an image that they are coming up with cures and treatments, and that you should be having them. This is part of their origins. This isn't just by the way. This isn't a point at which I can say to you, "You know what? They were selling heroin, but they were also selling these six other drugs that were helping people on a whole range of things," because they didn't know what those drugs were, and they didn't know what was causing illnesses.
- 1.43** This is where the industry gets its start: the barbiturates boom. Discovered in 1903 by Bayer—how do they market barbiturates? I think barbiturates are very, very powerful drugs. Treatment for insomnia: I suppose that will help you get to sleep if you take barbiturates. Anxiety: will make you less anxious because you'll be zoned out completely. Nerves and depression.
- 1.44** Phenobarbital, sold as Luminal, was marketed widely. Without a prescription, by 1908, it was a big seller in the US, and followed the same pattern as cocaine and morphine.

Highly addictive substance, minimizing its dangers and finding profits in this, phenobarbital is the first one marketed. Bayer comes out; the rest are copying it. The same profitable pattern continues, as I always say, as you find with the other, if it's a moneymaker, once one company comes out with it, then other companies are chasing the same type of drug because they want to cut up part of that same pie.

- 1.45 Now, finally, regulation. Oh, my goodness. We've been waiting for a while for it. One thing to keep in mind as we go through this history all the way up to today, through the opioid crisis, the government, the federal government, has almost always reacted late. It's not surprising, but there's a buildup. There's such an outcry. There's a crisis. There's a lethal prescription crisis. There's an abuse crisis. Later, we'll talk about Valium and amphetamines way down the road. There's problems, and then the government reacts to a public health crisis.
- 1.46 That's very much what happened here in 1906. You end up getting Harvey Washington Wiley, who is running the Bureau of Chemistry. He is a government worker. He's a real crusader. He's fantastic. His concern are preservatives. He is worried about preservatives in food. He's been trying for a number of years. There are some members of Congress who are submitting bills that say, "We should really go ahead and get some regulations on these preservatives." But nobody could agree on which preservatives were good or which ones were bad, which ones were dangerous. People were using a series of preservatives, formaldehyde, for instance, to make sure that meat could be sent across the country safely. Not the safest type of preservative.
- 1.47 What Harvey Washington Wiley does is he creates inside the Bureau of Chemistry an experiment with people nicknamed the Poison Squad by a local reporter, and it stuck. Every night, those volunteers would eat different meals that he would prepare with preservatives that he wouldn't tell them what was in it, and then they would see who got sick and who didn't over the long term. After one died, they stopped that, but they had a pretty good idea that they had to have regulations. On top of that came a book called *The Jungle*, and that book, although it was a novel, had some stomach-wrenching pages about the meat markets in Chicago. That got politicians up in arms as well.
- 1.48 So they end up finally passing in 1906 the Pure Food and Drug Act. You think by its name, it's going to be fantastic because "pure food and drug"—we're going to finally regulate it. Here's what happened. The companies, the patent companies, and what I call the ethical pharmacists—that's what they called themselves, the Mercks and the Pfizers—they fought over regulation. They said, "We're a new industry. We're helping people. Don't overregulate us. Don't create something where we can't create all these great things."
- 1.49 This is the beginning. This particular battle is what I call the first successful lobbying effort by the drug industry. They're unorganized. They don't have a lobbying group as they do today, called pharma. They don't go through a central outfit. They don't have Madison Avenue people, but they know how to reach politicians, and they're hitting

away at it. So the final, you see the law, the proposed law for the 1906 Pure Food and Drug Act, changes as it goes along in these drafts. What comes out eventually pretty much leaves, in some ways, the drug industry, the patent drug industry alone.

1.50 Here's what I mean: the same addictive products stayed on the market. There was still no requirement for a prescription. There was no requirement that you had to prove that they were safe or effective for what you were selling them for. You didn't have to take any ingredient out. Here was the change: you had to list what the ingredients were. For the first time, if they're selling you a product, they have to tell you what's on the label. They don't have to tell you the amount, but they have to tell you what's in it. That's a big step forward. I agree, no question about it, but that really doesn't stop the industry.

1.51 I'll give you one example. There was a very, very popular product at the time called Vin Mariana. It's a wine, a Bordeaux wine, and it was extremely popular. It was marketed as a wine that would give you more vigor. It would make you stronger, it would make you more alert. It had 7.6 milligrams of cocaine per ounce. That's what it had. So Vin Mariana was an unusual wine. It had an endorsement, by the way, and they would sell it from Pope Leo 13th, and thousands of doctors, not all of them doctors, but they listed themselves as doctors. So it really, really was popular. Vin Mariana, afterward, put on the label that it had cocaine in the wine, and it didn't affect its sales at all. So the disclosure of that, it wasn't as though the public said, "Oh, my goodness, cocaine in the wine, we're not going to have it." Maybe it actually brought more customers to them because they said, "That's really a good item."

1.52 So the Pure Food and Drug Act doesn't have the effect that you think it might have. One of the things is this lobbying in what I call regulatory capture. In the early 20th century, the successful effort to lobby and weaken, undermine regulatory and safety standards in the industry, is going to be part of the DNA going forward. You're going to see this time and time again. The government gets involved, says, "Maybe we should do something." The industry fights back and pushes and says, "Don't overregulate. Don't stop innovation and science." Somebody comes up with that line. I've never found the first person who developed it at which company, but they should give that person a special medal in the pharmaceutical industry because it's the best line they've ever had, which is: "we're science and cures and medicine, so don't hurt that by overregulating." That's what you see here.

1.53 By the way, the first case, Harvey Washington Wiley, was a fantastic guy. He's one of the heroes in terms of getting this passed and doing everything in the government. He decides to go ahead and bring a suit by the federal government against a manufacturer to show the power of this new law. Who does he choose? You've got a lot of companies out there: Merck, everybody else. He goes against Coca-Cola. He decides to sue Coca-Cola because he says, "You didn't disclose that there's part of the coca leaf in there and that there's caffeine. You didn't put that on the label." Coca-Cola fought back and said,

"We took the coca leaf out like 40 years ago, and the caffeine, we'll put it on."
Government lost the suit, but it ended up taking over a year of government time. Not the best way to start in the Pure Food and Drug Act. It didn't scare anybody at all.

- 1.54** So the real rival now, even after this, becomes these what I call two main areas: patent medicine makers—the wild, Lydia Pinkham and Kopp's Baby Fr and all of those—versus the drug manufacturers of Pfizer, Merck, and the rest. In the public's mind, these elixirs, tonics, remedies, what the Mercks, Pfizers, Lillys had was that they were saying already to the public, "We're better than the mom-and-pops. We're purer. We have standards. We have manufacturing facilities. We can do it a little bit better than they can."
- 1.55** Transition to ethical pharmaceuticals— Dr. Winslow's, by the way, was a big opium-based cough syrup that was very, very popular. The Harrison Act was passed in 1914. The Harrison Act is one of those footnote laws in American history, but a major law. It's the law that says, "Hey, no more. We're not having cocaine and opiates and morphine. We're trying to control these addictive drugs," and it essentially bans them.
- 1.56** That came in out of left field to the drug industry because they're worried about the 1906 Act. They're worried about the disclosures. They're not focusing as much on this effort inside of Congress that might ban these addictive products. And so when that comes in, and they essentially ban it, you have the pharmaceutical industry, including the patent drug manufacturers, go cold turkey. They have to go overnight. They have to say, "What do we do? What do we do with our opium-based products? What do we do with all our tinctures? What do we do with our elixirs?" What about the fact they can no longer sell heroin? All of that happens. You literally have an industry that was heavily relying on these addictive products that now has to focus on how they come out of that and sell products that are different.
- 1.57** That is where the traditional companies, the Pfizers, the Lillys, the Burroughs & Wellcome, emerge stronger because they don't really have many products to sell. They have to get rid of a lot of their own. But the patent drug manufacturers are wiped out. They had single products. They were one-hit wonders. That's pretty much the end of them. You start to see them after the Harrison Act lose their strength. Now you have an industry that is looking for—
- 1.58** By the way, you should also know that we talked before about the discovery of cocaine, the discovery of heroin, the discovery of morphine. The industry is also attempting, once they realize that these are very addictive products, to come up with less addictive products. In 1916, they had a synthetic version, a different variation: oxycodone. They thought, "Oh, that's better. That's better than morphine, heroin, and everything else. Oxycodone. We can really market that, and it'll be better." We know about that in OxyContin, that hydrocodone, 1920, a variation of heroin. So, they're looking always for what I call the "me-too" versions of addiction to see if they can come up with something new that might get around the Harrison Act and still be new to market.

- 1.59 But as a practical matter, they have very few products. The products after the Harrison Act until World War II and penicillin really become a series of laxatives, malt, headache adjustments. It almost is like a list of an industry looking for a real product. There was a real product. In 1922, one of the greatest discoveries in medical history takes place in Canada: two researchers come up with insulin. Boy, you talk about a game-saving drug.
- 1.60 Now, a sidelight on the pharmaceutical end of this, we talk about this. Those two researchers, they said, "You know what? This is such a great product. It's so important. We should essentially give it away." You come up with medicine that changes people's lives, you shouldn't really make money on it. So they licensed it for a dollar to Lilly, to Eli Lilly. Lilly taught them a lesson. Lilly kept the patent on that, and we'll talk about it later, for 40 years. The patent is only good for 17 [years] in America, but companies learn how to game their role, how to make slight adjustments, therapeutic dosages, evergreening, make a difference on how it's done, and you keep that pattern for 40 years.
- 1.61 Insulin prices—we've been talking in the last few years in this country about making insulin less expensive and more affordable. Administration is talking about \$35 a cap, for instance. We're talking about insulin prices today on a product invented in 1922, given away for a dollar by its inventors. And an American company said, "Oh, yeah, we'll figure out how to lock that up." Lilly didn't have the patent on the item. It had the exclusive distribution and worked that into a moneymaker.
- 1.62 I understand. I'm a capitalist. I appreciate companies that make money, but we are talking about the drug industry, and I say always that it's an unusual industry because it's at the intersection of public health. So it raises these uncomfortable moments sometimes about always maximizing the bottom line at the cost of public health.
- 1.63 Insulin is a perfect example. When I say that they had nothing to sell, and insulin was their one big product at this point, I'll give you the best evidence of that. In 1909, Moody's, which is the company that ranks American companies; it gives you the size of companies and industries. 1909 is when Moody's started to do that. They put out their first listing of American companies by category, and size and influence in the American economy. You know where the pharmaceutical company was? The industry? It didn't exist. Moody's didn't even list the drug industry as a separate industry because it wasn't big enough. It was a subset of the chemical industry. That's what they thought of it. It was 20 years until 1929 before the pharmaceutical industry had even broken out as a separate industry.
- 1.64 So it's hard to imagine today, we look at Big Pharma, and we think of it as this monster, this Goliath, and it is, but hard to believe that just 100 years ago, it was this group of mom-and-pop and small stores and manufacturers with a few addictive products who weren't even considered a real industry. The growth has been spectacularly large.

- 1.65 With the drug industry, going into the future, time and time again, we get a product that turns out to have a problem, and then people will say, "Well, we've got to come up with a better version of that product, not give it away." So when Valium becomes the most popular drug, and we'll talk about this in another lecture, for 15 years in a row, and then it is heavily abused, everyone says, "Oh, benzodiazepines, we have to get away with that. Let's come up with something else. Xanax, that'll be better." So you don't necessarily come up with a different alternative. You come up with something that looks better at first, and that's what they were trying to do here.
- 1.66 Refining dependency—they're still not discovering cures, but they are refining and profiting from what I call this chemical dependence by people. Best example, barbiturates like Phenobarbital. That's the business model. In the 1920s, we're getting a pretty good idea of it's firmly established. You sell potent medicine, it's often addictive, miracle solutions, and you don't have to disclose harms, nothing about cures. You don't have to prove it, and the regulations are still pretty weak. So sell the potent psychoactive substances as miracle cures, avoid disclosing ingredients or harms whenever possible—they did that time and time again—and exploit the regulatory gaps. This is one of the things that I think any industry has become good at this. I understand this. But the drug industry was good from the start, which is they look at the law that's passed, and then they figure out how they can go around it. That's what happened here in this.
- 1.67 The use of advertising to shape public belief— long before there was direct to consumer ads, which we are bombarded with billions of dollars of TV ads in the drug industry since 1997 and on, in this era, there's no limitation on advertising. Here you go in, and you advertise. It was basically newspapers because people actually used to read newspapers. There were billboards. There's a British writer who came to America around 1900, who wrote a book later about travels through America. One of the things I loved that he talked about was the billboards that he would see for miles on end about patent drugs. They didn't have that in Britain at the time. They had patent drugs, but they didn't have the billboards because they didn't have—and he would see them forever around the cities and that. They were advertising heavily.
- 1.68 Focus on the symptoms, not the cures—you're treating dependencies. So somebody has pain, they have asthma, they have a cough. You're giving them something that you think might be able to alleviate that, not necessarily cure it. Would you have trusted these medicines? Would I have trusted these medicines? I ask myself that sometimes. I don't know. If I had a hacking, terrible cough and I couldn't get rid of it, and I tried one of these elixirs that got rid of the cough, and I didn't know that it was morphine, would I be worried about it? Maybe not if I didn't know what the downside was.
- 1.69 We look back at it now, and we're shocked and thinking, "How is that possible?" But it's hard, and that's why I say to put your minds back into that. It's not as though these

companies were selling to a public that had to be forced to buy them. The public was enthusiastic and had loyal followers often to each of these medications.

- 1.70** Modern parallels—just as a patient in 1900 might have avoided some cocaine-laced topics, I think today that as consumers, we are the ultimate, we're the patients and consumers, we have to have this critical eye about medicines involving. It's important to remember the historical part of the profit-driven business. They will always exaggerate the benefits while downplaying the risk. That's part of the industry. I understand that. That's not to say that there won't be, as we go through this, some really important life-saving drugs that have changed the course of human history. I know that may sound dramatic, but penicillin is one of them, with all the downsides afterwards, but it saves lives across the board. So, there are some really important discoveries. For that, the industry is to be rewarded and congratulated.
- 1.71** Then what happens is, I say, we have people working inside laboratories who are really dedicated, often scientists, working on trying to find a cure or better medicine. They are the real nerds of science. Then they invent it. It goes on. The people, by the way, who invented some of the biggest drugs that were used later, like Valium, which made Hoffmann-La Roche so much money over a period of time, the scientists who invented it got a few thousand dollars as a bonus, won an award from the company, but they got nothing else. That never bothered them. Those scientists are working as employees to try to create the product.
- 1.72** Then, it goes over to the marketing department and the sales department. That's where a little bit of the over-exaggeration comes in. They know what they can push it for, but they're pushing it for even more. Remember, back here, they don't even have to prove that it works. Not yet. Nobody's telling them that they have to prove that what they sell it for is accurate. Nobody is telling them yet that they have to tell you what the harms are. They just have to tell you the ingredients. This, in some ways, for an industry that is selling things to patients, is the golden age. The real golden age comes later, but this is a golden age in terms of not having to do something.
- 1.73** What are the summaries and takeaways from this lecture? The roots of Big Pharma are unregulated, profit-driven, often dangerous medicines with little or no regulation. We know that. The business model is profitable for public health, but you will see later, as we talk about this, that this persists today. Understanding this history, I believe, is the first step to becoming a smarter patient, because if you can recognize the historical patterns of what I call the deception or exploitation, you're a little bit more attuned. Don't become cynical. I always tell myself, "Don't become cynical about it." I'm going to have to take some of these medications as I go along, and hopefully, we'll live long enough that at some point I'm going to need some medication, but I don't want to become cynical and say, "It's all bad." But at the same time, you have to be a discerning consumer. You have to take control of your own health in this way. By grasping those early origins, I do think we understand a little bit more about the industry today.

- 1.74** Profit, power, marketing, and shaping American medicine are the highlights of it. Profit-driven motives start from the very beginning of the industry. Strategic marketing campaigns and impact on patient welfare—the focus is often on: how much are we selling? What's the profit margin? What are we taking in? Patient welfare becomes important if it starts to hurt sales, no question about it, but it isn't always the first item on the agenda.
- 1.75** The roots lie in these addictive substances. I also think there's something else on this. It's not a coincidence that the industry grew up on addictive products. Addictive products, if you are the seller, the manufacturer, are good products to sell. I don't mean good in terms of the effect, but good in that you get a patient, a consumer, that's going to be there for your product. The drug dealers know that. I'm not talking legal drug dealers; illegal drug dealers, cartels know that. There's a power to that addictive product.
- 1.76** These companies, in the very, very beginning, without any doubt, underestimated, pushed aside, were in denial about the addiction for things like morphine. It took years for them to realize that morphine was, in fact, addictive and that people who stopped using it for pain went into withdrawals. That's why they developed things like heroin. Then, they realized that heroin was addictive. So then, they came up with oxycodone, maybe hydrocodone. Those could be better. So they realized it. But when they realized they were addictive, cocaine with physiological and physical addiction— not the same as heroin, but different—when they realized it, they didn't stop. It's not like they said, "Oh, my God, an addictive product. We can't market that. We don't want people taking barbiturates all the time with that." They just said, "Oh, yeah, addictive. What do you know?" They kept doing the same thing. That's that origin of it in terms of that.
- 1.77** Always question the marketing story behind a drug. Don't take the pharmaceutical company's claims at face value. You're going to hear them in America and in New Zealand, the only other country on Earth that allow direct to consumer ads. We're going to hear them in ads all the time. Examine if there's motivations behind the tactics, the promotional efforts. Remember, promotion is being had here.
- 1.78** Read the labels carefully. Review the ingredients, the warnings, the disclosure. That sounds like common sense, but at a later point, you will see in another lecture that when companies had to disclose the side effects or the harms, then they had to put in a packet, a warning packet of what side effects might be, they fought with the government to make sure that initially, those packets would go to the pharmacist who was dispensing the drug, not to the patient. The idea was that this drug company said, "Oh, by the way, you know, you're going to force us to give these disclosures out. Okay, we'll send them to the pharmacist. The pharmacist will do what the pharmacist should do. They'll read this list of side effects, and then every time it's dispensed to a patient, they'll tell the patient all those side effects." Well, that didn't happen. Maybe there are

a few pharmacists who did that the right way, but for the most part, even when they had to do labeling and include those initially, they didn't do it.

- 1.79** Then, when they had to give labels, finally, when you picked up a prescription of patients, they decided, "You know what?"—clever marketing—"let's not just tell them the four or five things that might go wrong. Let's give them copies of backgrounds of the studies and everything else. Let's make them these little foldouts that when you fold them out, they're like four feet long and in little tiny print, the smallest font possible for the human eye to see, and people will say, 'Oh, forget that,' and they'll just take the pill." But that's the sidelight. Here, you didn't have to even do that.
- 1.80** Ask what evidence supports any claimed benefit. All right. Now, I understand you're not going to be able to do this all the time, but if your doctor ever tells you that they have a brand new pill that's going to treat a condition you have, whether it's migraines or whatever else, can you say—if you do, your doctor I don't think you're probably very strange—but, "Can you point me to the clinical studies that show that or some studies that show it might benefit me or back up this new drug?" It's always nice to know.
- 1.81** Investigate whether a drug treats the underlying disease or just manages the symptoms. This is a problem. If you have terrible symptoms, you have migraines, let's say, you may take a drug that alleviates those migraines but isn't treating the underlying cause because we don't know what's causing the migraine. You know what? If that's helping get rid of your migraines, that's fine. But you just want to be able to ask the doctor, "Are you actually treating the cause of what's happening with me, or is this just treating the symptoms?" Sometimes treating the symptoms will be fine. That's all you need to do. But you need to understand it because I think a lot of people take drugs from doctors, prescribed, thinking that they're getting rid of the underlying cause of their It's a problem. Not necessarily so.
- 1.82** Be wary of medications positioned as quick emotional fixes. Boy, I'll tell you, mental health, stress, and mood issues, it's a wide open area, and we'll talk about this at another lecture, but if you're marketing a drug for high blood pressure, you can do a test to see if somebody has hypertension. You can see if their blood pressure is high. During diabetes, you can do a sugar test to see what their glucose scores are and everything else. With mental health, it's observational.
- 1.83** People are determining, doctors are making decisions on your mental health, whether you're clinically depressed or whether somebody has an anxiety that needs medication. Think of this: pharmaceutical companies are selling products to alleviate those conditions. So there's a tendency to overprescribe. Do we have an increased incidence of children with attention deficit, anxiety disorder? I'm sure. So, it's being diagnosed more. But is the drug Ritalin and others to treat it being overprescribed? Absolutely. Do the drug companies say, "Oh, God, we've got this product Ritalin. It's being overprescribed. We don't want to sell more." No, they always want to sell more.

- 1.84 Note the ingredient, what I call opacity and exaggerated claims, the fact that they weren't opaque, that they didn't have full disclosure, none of that disappeared. It evolved. That's what we're talking about with the side list of things that are so long.
- 1.85 Preview of Lecture 2: The Penicillin Project. You think going from morphine and heroin to penicillin, it's a good move. We're moving in the right direction. I tell you, it's groundbreaking. The antibiotic revolution. It's the 1950s. It's golden age, another golden age. There are a lot of golden ages of the pharmaceutical company, another golden age of the pharmaceutical company, the 1950s and antibiotics, and the birth of the modern pharmaceutical empire. Now you think I've misled you because I told you the birth was morphine and heroin and everything else. The birth of the industry as we know it really takes place in this era coming up. We'll get insights into how pharmaceutical industry was transformed in World War II by that. Thank you very much.

Lecture 2: Post-War Drug Boom

At a glance — The post-war drug boom.

- 2.1 Today is the story of science, because we finally get to a stage in which you have real, true breakthrough science, but also the story of how science meets profit.
- 2.2 We were talking before about the fact that this is an industry that is at the corner of public health. Boy, this is a great time as an example of it. World War II had created these unprecedented conditions in many ways, just like the Civil War had, in terms of what was happening on the battlefield. What World War II did was create the possibility for a government-academia collaboration because everybody is chasing the same goal.
- 2.3 On steroids, in an emergency, we saw it with COVID, the government found billions of dollars all of a sudden to be able to pump into vaccines and manufacturers, and what was going on. The government finds the money, and they're fighting a war, so they were definitely doing this. This process really reshapes the pharmaceutical industry. It catalyzed in many ways the globalization of the industry because they come up with a product that everybody can end up using.
- 2.4 I'm going to tell you in this lecture, telling you what I'm going to tell you, and then at the end, tell you what I told you. But what I'm hoping to tell you in this lecture is how World War II transformed the industry. The Penicillin Project becomes a blueprint, a template that the drug industry will use time and time again on other drugs going forward. The rise of these corporate empires through the antibiotic revolution—and I do mean empires. We're going to take these little companies like Pfizer, Lilly, Merck, that really weren't even on the radar, and we're going to make them grow into these mammoth corporations that they are today. So it's that impact of World War II and the transformation leading to Big Pharma and the shift from cures you'll see eventually to chronic conditions.
- 2.5 So World War II is this unprecedented effort to treat battlefield infections. The rise of mobilization is just coming in at the right time for the pharmaceutical industry to be able to have this happen. There is always a story about the chance of how this happens, meaning that it takes a series of events. It's not just as though World War II gets underway and a group of drug companies get together and ask the government, say, "Can we do something about this?"
- 2.6 But there's this little chance discovery of how penicillin comes around, and it really is unique. That is what I call the Penicillin Hunters. That's at Oxford. The Oxford turns out to be key in this story. Part of the story is— I took that picture at Oxford this past June. That's Trisha. I'm in a tuxedo, not because I was going to Oxford. I was going to Oxford to participate in a debate on the pharmaceutical industry and profits, and I debated on one side. On the other side of the debate was a former head of the FDA and the chief

medical officer from Pfizer. They didn't like any of my ideas for reducing profits. But I am happy to report that in front of a group of students at Oxford, the side that I was on, the two of us that argued for taking excess profits out of the drug industry, we won by a 10:1 margin. But it's only students at Oxford. They may not realize the real world, that the head of the FDA and the head of marketing in the drug empire at Pfizer may carry more weight with Congress.

2.7 So the way this ends up happening is interesting because it goes back to 1928, this Scottish bacteriologist who is working on staph infections, little infections that can take place. He had been a surgeon in a British military hospital in World War I in Northern France. So he knew firsthand that infections cost lives as much sometimes as being shot by a bullet on the battlefield. So he saw men, sometimes women, but basically men, but he saw even in terms of nurses and others, get a cut, and they couldn't treat it, and it would get infected, and it would cause a staph infection and a blood infection, and people would die. So he came out of that World War I experience understanding that. His area of study is staph infection to see what he could do with it.

2.8 He goes on vacation in 1928 for a couple of weeks, and he has these petri dishes that he decides have these staph microbes on them, and he's going to clean them up when he gets back. When he gets back, the weather had been unusually warm at that time in Scotland—in Scotland, if any of you know the weather, warm is unusual, almost on its own—and that warm weather, he comes in, he looks at the petri dishes to go to clean them, and there's this greenish-looking mold around the edges of the petri dishes, and he hasn't seen that before. Not only is there this mold, this greenish mold, but the staph material that had been left on the petri dishes had largely disappeared. It's like this mold had grown onto the dish and started to eat the microbes that were the infectious microbes. And he thought, "Wow, that's unusual."

2.9 That leads to a year of study with his lab assistants on whether they can do something with that. They find out that they can start to grow a variation of that mold by letting it sit there. It actually ends up eating in terms of the petri dishes, other types of staph infections. But what they can't do—it's from the penicillium family, he finds out; it's from this line of germs that comes from that, that's where it comes from this growth of mold—he cannot extract, pull a pure alkaloid out of it. Remember what we were talking about before in Lecture 1? You have to get the pure alkaloid pulled out of it to create the basis for the drug. That's how cocaine comes around. That's how you ended up getting morphine when it's finally done. Here, he can't do that. He can't isolate that.

2.10 So he publishes a paper in 1929 about this unusual little mold that he calls a mold juice that is eating up staph bacteria. But he downplays its potential because he really doesn't know. He doesn't have the PR. He's not a Barnum & Bailey. He doesn't have that thing. So he downplays what the potential is, and he says, "For an English audience"—Penicillium, in Latin—"I'll call it penicillin." So that's what he calls it. That's where penicillin comes from as a name. And it disappears. In 1929, nobody pays

attention to the paper. It's one of another—there's hundreds of papers being published. There were a few researchers that read it and tried to pick up his research and do something with it. They never could.

- 2.11** In 1936, seven years after he publishes, a group of researchers at Oxford decide that they might be able to do something with it. It's really interesting because it's a Rhodes scholar, Howard Florey and Norman Heatley, and they take in this young 29-year-old doctor who's come in Berlin—his Jewish family had fled the Nazis in '33—Ernst Chain. He's this brilliant member of the team. They put them together, and they start to see if they can do something with penicillin. Now, they are able to pull the alkaloid out. They are able to isolate it. They start to do human studies on penicillin only in 1940.
- 2.12** Now, think about this. This team at Oxford, they realize the potential of this drug. They believe that if it can kill staph bacteria and microbes, it might revolutionize medicine. But Britain is already at war. They don't have the money at Oxford to be able to do the study they want to do. They go to the British government, and the British government essentially says, "We're spending every penny we can fighting the Germans," because the war had already started after Germany had invaded Poland. They say, "Let's see if we can make some progress on it with human experiments. Then, if we can, we'll figure out what to do next."
- 2.13** So, they start the human experiments in 1940, late. This is an age in which we don't yet have rules about what you can and cannot do by asking somebody to participate in a clinical trial. So they went to terminal cancer patients because they weren't sure if they gave the concoction in an IV—about 300 milligrams coming through of penicillin as they had created it—to a person, whether it was going to help them or kill them, whether it was going to cause them to break out and be unable to breathe or whatever else. They just weren't sure. They needed to see the effect on people.
- 2.14** So they went to a number of these cancer patients who were literally in their last weeks and asked them if they would be willing to have this drug administered to them. They said, yes. It was really a remarkable moment, this history in which you see that. They gave the drug to a number of those patients, and the patients ended up dying of the cancer in a short amount of time. It didn't make any difference really on their treatment, but they didn't develop any side effects, in which those three researchers would say, "Oh, my goodness, we're giving a drug to somebody, and it's an absolute disaster."
- 2.15** Then comes the critical moment. There's a policeman in his 30s who's been admitted to the hospital, who had been pruning roses, and he had cut himself on his face, from one of the thorns, and it got infected. That infection had spread. He had lost an eye by the time they got to him. He had abscesses across his arm and on his chest. He was on the way from probably being a few days away from dying. They asked if they could administer that to him, penicillin. His family agreed, and they gave it to him. It's one of the great stories in medicine, because in a day, his fever had broken. In three days, he

was able to stand up and start to move around. In five days, he was essentially, they almost thought he was healed.

- 2.16 There was a problem. They had only created a small amount of penicillin. They had no more to give to him, and they watched over the next couple of weeks as the infection returned and he died. In an odd way, they had their ultimate confirmation that the medicine that they had given him was life-saving. They just needed more of it to continue until the last vestige of the infection had been eliminated, and they weren't able to do that.
- 2.17 At that moment, they decided, two of them, to come over to the United States, and they made an appeal to the US government to become, as I call, the government as a drug incubator. Not an easy sell because by the time they're coming here, we aren't yet at war with Japan and Germany, but the eye is on it. The US doesn't have necessarily a lot of extra money to be able to provide, but it says, yes. They get sold on the idea of this medication being literally one of the biggest changes that could take place if there's a war.
- 2.18 Roosevelt makes a decision to have a secret program launched, which becomes the second largest secret program during World War II, the first one being for the atomic bomb, the Manhattan Project, and the second one being for penicillin. They created a lab out in Peoria, Illinois, and they bring in the British researchers, and then they need just one other thing: they need American pharmaceutical companies who might be able, if it becomes successful, to manufacture it and manufacture it in quantity. They don't have that ability, but they're thinking of how they can do that.
- 2.19 So they ask for volunteers from the drug industry. Twelve companies end up coming in, including all the big names that we talked about before. They'll be part of this process going forward. This is really it. It's Pfizer, Merck, Squibb, the Oxford, and the American labs. You have academics working on it here, the Rockefeller Group, and they're all trying to get these high-yield strains of the mold. They're looking for mold that could come in that might be able to help. You'll see this at another point.
- 2.20 The Peoria lab that's trying to create the best, most potent form of the penicillin mold decides at one point, "We should ask for samples from all around the world. Maybe there's a sample of this mold on some moldy piece of produce in India or in Argentina or somewhere else that's better than anything we're working with here." They literally had samples sent from American diplomatic missions and everything else from around the world. It turns out to be that a moldy cantaloupe bought at a farmer's market a couple of miles away from the lab in Peoria, it turns out to be the most potent strain of the mold, and they're able to work with that.
- 2.21 So wartime collaboration, government, academic, private industry, working together to rapidly develop and mass-produce penicillin for World War II. That's the backdrop. The government funding, big time. The US provides all the money, and we're talking the

equivalent of hundreds of millions of dollars today, but millions of dollars have then. They recognized the importance of this. The academic contributions—there are unsung heroes in terms of the penicillin story of people who are working inside these academic institutions to provide advances in terms of everybody working together on a very collaborative process. And the private sector expertise— in comes the private pharmaceutical companies looking to say: how can we help? That was discovered pretty quickly. That collaboration between government, academics, and what I call the drug companies becomes a formula that you could replicate in the future, sometimes done, but not always.

- 2.22** So the Penicillin Project: The US government partners with the commercial manufacturers to treat battlefield infections. The Peoria cantaloupe, as I said a moment ago—I know it's hard to imagine, but you now know that if anybody should ever ask you in a trivia game, "Where's the cantaloupe from that made the difference on the Penicillin project?", you will know where it's from. That is the one that they end up finding. Then, mass production leveraged deep-tank fermentation. So, what they're looking for is they're looking for deep-tank fermentation for penicillin.
- 2.23** Pfizer comes up with the innovation. They come up with this citric acid, leveraged. It makes penicillin faster and makes it as potent and pure as possible. Pfizer's fermentation breakthrough is the one that leads everybody else. It revolutionizes mass production of penicillin. This technique allows for these large-scale tanks that can manufacture the drug at quantity needed to spread it around the world.
- 2.24** Problem? It's really costly. They don't have them. The government says, "We will build them." So, the government not only builds the fermentation tanks and the manufacturing facilities at taxpayer expense, because taxpayers will benefit, for Pfizer, but for all the other companies as well. We're building essentially the infrastructure for what I call large-scale manufacturing of antibiotics at government expense during this period because these companies are going to provide the material that is so important.
- 2.25** Now, one thing: they do succeed. Units of penicillin being made in 1941 and 1942 are small. In 1943, it's enormous. By 1945, they're providing hundreds of millions of units of penicillin. Something that the government did on this that didn't go unnoticed but didn't seem important at the time, it said, "We're buying it, we're providing it, we're putting everybody together. It's our money, taxpayer money, and you, the drug companies, thank you very much for being part of this. You're patriotic companies, but guess what? Nobody gets to patent penicillin." It was a little bit different.
- 2.26** As a matter of fact, as all this was going on with penicillin, there had been the discovery in Germany by a German chemist of sulfa drugs. Sulfa drugs are very powerful, and they're used for infections often. They hadn't been invented until 1936. Those are patented, and they are being put out by companies, their treatment. But penicillin is now available to treat infections across the board. Nobody owns the rights to it. What does that do to the price? A pretty good example of a free market. You love the free

market. When you have a dozen companies making the same product, and everybody wants it, the price drops and the price of penicillin dropped and dropped and dropped and dropped. Drug companies hated that because they had, in fact, had this process, and they were selling a lot of penicillin, but they were not making any money from it.

2.27 That was the interesting thing. That was the template where, if you had a major crisis and the government got behind it and brought drug companies in, you might be able to get them to do it for the good of the public and for public health and not necessarily make outsized profits on it. That's why, and we'll talk about this far down the road in another lecture, when COVID comes around and was happening, I thought naively, "The Penicillin Project. The government's going to do, and all the governments are going to do, what they did in the Penicillin Project. They're going to put all this money into finding a vaccine and treatment, and they're going to make sure that nobody owns the patent, so the price drops to near zero." They forgot about the "nobody owns a patent" part, but we could talk about that later.

2.28 So, postwar surge—think about this. The streptomycin had been discovered in 1943. That's the first effective treatment. It's an offshoot, as we talk about antibiotics for tuberculosis. So now you have not just penicillin helping in terms of infection, you have streptomycin helping in terms of TB. Aureomycin—I always love that—arrived in 1948. Tetracycline comes in the 1950s. These drugs aren't just medical miracles, but they are really what will fuel the American drug company in the '50s, and here's why.

2.29 So you have streptomycin discovered, the first effective drug against tuberculosis. Aureomycin, the first broad-spectrum antibiotic introduced—broad-spectrum, they love this because it affects a whole range of different bacteria. So you're not sure what somebody has when they present themselves at the office and they're sick with something? Give them a broad-spectrum antibiotic and you assume that some element of it is going to work to help. In '55 comes tetracycline, very versatile antibiotic. And from '45 to '55, the end of the war, pharmaceutical revenues increased 600% fueled only by antibiotic sales.

2.30 Now, here's why. At the end of World War II, German pharmaceutical companies like Schering and Bayer, they had been providing one out of every two prescriptions in the world. The German pharmaceutical companies, you notice it's usually German chemist or apprentice who's discovering morphine or cocaine or doing mustard gas. The German scientists, they discovered with Bayer, acetaminophen and aspirin, and heroin, and barbiturates. Their scientists were at the forefront in many ways, and their established companies were churning out and selling the most pharmaceutical products.

2.31 What happened at the end of the war? The infrastructure for those German companies had been destroyed, had been bombed out. In the closing days of the war, the American and British bombing campaigns on Germany had taken out the physical infrastructure. It would be as if Pfizer and Merck lost their fermentation labs and had to

build them from scratch. So essentially, the American companies end up with an unexpected opportunity because the greatest competitors in Germany have been for the the moment put on the back burner. They will have to rebuild.

2.32 And what happens by 1950, five years after the end of the war, 80% of all the drug sales in the world were coming from American pharmaceutical companies, almost all of those from the top 10, and 95% of all the profits in the drug trade were coming from America. So you've had a shift in terms of an industrial weight that in history is almost unparalleled. American pharmaceutical companies would have competed. They would have continued to gain market share. They would have continued to increase their profit margins. But their biggest competitor was on the losing side of the war and lost their infrastructure. It put them into a position that they could not have imagined before the war took place.

2.33 By the way, in 2005—just jumping forward to one thing, we may not come back to this—the top 10 pharmaceutical companies, they all traced their origins back to the penicillin program. The top 10, literally, when you're talking about it 60 years later, all came out of the penicillin program. So it's hard to overestimate the head start you got based upon this.

2.34 Antibiotics become a gold mine. A moment ago, I said to you what? That Pfizer's revenues increased, for instance, 600% after World War II. Eighty percent of those revenues came from tetracycline alone. Talk about a one-hit wonder. This was the era in which the drug company set something interesting: astronomical profits. Eighty percent of Pfizer's profits coming from just that single drug. The profits funded the earning multinational subsidiaries.

2.35 If you go through and see where drug companies in America are expanding and opening branches in the UK, in France and South America, it's in this period in the '50s. New research labs, they're expanding. They aren't just taking the money of sales and going on holiday and spending it on yachts and cars. They are putting it back into the business to make the business expand, and they're doing it very aggressively at this time. These are the corporate structures. These are the early days of the structures that seem familiar to what we see today. This is the period from which the 10K reports and the books put out about the history of the companies by the companies themselves will take you back to, because this is the good story of their start. This created the corporate structures of pharma. It's a hybrid of scientific discovery we talked about inside the lab, together with this industrial scale, and then the selling process.

2.36 What happens here? You couldn't patent penicillin. Everybody understands it, but you could patent every other variation of every possible antibiotic you turned out: streptomycin, Terramycin, Aureomycin, all of them end up getting patented by the companies that first discover them. And then, everybody's looking for their version of it. How is it possible that I might compete because they have it?

- 2.37** You get into the first wave, and boy, this becomes part of the drug industry over time. All allowed at this point in the 1950s by the US lack of regulation, still allowed today to some extent, what I call "me-too" drugs. Me-too drug is you take somebody, a competitor's drug, they've got a version of an antibiotic, Aureomycin or that, and you say to your lab guys, "Gee, can you work up something similar?" And they change literally, in some cases, a molecule or two. Now, is it a better drug? We don't know because no clinical studies are being done. Companies don't have to prove it's effective. They just have to sell it at this stage.
- 2.38** No requirement to prove what you say it can do is what it does in the 1950s. Pretty unusual. There had been a law passed. We'll talk about it later in 1938, the first drug law, but it really wasn't very effective. What that drug law said is you have to tell us if you're really harming somebody. It's the first time you had to say if your drug was going to kill somebody. That's really what it meant. You didn't have to list all the side effects, but you had to let them know if that drug might really be dangerous. Other than that, you didn't have to show it was effective. So, they're changing sometimes the chemical structure very, very lightly and no therapeutic benefit, but they're able then to patent it. Once they patent it, they have to hype it as better than another a drug.
- 2.39** Pfizer and some of the big companies say, "Hey, we're not going to put up with that. We're going to stop you over there and other companies from copying our drugs." They take them to court. They say, "Patent infringement." That's why you have a patent. It goes up through the courts, and the courts, good judges, but they didn't understand the science; they couldn't get a handle on it. They eventually come down with a series of decisions that say that if you change the chemical structure of the drug, you can patent it.
- 2.40** That opens up the Wild West, in a sense. Every company in America that's selling drugs in this period does what I call dozens of me-too variants. They all are marketing with a promise: broader spectrum, fewer side effects, better cure rates. They're selling to doctors—not selling the actual physical antibiotic, but selling the idea of what it could do. This becomes the period at which, and we'll talk about this in a later lecture, nobody knows it, but bacterial resistance is forming to antibiotics, the overuse of antibiotics, because in the 1950s, it was dispensed like candy. Doctors gave it away if you had the sniffles. They gave it away if you had a sore throat. You came in and complained you had a bad head cold, or that, well, why don't you take this antibiotic for five or six days?
- 2.41** There was a study in the late 1940s in Japan, in which they said, "Gee, some of this penicillin and that, we see people who have been using it for a number of years now, two, three years, and it seems as though they are getting some resistant types of bacteria that it doesn't respond to any longer." Doctors in the West said, "That's the craziest thing in the world. That must be something that the Japanese got all messed up. It must be something peculiar to Japan because it's certainly not happening here."

Little did they know that it was, in fact, happening, and the overprescribing of it was, of course, one of the problems.

- 2.42** But the overprescribing is part of why they're making so much money. So there's no incentive for the companies to cut back on that. This antibiotic boom is the most profitable, makes pharma the most politically influential foundation for the modern company. And once the pharma companies understood that curing a disease and antibiotics could do that was profitable, there was no turning back. They were going to be chasing that as hard as anything. They didn't have to worry anymore. That was the good news about the narcotic end of it. They seemed to have finally found a series of products that would work.
- 2.43** The profits from all of these antibiotics, they fuel the expansion, and the companies used it to establish not just international supply chains, which is something that I think becomes incredibly important in the profit end of it, but they buy out smaller competitors. So, you're a mom-and-pop starting up right now. You want to try to make drugs. You've got something that you think might be interesting. If Pfizer or Merck or anybody thinks you're doing something interesting, they'll pay the money to take you out. So you're not really seeing new companies start here that are becoming giants 20 or 30 years later. They're keeping that competition off. And by having the international supply chains, they're becoming a bit of what I call a vertically integrated institution. They understand how to make it, discover it in the lab, patent it, manufacture it, and distribute it so that the middlemen are cut out.
- 2.44** Dark side of the antibiotic boom: overuse and resistance, no question about it. There comes a point in the 1950s in which, boy, they are fantastic, the drug companies. Excuse me. In 1945, the end of the war, on a visit to America, the Alexander Fleming, person who had discovered penicillin and couldn't quite make anything of, he was worried from his own understanding of how it works, that eventually microorganisms and staph bacteria, their sole purpose is to survive and spread, will find a way to morph and change and become resistant to the same drug you're using to kill it. He warned American companies when he came here in 1945, "You should worry about this." They said, "Oh, yeah, thanks for discovering it. See you later." I think that that's part of it is overuse. Doctors are encouraged to use it subtly and explicitly, and they do.
- 2.45** They also come up with something—there were efforts in some of the legislatures to talk about, should we cut back on antibiotics? Nothing was done. We talked about the 1938 Food, Drug, and Cosmetic Act, which had limited powers. One of the things the drug companies were doing here in the '50s is not only are they doing me-too drugs and variations and dozens and dozens of antibiotics all doing the same thing.
- 2.46** But some clever people, as a matter of fact, get the idea—Sigmamycin is the first of these, marketed by a guy called Arthur Sackler, and we'll talk about him later—to do a combination dose antibiotic. So inside the companies, clever marketers said, "You know what? We've got this drug. It's selling well: Aureomycin, tetracycline. Why don't we

combine it with another antibiotic? And then the two of them must be more powerful, must be better. Two are better than one. It's more. Let's do that." And they combined literally over time, hundreds of antibiotic combinations.

- 2.47** It would take years, and you'll see the fight over this before Congress and everything else, to discover that in almost every instance, there was never a case in which the combined therapy was better than just a single antibiotic on its own, but the risk for side effects multiplied many times over. So, fixed dose, multiple combination antibiotics: big seller, no science, but eventually a downside. A little bit of this comes in here, but we're going to see more of this over time. Right now, the companies are—the '50s is the antibiotic period. It's also the period, by the way, I should say that not all companies are just making that. They're not all doing it. It's the big money driver.
- 2.48** There's some development also on drugs for conditions of depression. The big breakthrough drug was one called Miltown. You may have heard of it, maybe not. That's the predecessor to Librium. It's the predecessor to Valium. Miltown was an anxiolytic, rudimentary first generation that came out. There were copies of Miltown that were later done. It was marketed as a new age anti-anxiety medication and became incredibly popular. In places in Hollywood, people would line up at a local pharmacy to try to get copies of Miltown, which made it even more famous. People would talk about it on television. It became an image.
- 2.49** Miltown had replaced another drug, which had been out for a little bit of time, called Thorazine. Now, you may have heard of Thorazine. Thorazine was viewed as a breakthrough when it first came out because it was a pharmaceutical version that replaced lobotomy. We're moving along here in terms of the world, so lobotomy. By the way, the doctor who invented lobotomy, some of you may not know this, won a Nobel Prize for it. Sometimes it takes a while to discover that something isn't as great as we first thought. So lobotomies were done. That would be a cure for people with this deep ingrained type of depression you couldn't cure.
- 2.50** Then comes Thorazine, which is marketed by a French company initially as a chemical light. It's lobotomy light, but people are still like zombies. Then comes Miltown in the '50s. With Miltown, you're no longer a zombie. That's the good news. There are other problems with it, but Miltown, again, becomes the first of these drugs. There are a few people who are on the marketing end who see something with that, and they say, "You know all this rush to antibiotics? It's fantastic, but people take antibiotics for five or seven days or something, so they don't take it all the time. They take it when they're sick or that. But Miltown is pretty great. They take it every day. Oh, we like that. We should be maybe doing drugs for chronic diseases. If more drugs could be developed for chronic diseases, we have a steady income stream coming forward."
- 2.51** Antibiotics, you're not sure. If you get a pandemic, you get a terrible flu season, your sales go up. That's fantastic for you. But you're dependent on what's happening in terms of how sick people are. But if you can treat chronic diseases, you can treat them

for the long term on a different thing. Miltown was one of those, the first ones that really comes in mind, the incentive for chronic treatments. They realized drugs for chronic treatments, lifelong conditions, are far more profitable than the one-time cures, no question about it. What you're looking for, what are the dream drugs here that you want to try to develop? They're in the early stages for blood pressure, diabetes, depression is leading the way.

- 2.52** This industry begins to focus more on developing these drugs, starting in the late '50s and early '60s. This is the time in which they're now saying, we're in for the long term, the long game, as I call it— increased revenue from increased prescriptions. Very interesting here. Drugs for chronic conditions are fantastic because patients have to refill their prescriptions. You know when they're coming in, and it's reliable, and you can continue to push it all the time. The drug industry would say in relation to this last point: incentive to cure the profit motive creates an incentive to maintain chronic conditions rather than find causes for cures. Some of the industry might say, "It's not our business to cure the disease. We're treating it only." I'm not necessarily sure that's true, but there is no incentive for them to actually cure it. But to treat the symptoms, there's a lot of money to be made. I think that those implications will affect, as you will see, the development of new drugs as we go forward.
- 2.53** The globalization of the business—by the mid '50s, Sackler, some of you may have heard that name, the owners of Purdue Pharma, creators of OxyContin later, in the 1950s, they are three psychiatrist brothers. Two of them are working at a state mental institution until they lose their security clearance because they're also card-carrying members of the American Communist Party, unusual in the 1950s. The third one, Arthur Sackler, the oldest of the pharmacist brothers, is a visionary in some ways. He's a visionary in terms of thinking of how to take a drug and to sell it. He has a small pharmaceutical company. They really don't a drug, they don't have a product yet. But he's already creating what I call the foreign subsidiaries for what he hopes will be an international company. Forward thinker. No one's buying them out because he doesn't have a drug that they really want yet.
- 2.54** The mid 1950s, tetracycline has become the first globalized, marketed, litigated across multiple countries. It's controlled, and Pfizer is able to make more money on that than any other drug as they sell it from country to country. By the early '60s, you really did have these what I call sophisticated end of distribution networks that hadn't existed just a little bit before.
- 2.55** Pharmaceutical identity crisis—how much are they going to put into scientific research into new labs and how much into marketing? That's an issue that you see come up time and time again for the 1950s and on. This is a snapshot of the end of the '50s as we go into what I call this identity crisis. These are rough numbers I was able to pull out from all of the companies that report them.

- 2.56** Academic research funding, 15% of their money is going into providing grants to different academic institutions to help them develop new drugs. It's on average. Some of the companies are providing more, some providing less. About a third, 30% going into in-house R&D, and so that's important, their own research and development. Five percent going into regulatory compliance. Thank goodness for the industry. They didn't have to comply with a lot of regulations in the 1950s, so they didn't spend that much money on that. Then 50% into marketing and advertising. That figure doesn't change much over time.
- 2.57** One of the things that I get so aggravated with is when the drug industry successfully argues against lower prices today, in the last 10 years, whenever, in modern times, in our era by saying, "We need these profits for research and development," for the R&D part, the 30% part. But they don't say 30% part. They make it sound like the R&D is the key. There's no question, and I'll talk about this later, that developing drugs have become more expensive because now there are all the clinical studies and everything you have to do. A manufacturer can spend a billion dollars getting into the pipeline to get a new drug out. Big money. And not every drug succeeds. So when you fail, you fail.
- 2.58** But even today, if you take the top 10 drug manufacturers in the world, they spend more on average a year on marketing and on promotion than they do on research and development. If you add in share buybacks on the marketing—the marketing, the advertising, and share buybacks exceed by far the amount spent on research and development. So, I always say, do a little bit less on the share buybacks for public companies and a little bit less on marketing and development, and you could lower drug prices.
- 2.59** The odd thing about that marketing and development is, this is an odd industry. You have manufacturers of pharmaceuticals who really don't have yet prescriptions widely required, who are selling a product but not directly to the user. Where are the users? Where are the patients? They have to go through this middle section: doctors. It's an odd industry because doctors are assumed to be able to have the absolute knowledge of who should get the drug that's being developed. So the research, the marketing, and the advertising at this stage is being done to doctors. It's not direct-to-consumer ads.
- 2.60** The doctors are being bombarded with everything from pamphlets to booklets to little books that are put out. Valium, they had put out actually hardcover books about the benefit of the drug. Very fancy things. They run them in medical journals. They pay for studies that are more likely to be favorable about the drugs, so the doctors will be persuaded, and they convince doctors that the latest and the greatest is the one to give.
- 2.61** One of the great things that they knew from an early time on this is that doctors, for the most part, have no idea and don't care what a drug costs for the patient. They're not out there pricing it. They don't know, especially as private insurance started to develop in the '50s, '60s, and '70s, they don't know what your private insurance is. They

don't know if you work for a company. They don't know later if you were on Medicaid, if you had drug coverage in the 2000s on Medicare. Doctors will say, "Well, I think this drug is the one to treat your condition." They don't know if you have a \$200 copay or you're getting it for free. And so as a result, doctors aren't making the decision based upon dollars and cents.

- 2.62** Drug companies can be charging a high amount, and therefore, the doctor will still prescribe the drug. If the patient goes back and says, "Gee, I can't afford that drug," the doctor will say, "Well, it's only your health, but if you can't afford it, we'll give you something that maybe isn't as effective or as new or as scientific at the edge of cutting. We'll give you something a little older. It might work almost as well." So there's a psychological part that makes people want to sometimes, as we know, and on a life-saving drug like insulin, you don't have the choice. You have to find the money. And so the pharmaceutical company had an interesting thing in that they're advertising to the doctors. They're trying to do the salesmanship to make the doctors prescribe more and prescribe a lot and prescribe the latest and greatest without focusing on price.
- 2.63** And the laboratories you think about where work is being done, it's taking place and being built out in the '50s with the profits that are coming in. The global supply chains are established in the '50s, and they change over time and different companies come in. But the basis for it is here. The marketing arms for the sales to doctors are good. They will get better and more aggressive, but they're pretty good at this time. There's this hybrid model of what I call science, scale, and marketing, and it's taking place.
- 2.64** Think about this. This is an industry that in 1909 didn't exist, according to Moody's. It wasn't around. In 1929, they're on, but they've got one product: insulin. And then, in comes penicillin, and they become patriotic members of this massive wartime project. But it teaches them. They learn. They learn how to make drugs in massive and big amounts. The drug that they learn, ready to make is, since they've learned on penicillin, all the wave of me-too antibiotics, and that fuels their discoveries into other drugs that will treat other conditions.
- 2.65** Science meets salesmanship—one of the things coming just around the corner, in the next lecture, we'll talk about it, is this great collision between the federal government responding to a public health concern and regulation. As the government goes to its next great phase of regulation, this industry—remember I said before when we were talking about patent drugs, that in 1906 on the Pure Food and Drug Act, they had lobbied to make sure that they weren't overregulated. That's child's play coming up to what they will do on the next round of lobbying as they become very effective on the political end.
- 2.66** By the way, a footnote, before I forget, in 1938, on the law that had been passed. That law, which said you have to tell us if there's a deadly outcome, if your drug is really unsafe, also had an interesting provision fought for by the drug companies, which was, drug companies said, "You know what? Doctors should be able to prescribe a drug for

anything that they think it can be used for. So, if we're prescribing penicillin and we think it's good for an infection, the doctor thinks it could be good for treating, for instance, nobody was saying this, but treating depression, they should be able to prescribe it for that. If they think it could be good for growing hair, they could prescribe it for that. The doctors know everything about medicine, so they should have the discretion to use our drugs however they want to."

2.67 At the time that was done, there were just a few dozen drugs in terms of classes that were available. It wasn't that difficult for doctors to stay on top of what was happening. Now, today, and we'll talk about this in a later lecture, there are thousands of drugs that have patents and uses, and doctors do use off-label prescribing for a whole host of things. That power is still unlimited, and drug companies love it because it lets drug companies make sometimes, literally, I'm not exaggerating, billions of dollars in sales for the use of a drug for which it was never clinically approved in safety standards or anything else before the FDA. It's one of the greatest loopholes in American pharmaceutical manufacturing, and it goes back to 1938 law.

2.68 By the way, at this time, when I say science meets the pharmaceutical error, companies like Merck and others, they had had the benefit of academic institutions helping them during the war on penicillin. So they put together their research labs, reaching out and making alliances with academic institutions: Harvard, Yale, Johns Hopkins, and others. They wanted those involved. And those doctors, in many ways, they were serious academic researchers, but they also liked the benefit of having their name associated with the new drug discovery. And so there was a real symbiosis in the '50s and early '60s between the drug companies providing some research funds to academic institutions that then came back to them.'

2.69 So this legacy of this antibiotic era through the '50s revolutionizes the treatment of infectious diseases. That's fantastic. It's a great change in terms of medicine. It changes the practice of medicine in terms of how doctors approached things. It fueled the industry growth cycle, made them into global multibillion dollar empires, increased their political influence without any doubt. They understood it. We know.

2.70 We saw companies in the last 40 years form like Google and Facebook, companies that when they started out were worried just about their product, and then they learned something as they went along. "We've got to be able to lobby Congress. We've got to be able to lobby the politicians. We need to get up and testify there. We need to have lobbies." They all learned that lesson. The pharmaceutical companies were learning it here in the 1950s very well.

2.71 The shift to chronic treatments—companies realize that chronic treatments are more profitable for the long term. They're starting to focus on that. The rise of antibiotic resistance, the super bugs, hasn't become a major public health crisis, but it will. And what did the postwar gold rush do to the industry? Massive research facilities, global

distribution, heavy lobbying power, sophisticated marketing arms, strategic relationship with regulators, and business model favoring chronic treatments.

- 2.72** The one on the bottom right there, the strategic relationship with regulators, I think one of the most interesting parts is that at this stage, we all know intuitively what a conflict of interest looks like. But pharma was pushing the edges, and I'll talk about this in the next lecture. They were able to go to the FDA, and they were able to take the chief of the FDA for antibiotics approval, the person who had an oversight on it, even though there were no clinical trials yet, but the most powerful health regulator in the US government on antibiotics. Arthur Sackler and a few others said, "Why don't you become part of our magazine that we put out about antibiotics? Why don't you help us out?" Nobody thought there was a conflict of interest in that. So, they were able to have a relationship with regulators that today we would frown on.
- 2.73** Although I say that, there's still an open door in all of Washington between your defense contractor and then somebody leaves the defense department and they go to work for a defense contractor company. When we eventually get up to OxyContin, the FDA official responsible for approving the label on OxyContin, which went further than any other label, goes to work a couple of years later for Purdue Pharma. I hate that because I always think that's wrong. I still can't believe that we haven't had either side of the aisle.
- 2.74** And that's because pharma is very, very good. People ask me sometimes, "Who's worse for the drug industry? Who lets them get away with more? Democrats or Republicans?" I say, it depends on which area you're talking about. Sometimes Democrats have let them get away with more. Sometimes the Republicans have because they buy compliance on both sides of the aisle. They're very, very smart that way. And as a result of that, they're able to have an influence that stays persistent long past the time that it should.
- 2.75** Summaries and takeaways—I told you that penicillin changed the industry. It did. A blueprint for future drugs, no doubt about it. They all want to replicate penicillin success with the only difference being they each want to own the patent on it.
- 2.76** The rise of the corporate empires, antibiotic revolution is off in the distance. The shift from cures to chronic, the antibiotic overuse and resistance, and this lobbying power and structure. I think that moving forward, it's crucial for patients, for us, for consumers to approach these products with a critical eye because if you understand the interplay, they aren't just scientists working on cures. They aren't just research labs creating drugs that are going to alleviate your symptoms. There is an interplay between how they lobby in Washington, how they get approval and what they do, and the business of public health.
- 2.77** How to be a smarter patient other than not just taking antibiotics for sniffles? Who paid for the original research? Innovations come from taxpayer-funded labs, not corporate

R&D. I used to think that that was critically important. After COVID, I'm not always sure that that is as critically important because you can still have innovations come from taxpayer-funded research. It may not always be as good as possible as well. So, the research you have to be careful for.

2.78 Why is this drug being prescribed to me? The reason I ask that, suggest that, is it part of a trend? Is it Miltown? Is it Valium? Is it something that is so popular that at that moment doctors are prescribing it? Is it GLP-1 drugs for weight loss? Or are they prescribing it for you specifically? Because not just you tick off one or two of the conditions that might fit for a trend drug. You don't want to fall onto a prescription because it's a trend and they think you might benefit from it. Because I guarantee you, there's always a benefit, but there's no free-lunch in pharmaceuticals. There's always a downside somewhere down the road as well.

2.79 Is this drug treating an infection or being used as a convenience? I don't want to take antibiotics all the time. Do I have an infection? Can you tell? "By the time we take a sample and send it out to the lab and get it back, you'll probably be over it. So it's not worth it." They don't tell you. They don't want to do it. There's no business in actually trying to winnow it down. It's completely frustrating. Now, I'm sure if you had elitist medical care, a concierge doctor or something like this, they might be able to do it if they had the incentive to do it. But the fallback is easier for them, the doctors, and the drug companies love it. Just prescribe the antibiotic. "It's not going to kill you. You probably don't take it often enough to create too much resistance. Hopefully, you won't get a super bug if you travel internationally, and you'll be okay." So they just write the prescription. That's unfortunate because I do often want to know: is it really treating an infection or not? You can't really get an answer to that.

2.80 Are there—don't ask this because you'll really be considered a troubling type of patient—are there non-pharmacological options that work as well? Lifestyle, less profitable. We know with GLP-1 drugs, there are people who are morbidly obese who have tried to lose that weight over a period of years and have been unable to do it for any number of reasons. That morbid obesity leads to potential heart disease, diabetes, all types of problems. The people who are treated on GLP-1 drugs and lose 150, 200 pounds, it's life-changing for them sometimes. There are side effects, but it's life-changing in terms of those chronic conditions that disappear. But is there somebody who wants to lose 20 pounds because they're going to a class reunion and they want to use a GLP-1 drug? Maybe they could cut back on food and exercise a bit more, but I don't know if doctors are actually going to prescribe that for them.

2.81 What are we looking at next? Selling sickness, how human experiencing conditions is creating new markets for drugs. Pharmaceutical companies realize something. It's great to be able to treat an illness, but what if we expand what those illnesses are, the description of them so we get more patients in as it is? See how they do that, expanding diagnostic boundaries.

- 2.82** This, it seems to me, is one of those things that you'll see repeatedly in medicine. There becomes a collaboration between the drug companies and the medical associations and the testing. You have drugs that are being given, let's say, for high blood pressure. The normal for high blood pressure might be 140 over 80. Then over time, pharmaceutical company is saying, "You know what? Not enough customers. Let's have some doctors at a prestigious institution do a series of studies and write some papers about the fact that 140 over 80 is too high. The number should be better. Then we can reduce it to 130 over 80, and then it becomes 120 over 70. Oh, my God, that's fantastic. We just have millions of new patients."
- 2.83** Now you go to your doctor for your annual checkup, and your blood pressure is what had been normal two years ago, but now it's considered borderline. "Oh, you want to take the chance of a stroke down the road? Maybe we should give you a low-dose hypertension medication now to control it." It wasn't even treated with medication just a few years ago, but expanding the diagnostic criteria is absolutely critical.
- 2.84** You'll see this later in another lecture, there's a thing called the DSM, the Diagnostic Statistical Manual—I'll try that again, the DSM. It becomes what I call the Bible for psychiatric diseases. It expands over time what's considered a treatable illness. Sometimes a drug gets lucky when it's released. When Xanax came out, it was the first one to be able to use the latest part of a DSM condition that hadn't existed before for panic attacks. Fantastic. They were able to market the hell out of that. So I assure you, sometimes the new diagnostic boundaries fit for the sale of a drug.
- 2.85** Direct-to-consumer ads—we'll talk about this going forward. Just know that if you turn on the TV and if you watch an ad, you are one of two countries in the world that is being subject to that, unfortunately.
- 2.86** Key opinion leaders and the influence of experts—something very important. It happens in opioids, it happens in other areas. Pharmaceutical companies are smart. They go out to leading doctors in the field on those areas that they are doing drugs for, and they make those doctors partners with them. Not partners in terms of having a piece of the drug success, but saying, "You know what? You've written papers about this. We'll fund some more research for you. You can go on our lecture circuit and speak to medical groups and our convention, and we'll pay you for that." Those doctors become, in essence, partners with the drug company, and they become influential thought leaders in terms of other doctors who look at them and say, "Oh, if that doctor thinks this is a good drug for this condition, I'm on board." That happens all the time.

Lecture 3: Selling Sickness

At a glance — Selling sickness.

- 3.1** Two points I want to make in relation to what we just covered was— we were talking about putting patents on medicine and how they make it more valuable. The three British researchers, the two British researchers and the one German researcher who had come in, Ernst Chain, who was then 29 years old, who worked on penicillin. During the process of developing that, when they realized it was going to be a real drug, Ernst Chain, the kid of the group, the German, said to his two British colleagues in this meeting they had, "Maybe we should put a patent on it, on penicillin." And the two British academics at Oxford were horrified at that thought. Chain was a Jewish immigrant, and they actually said to him, "This would reflect badly on the Jewish community. If you asked for a patent, it would look like you were trying to make money on a drug that was really for everybody."
- 3.2** What do American companies do? They're so smart. So you have a patent on a drug. That's one thing. You own the drug. But you could also get use patents. So, a use patent is when, as a separate pharmaceutical company or manufacturing company, you come up with a better way of manufacturing that drug. So you're able to show that you can manufacture it faster, you can streamline it, the purity levels might be higher. American companies had, within two years of the end of the war, 250 use patents on penicillin. The British, much to their great anger, had to pay for the right to use the processes that became the industry standards to make the drug that they had discovered. So Ernst Chain's idea of putting a patent on the drug wasn't the craziest thing ever. It didn't happen, but the idea of use patents became predominant.
- 3.3** Also, when I was talking about antibiotic resistance, it is true that Americans were using more antibiotics than all other medications combined. But antibiotic resistance also started in the 1950s. We were ground zero in America for it because the 1950s is when we started to use antibiotics encouraged by the companies who were marketing penicillin, streptomycin, and other drugs—to use it on farm animals. So we started to use it in the 1950s on chickens, cattle, turkeys, and other animals to fatten them up sometimes, and as well as to protect fish in overcrowded farmed fish settings in which disease is spread. We are providing, by the mid '50s, 30 million pounds of antibiotics a year toward animals and toward fish farming.
- 3.4** Now, you don't care—if you are calculating the profit numbers at a pharmaceutical company, you're selling a drug, you really don't care if it ends up in a human or an animal, if that makes sense. You want to see your product moving. So the expansion of this for 30 million pounds, much larger afterwards, also helps to do it.
- 3.5** In addition, they became in the 1950s, the start of the use of antibiotics—very smart by pharma companies—in plant agriculture to protect crops sprayed over a two-week

period, normally, to prevent the growth of fungi and other bacteria that might damage crops. So you get crops that are having it as well. And I really do believe that at this point, this constant exposure of what I call the antibiotics from animals, they come into the food chain, from fish that come into the food chain, from the use on agricultural products, it adds to what's already happening to make this ground zero in terms of antibiotic resistance.

- 3.6 So, the pharmaceutical companies are providing the product, but it's not just being distributed by doctors to patients. It's also ending up in other areas. So, even if you were saying to yourself—and nobody was saying it in the 1950s—you were some advanced thinker who thought, "I think antibiotics, I shouldn't be taking too much of it," I bet nobody was thinking about what they were eating, fish or poultry, or what was happening in terms of that entire food cycle.
- 3.7 Now to Lecture 3 and how pharma turns marketing into medicine. Antibiotics had shown that you could transform medicine into actually doing something, but I really think that marketing could transform science. That's what you start to get into in this part. The idea that you could convince doctors to prescribe drugs, you can convince patients to take drugs, that idea is incipient, a revolution that's really going to take place in the '50s.
- 3.8 Here's what I hope you will start to pick up in this lecture: we'll examine the shift from science-driven to marketing, the growth in the pharmaceutical industry and how that's taking place. The rise of Madison Avenue, and this explores Arthur Sackler. He's the pioneer. He's acknowledged inside the industry as the pioneer, but there are many people that have followed in his footsteps. Inside the business, if you talk to doctors, they call it Medicine Avenue. All right, close, but that's their payoff of what they think was the marketing part for pharmaceutical companies. And the creation of demand and the blockbuster era. So blockbusters take place, but do not underestimate the ability of drug companies to help create a demand for them, and they become effective in doing that over a period of time.
- 3.9 Before safety regulations—you know this, no safety regulations for pharmaceutical products. The 1906 Act doesn't do much in terms of slowing it up. The heroin cough syrups and opium-laced soothers, boy, they suddenly seem like old school. In the beginning, we're shocked that that's the start of the industry, and now you have an entire new generation that just knows it from terms of antibiotics and penicillin, and they're doing different research. The companies are different. That past, they've closed the door on completely. Barbiturates and cocaine inhalants readily available to consumers—those days are over. And the lack of oversight-enabled profit-driven practices, that's still very much a part of the business in the 1950s. But the pre-1906 pharmaceutical landscape was rife with all types of problems.
- 3.10 In 1937, there's what is known inside the industry as the Elixir Sulfanilamide catastrophe. Here's what this is. A manufacturer from Tennessee is putting out a

product as an elixir, as a cough syrup. He mixes diethylene glycol, which is used as a product in antifreeze, to sweeten the cough syrup. That's the only reason it was used. So it doesn't add to make it better as a syrup or a cough or anything else. But in testing it, he finds out that if you add a little bit of that, it makes it sweeter, and so it makes it taste better. Puts it into that. No safety test, of course, under the law. So he's not doing anything wrong. Just put it out, and you can say that it is an elixir that helps for cough. The company does nothing illegal at that point.

- 3.11** One hundred and seven people, most of them children, died from that batch that went out onto the market. It caused, as you might imagine, a national uproar. The amazing part of this—I never cease to be surprised, and then I am surprised when I get back into these—the manufacturer, the person selling it, this Tennessee manufacturer said, "By the way, nothing illegal here, no safety laws, nothing else violated. I didn't do anything illegal." The government looked for something that they could find wrong with him. There was no legal requirement to test the drugs, to find out if they were safe before they were drug.
- 3.12** They finally found the company and the owner guilty on a minor violation: an elixir, by law, had to contain alcohol. That is a little footnote story. That elixir contained no alcohol. So they got him on the violation. Remember, the 1906 Pure Food and Drug Act: you had to put everything on the label. He had it on the label, but he didn't include alcohol, which nobody noticed as a technical violation. That got him a slap on the wrist. Nothing else, which is pretty remarkable.
- 3.13** So what does that do? That preventable tragedy—we talked before about how the government often reacts in response to tragedies—that tragedy is what passes that first law that we had mentioned before: the 1938 Act. And that Food, Drug, and Cosmetic Act becomes the first act that says, "We're going to follow food safety a little bit and drug safety."
- 3.14** Now, think about this. The FDA had been formed in 1906. So the FDA was formed in the Pure Food and Drug Act, but it wasn't its own division. It was put under the Bureau of Chemistry, and its purpose was to really just watch over food additives. So it isn't until 1930 that the FDA is created as its own stand-alone organization. It still doesn't have much in terms of personnel. It doesn't have any real power. But now in 1938, in comes the new law that says, by the way, drug companies have to show you that for safety, they're doing something.
- 3.15** Here's how Elixir Sulfanilamide finally forced action. In '37, you have the tragedy of over 100 deaths, 107, most of them children. In '38, you get the Food, Drug, and Cosmetic Act. First law, it's the first time ever that they were asking for proof of safety, major safety. You'll see how this varies over time. Prior to '38, nothing at all. So this is a turning point in American drug regulation. It came at a terrible human cost, and it really is only the first—I'm still amazed at times, no matter how many times I go through this history, look at it, write about it, study it, that until 1938, there had been nobody in the

government saying, "You know what? Millions of people are taking these drugs. Maybe we should test not to see if they work, but just to see if they're safe." So this is the first effort that you get to see something like that taking place.

- 3.16** In the mid-20th century, I wrote this in *Pharma*, the pharmaceutical industry reached a turning point. Some people, some pioneers in advertising, realized that marketing could transform science. It doesn't make the science better; it makes the science sell better. That's what they're really looking for at some point.
- 3.17** So here's the old guard. Now, the old guard: George Merck, great-grandson of the founder of Merck, a big German company, he truly believed that pharmaceutical companies and owners of pharmaceutical companies had a public duty, and you had to balance that. In an address to the graduating class of Medical College of Virginia, 1950, this quotation sums up much of what I've read and seen about him and what he's published. "We try to remember that medicine is for the patient. We try never to forget that medicine is for the people. It is not for the profits. The profits follow, and if we have remembered that, they have never failed to appear. The better we have remembered it, the larger they have been."
- 3.18** There are some people who would say to you, "You know what? That's just written by a PR person. They just got up there, and they said it because they know that sounds good." But George Merck believed that. He thought that was the way you do it. You didn't price drugs at the most expensive possible. You didn't push the limit to see what could take place. You tried to figure out how large the market was and if people needed it, and then you worked to get the right price. That's what I call the old guard.
- 3.19** In between, there were some people that came forward who embodied that idea, very few, not many. That's Jonas Salk. Some of you may know Jonas Salk as the inventor of the vaccine that's going to cure polio, or at least stop polio. Polio was a disease at the time that left thousands and thousands of children crippled, primarily across the United States and worldwide as well. The Salk vaccine, once it was developed, and Sabin, another doctor, had his vaccine.
- 3.20** But the Salk vaccine was an amazing public-private partnership: 20,000 physicians around the United States essentially contributing their time at a time when there were only like 35,000 physicians, public health officials, 65,000 school personnel, a quarter million volunteers, and 1.8 million school children—I remember being one of them—who participated in the trial, essentially, as they rolled that out.
- 3.21** A 1954 Gallup poll—I love this—showed that more Americans knew about the polio field trials than knew who was the President of the United States. So polio field trials, drug, big thing, potentially big money maker. And on a radio show, Edward R. Murrow, one of the great newsmen of his time, asked Jonas Salk, not knowing who owned the patent. And Salk said, "No one. It's like, could you patent the sun? How could you patent it?" That was his view of it. That's the old guard: George Merck, Jonas Salk.

- 3.22** The new guard is Arthur Sackler. I won't tell you just Arthur Sackler; it's not just him, but he's emblematic of it. You know what the new guard thought? They thought the old guard, they were ridiculous. They thought the old guard came from another century; they might as well. This idea that this is a business; that's how they viewed it. You could help people with drugs at the same time that you maximize optimized profits.
- 3.23** He really is the architect of pharmaceutical persuasion and introduced innovations—I use the word innovations. Innovations aren't always good, but psychologically-informed advertising, education is a form of advertising, amplifying the sales force. You'll see in a minute how he is the one who creates the idea of sending sales reps to doctors, which had never existed before, that it could be done. His strategies really rehabilitate, they reshape the industry, and they create the basis for blockbuster drugs and an increased commercialization of medicine that they hadn't thought of before. Sackler understood that you could do psychologically-informed advertising. Doctors were human. They were subject to authority, repetition, prestige branding. All of that worked with doctors. You wanted to convince them that they had a product that was really good.
- 3.24** Education—boy, there's a list of things I could go down and give you and show you where he had perfected the art, and then it became widespread of ghost-written articles. So something would appear, and it would be ghost-written. This was the time in the late '50s, early '60s, in which you had Time magazine, Newsweek, and others having an editor who was covering health. Arthur Sackler knew that those health writers at the smaller newspapers often were under deadline pressures and would provide for them, for the companies that he represented, ready copy, ready to go. He was the equivalent of asking ChatGPT if it could give you a draft of an article. Arthur Sackler would give you the draft of the article. All you had to do was just modify it a little bit to put it into the paper.
- 3.25** What you were really running, as opposed to news, you were running a PR piece, described as news. That's why from the '60s and '70s on, we are flooded with examples of newspapers, magazines talking about the next big miracle drug, the breakthrough drug, the revolutionary drug. Time and time and time again, it's the language sent over from the PR companies. Pseudoscientific brochures, independent educational materials, thought leader endorsements—all of it amplifies, as I call it, the marketing that used to be just merely informational. So this is also the birth time of the modern drug detailer, the drug sales teams, the sales teams that go out, and he uses it in particular on one drug, which we'll see in a minute.
- 3.26** So there was a time in which the drug company was selling it in the old ways by word of mouth and then by doctors doing it. What Sackler comes in and does really with William Douglas McAdams, his advertising company that he has started, which is medical advertising only, is to say—and he gets an award, a lifetime achievement

award and everything else after for having revolutionized the business—is that you can apply the techniques of Madison Avenue even though you aren't selling directly to the consumer. You can generate interest and you can persuade doctors that they should do something.

- 3.27** A case study: tetracycline marketing wars. I'll tell you. Tetracycline—a blockbuster. Variations and me-too's of tetracycline, and we'll talk about this, but they show the monopoly power of scientific innovation. Companies are fighting to ban substitution of this and maintain their exclusive hold. This is an area in which they really go to war. So here's the birth of the modern blockbuster in some ways.
- 3.28** Companies like Pfizer increasing their revenues enormously after World War II on nearly two-thirds of their sales of all antibiotics. Tetracycline is 80% of their profits. In the '50s and '60s, Pfizer employs aggressive marketing tactics to promote tetracycline, and then it becomes a blockbuster drug, setting the stage for what I call Pharma's forward way of marketing. But there's something that comes into this that really, truly changes the business: Arthur Sackler's Terramycin triumph.
- 3.29** Terramycin was a drug that Pfizer developed. It was nearly identical to a competitor's drug: Lederle had this drug called Aureomycin. It differed by a single oxygen atom. That's it. We're talking about the smallest amount of difference possible. It did not change its therapeutic value; it did not change its use. And even inside the company, inside Pfizer, there are scientists changing papers and notes with each other that would say, "We haven't come up with it yet. We've got to go back to the board. We've got to come up with a new product. This isn't it." Because, essentially, this is just Aureomycin with a different name.
- 3.30** But they have a CEO at the time, Jack McKeen, a hard-charging CEO. Jack McKeen gets focused on Terramycin and decides legally he's going to protect it. He's going to say that one atom makes a difference, and he wants to make it into something that will sell. So he starts to interview people to see who can promote it. Jack McKeen took Sackler up on a challenge. So Sackler launched an unheard-of \$10 million campaign that transformed the pharmaceutical industry. Essentially, what happened is that when Jack McKeen was interviewing people to see who could promote his drug, Arthur Sackler went home and said, "If you give me a \$10 million budget, I'll make it the number one selling antibiotic in the country."
- 3.31** It's pretty nerve-y to be able to say it, because there's a lot of competition for that. Most people would say, "Thanks, anyway, see you later. I don't think that can happen." Jack McKeen said, "I'll give you the money, and it better happen." Sackler got the \$10 million and then launched a campaign that was really ambitious, including the fact that the drug—there was this hype beforehand, newspapers wrote about it, it was in magazines—that it was coming out. It was going to be the latest and the greatest.

- 3.32** The day that it's released, there was literally a group around the country of Pfizer marketing and salespeople who got a telephone call at a pay phone because they were waiting for that. The telephone numbers were in Pfizer's headquarters. And then, they went out to the major hospitals and clinics and started to promote the drug and hand out the materials. That's what I call the early birth days of visiting doctors and trying to create sales. What the amazing part is that the extra oxygen atom allowed Pfizer's attorneys to get a patent in the patent office, defend it in court, and Arthur Sackler made Terramycin the number one selling drug in America for antibiotics. It came through.
- 3.33** So what does that do? You know what it does. That means the old guard, the George Mercks, they're out. Forget about it. You want to argue that you should have some type of public benefit when you're doing drugs, or do you want to follow what Jack McKeen and Arthur Sackler can do with Terramycin? Everybody's chasing that. They all want the next equivalent of that. They're investing \$10 million to make much larger amounts. That to them is a win-win.
- 3.34** It's very interesting because pharma closes a loophole here. As Pfizer and Sackler are getting ready to launch their campaign, they realize that there's a problem, which is that you come out with a new drug, you get a patent on it, you send it out to doctors who prescribe that drug. Fantastic. Then the pharmacist had the ability to substitute the drug. Pharmacists could say, "Hey, you just prescribed Terramycin. I think Aureomycin is just as good. I have a large supply of that. I'll give you Aureomycin instead." But that destroyed the purpose of making the sale.
- 3.35** So what do pharmaceutical companies want to do? They wanted to stop the ability, prohibit, make it illegal for a pharmacist to substitute drugs. "The doctor prescribes it, you've got to give what the doctor wants. Even if you, the pharmacist, think that there's a better alternative that might be cheaper." They went to the federal government, they lobbied the federal government, and the federal government wasn't interested in helping the pharmaceutical industry. Hard to believe. So they did the next best thing: they went to state legislatures, one by one by one by one. They got 44 of them to ban pharmacists from doing that.
- 3.36** So industry unites against substitution. Pharmacist companies combine, they fight the effort, they say, "Stop distributing drugs for similar versions." They want to protect their market interest, but, of course, they don't say that. What do they say? "We want to protect the patient because the doctor is prescribing the best possible drug for the patient. The pharmacist doesn't know the patient's history. The pharmacist just knows the patient." When the patient comes into the pharmacy, hands in a prescription—the old way it used to be done, hand in a paper prescription— and then the pharmacist will fill it. They may not be aware of other conditions the patient has, the history of the patient, but the doctor knows all of that. Therefore, preserve the integrity of the doctor. So, lobbying for patient protection.

- 3.37** By the way, the AMA, the American Medical Association, had only been around for a few decades, so it likes the idea of protecting the integrity of doctors. They join with the pharma companies in saying, "Let's cut the pharmacists out of this." I really think that what good is that drug patent if you can't protect it on the sales? So they are able to do it in 44 states; six states hold out. They eventually get them around. But in the '50s, they had the bulk of them there, and they were ready to secure those exclusive market rights.
- 3.38** Fixed-dose combinations—we talked about that before. Marketing versus science, the idea that you're going to have these fixed doses come out. This became such a process, fixed-dose combinations, that there's one point as we get toward 1960, in which one out of every five prescriptions in America that's being given out, 20% of all the prescriptions are fixed-dose combinations of antibiotics. It goes from nothing to becoming a fifth of the market in no time at all under the idea that maybe we'll actually be able to make it better by combining something else.
- 3.39** Inside the drug companies themselves, we now know that they had studies that showed at some of the companies, only 25% could even make the claim that there might be one feature that made the fixed combination better. Another company was up to 40%. Over half of the companies had done studies that showed to them that there was no benefit at all. But they knew there was a benefit in sales. "New" meant they could sell, and that's part of what the drug industry has always had: newer, better—that means the best.
- 3.40** Sackler's drug agency is involved in the first of these drugs, as you see, promotes the combinations more aggressively than any other advertising agency. Skeptics warned that they could be combined and cause problems. All of those were ignored over a period of time.
- 3.41** I think that one of the things that's interesting: 240 pharmaceutical firms had sent truckloads of information at one point when a researcher had asked for information. He was trying to do a study on the fixed doses. They got together. They sent so much information to that particular researcher that the researcher never came up for error or published the studies. Burying a potential bit of bad news with so much paperwork that they can't come out of it was just another technique to make sure that the studies weren't done that might affect otherwise patient safety.
- 3.42** The dawn of me-too drugs. Drug makers realize how profitable they are when they're changing a few molecules. Nothing shows that better than being with Aureomycin and what happened with Terramycin, the risk of me-too drugs flooding the market with near identical options. So the market is literally saturated at a certain point, 50s into 1960, '61, with drugs that are so similar, you could have flipped a coin, and it would have had the same effect for a patient, but they weren't different except for price and the patent.

- 3.43** This is the part that, to me, is the most important in terms of looking at the drug industry. Me-too drugs and R&D—in 1960, 25% of the money put into research and development at the 10 largest pharmaceutical companies went into developing totally new drugs. Seventy-five percent went into me-too drugs. That should be reversed. It should be the other way around. You should be able to say, "Three-quarters of our money, we're looking for absolutely new drugs that nobody else has." Now, you can argue whether those drugs are effective or if they're not, if they're overpriced, that's a different story. But they're spending three dollars of every four dollars on just trying to game somebody else's patented medication so they could have their version out and sell it to doctors somehow. That indication of where the money is going will tell you why so much of the me-too drugs were coming out.
- 3.44** A closer look: they're slightly modified for profit. We know that. The incremental modification—and I say incremental with a capital I—they also found something very interesting at this time. You're running in the next few years, not immediately, into the expiration of patents on original drugs that will come on the earliest of the tetracycline drugs. And that opens the door, of course, when a patent is up, to generic competition. Generic competition: drug prices drop by two-thirds often on average within the first year, the first nine months, after a generic is out.
- 3.45** So what do drug companies do themselves? It's fantastic. We used to think of me-too drugs just as a way of knocking off the competition. So Lederle has Aureomycin, Pfizer comes up with Terramycin, changes a molecule and is able to sell it. But now you have Terramycin or Aureomycin, or tetracycline, and it's coming off patent. Now, you, in the company that owns the patent, you change the molecules. You adjust it molecularly just a little bit, and you go back, and you say, "By the way, we've got a better addition, better and improved, new and improved." How many times have you seen that? New and improved in pharma means longer patent protection. So if they can show that, they extend the patent right away. So me-too, drugs, they learn to do it to competitors, and they learn to do it themselves to game the system on patents.
- 3.46** Aggressive advertising—let me tell you, new and improved alternatives, it didn't matter that science didn't really protect that or say that. These were being marketed that way. Profits over innovation, time and time again. The New options weren't always better medically, but they were certainly better for the bottom line.
- 3.47** Now, what's the challenge for the FDA? Limited FDA power. No doubt about it. We have the FDA in existence, but it really doesn't have a lot of power. When the FDA first comes around, by the way, when it was first formed as a subset of the Bureau of Chemistry in 1906, the FDA has responsibility for medicine, has responsibility for tobacco and has responsibility for food. Their priority was really food. Tobacco wasn't a priority really at all; it was just tagged on there. Then medicines. Food was coming ahead of it.

- 3.48** In the 1938 Act, when they passed the law, and they gave it some authority to say you've got to have safety on drugs, they also gave the FDA medical devices, which they hadn't had before, some food-coloring agents as well to keep track of, and cosmetics. So the FDA picks up those in 1938. Its staff increases by 2% over a year. So the actual regulatory powers expand, but the size of the agency doesn't. So it's still fairly limited, not just in its power, but in what it's able to do.
- 3.49** The industry, meanwhile, floods medical journals with advertising, makes it hard for doctors to distinguish between what's an ad—I know this is hard to imagine, but I'll give you an example in a minute—between what's an ad and what is actually a peer-reviewed article or decision reached by a group of doctors. The courts side time and time again with pharma. So on a few cases in which the FTC or others would say that's false advertisement, the drug companies will go to court and say, "No, it's not. We weren't saying anything wrong. We're not saying anything illegal. It's not false advertisement. We'll adjust that ad." The courts, time and time again, sided with pharmaceutical companies, and the messaging outpaces oversight.
- 3.50** This will not surprise you at all. You will not be surprised to know that private companies with lots of money, able to hire large staff, are much more nimble than the bureaucracy, even when the bureaucracy is slow. They outpace them, they stay ahead of them, they outperform them. It seems to be that they are one step ahead of the oversight at almost every level.
- 3.51** Now, before '38, the FDA could do almost nothing. They get authority for safety testing, but it's under-resourced and politically constrained. The industry floods the medical journals. The line between peer-reviewed and advertisements becomes blurrier and blurrier. The courts side with the companies time and again— I forgot to say this to you, sorry—the standard was to have an action against a drug company for something that they put into a newspaper or an ad or promoted to doctors. It had to be intentionally false. It didn't have to be false, but intentionally false. So if you were negligently false, you could get away with it. Say sorry, adjust the writing, and go on to the next ad.
- 3.52** The predictable result? The FDA really struggles to keep up with the aggressive marketing tactics that are becoming widespread in the business. This is around the same time, what I call disease branding, starting to redefine illnesses. With Sackler's help, companies like Hoffmann-La Roche and Pfizer and others develop a strategy, and it's extensive when they sit down to do this, to expand the definitions of diseases, what applies as a disease, and it's in order to expand the market for what drugs will be.
- 3.53** One example is the successful reframing of mild anxiety as a medical condition that's caused by a chemical imbalance. Nobody's quite sure how that works at the time that it first comes out. But we're talking about the earliest days, like Miltown and others, where you're going to actually prescribe a drug for what they now call anxiety as a medical condition. Fatigue was another example. Companies literally worked to

position normal human fatigue, which we all get sometimes, we need that extra cup of coffee, this type as a medical disorder. So you're able to give drugs to be able to go ahead and figure out how for a patient, you can give something.

- 3.54** By the way, it reminds me of another point on patents, if I can, for a second, go back. There's a company that ends up getting very early on a patent on the drug adrenaline. Some of you may remember that they were able to get a patent with that name on adrenaline. This is the early 1900s. Adrenaline, a drug that could certainly cure fatigue and save lives at a given moment. Every other drug company that's put out a me-too version or has a knockoff of adrenaline can't call it adrenaline. So, when you hear the word adrenaline, it actually had a patent for just one particular company that was able to hold that for a long time. Everybody else could do a version of it, but had to name it something else.
- 3.55** Broadening indications without evidence—pharmaceutical companies would often expand the approved usage for a drug. There are no clinical trials at this stage. But later, even without conducting new clinical trials, you'll see cases in which they broaden out the reasons for what the drug was approved for.
- 3.56** Creating demand, not responding to it—this is where you have patients get the idea that they might need something else, and they're ready for the doctors when they come to prescribe it. So here's the process: pharma companies redefine illnesses to expand markets. Conditions like prediabetes and social anxiety are rebranded as medical issues. Boy, you thought you were diabetic or non-diabetic. How about being prediabetic? "Oh, that sugar test is a little bit high. Prediabetic, we better treat that." It's a fantastic way that over time, you start to see the drug companies expand out onto what I call the periphery of the disease. So you don't have the disease, you don't have the condition, but you might be heading toward the condition, so let's stop you from getting there.
- 3.57** Disease awareness campaigns pathologized normal human experiences. Well, I'll tell you: drug companies would end up spending big money on public awareness campaigns that would actually make people feel as, "Oh, gee, I think I have that illness. Oh, my God, I must have that." You watch an ad or you see something or you read a story. That's why I hate doing a Google search on any condition I ever have because every Google search eventually leads to death. You eventually get down there. It's true, but you end up finding out that scratch could be what? I never even knew there was such a disease.
- 3.58** Marketing overwhelmed clinical evidence and guidelines. Time and time again, the marketing becomes more important for the drug companies than what they're showing as their internal reports. Patients are persuaded by unmet medical needs, needs they always didn't need. And again, this idea that demand is created, not met.

- 3.59** Drug reps never existed. Sackler sends them out for the first time with Pfizer for Terramycin, and they work well because, I'll tell you what: this is a time in which pharmaceutical companies discover something. If you send out the same drug rep to the same doctors, they develop a relationship over time. They like each other. I'm not talking about anything untoward. No payments under the table to the doctor or things like that. Maybe the drug rep will buy the doctor or lunch or something. They know the office, they go in, and the doctor ends up building up a level of rapport and trust. That's the key with that drug rep.
- 3.60** So when that drug rep says, "By the way, I have to tell you, we've had some great drugs before coming out of Lilly, but this one just knocks my socks off. I've never seen anything like it. It's the best thing we've ever had. It shows inside—" The doctor is listening to that. Is the doctor sitting down and reading all of the clinical studies that later come out or reading all of the reports? The doctor is running the practice, and so they will rely on that drug rep often, those personal relationships.
- 3.61** The companies knew it because one of the things that Sackler taught Merck, and then went on to other companies, was that you measure the sales in the areas in which your drug is being ordered and dispensed. And therefore, you could see who the star drug reps are, and you reward those drug reps with bonuses. The ones who are really turning out numbers with doctors should get bonuses and be acknowledged as the star sales. It's the equivalent of commission. Nobody makes a commission on a drug, in essence, but it's the equivalent of that inside the industry.
- 3.62** Free samples—oh, my goodness, how could you never have thought of that before? No one thought about giving away freebies to doctors? Of course, Arthur Sackler did, and that became the idea. When they got to—I mean, antibiotics are cheap. They might charge a lot for them, but they're cheap to make, so you were flooded with free antibiotics. And afterwards, with Librium and Valium and others, doctors had all types of drugs that they were able to give out. They weren't able to give out controlled substances eventually, so they couldn't give out things like OxyContin or narcotic pain killers. That was different. But they could give out free drugs here, and free samples were great.
- 3.63** I don't know if any of you have ever gone to a doctor and had a situation in which the doctor says, "Here, I have a free sample of this. You might want to try it." You take it. Doctors are not trying to hook you. It's not like heroin, in which they want you to take the one first shot of heroin, and then they know you're hooked for the rest of your life. They're not doing it for some nefarious purpose. They actually think they're doing something good by giving out the free sample. But by being part of the machinery from the promotion end of the pharmaceutical company that gives the free samples, to think that a doctor is giving you a free sample of something as a medication to see if you like it, tolerate it or whatever else, is a fairly revolutionary type of idea in the 1960s. The

industry at this point was spending a majority of its budget, as I call it, on physician-facing marketing, so with drug reps in front of them.

- 3.64** The sponsored medical conferences—this is the time at which they're funding continuing education. As a matter of fact, one of Arthur Sackler's other companies was a continuing education company. They start to pass laws in the 1960s: the doctors should go back, just like lawyers and everybody else and architects, continuing education so you could be up to date on all the latest rules and regulations regarding your profession. It's a great idea, but of course, it became a business as well. Continuing legal education, continuing medical education, companies have come in and formed a real business out of it.
- 3.65** Well, Arthur Sackler and others like him thought, what a great way to pass along some of the ideas we want. They aren't telling you what drugs to prescribe, but they're making them very pharma-friendly. So continuing education inside the medical field becomes one in which doctors really think of drug companies as their allies. The doctors are the ones who are able to take care of a patient. The drug company is the indispensable ally who's doing everything it can to make your care better. So you don't think of it as an adversarial situation at all. You think of them as you have the same mission, the same allies.
- 3.66** Ghostwritten articles—they were excellent, disguising marketing as independent research and creating patient demand. The symbiosis was complete at this point. The legacy of this: perfected target persuasion. This is the beginning of times when they're starting to tailor pitches for particular markets, specific audiences. Not easy to do yet, but when I look back at this rudimentary effort in the early 1960s, '59, '60s, '61, to try to target a particular audience, I think to myself, no wonder the digital age has provided such power to marketers.
- 3.67** They know when you go on Google and do a search for a particular illness or disease. They're selling and sharing information all the time. They can cut and dice the market 100 ways to Sunday. They can cut and dice it so they're just covering people on Medicaid who have a particular worry about whether they're going to become diabetic or people on Medicare with Part D. That type of precision marketing back here looks like prehistoric days. It's very rudimentary. It wasn't very sophisticated, but it's the earliest time that they're starting to think about it.
- 3.68** Mastered reputation laundering through scientific language—I love this. The industry gets the credibility of these medical journals, JAMA and others. They carry weight. They're being created at the time. Arthur Sackler created his own, as you'll see. These journals really become the thought leaders on what's happening in terms of drugs and what should be happening. They're pushing for educational content, engineering the markets versus disease expansion. And so with the help of people like Sackler, the pharma companies are expanding the definitions of the diseases and the potential customers.

- 3.69** User educational content to shape prescribing habits—these are journal articles and educational programs that are taking place in the community. Building a promotional machine that regulators can't match. They're always running ahead of the FDA. Pharmaceutical marketing targeted persuasion markets reputation from glossy ads to branded swag. This is the period at which patient advocacy campaigns and all are starting to be used to drive sales and permeate every level of the healthcare system.
- 3.70** What are some summaries and takeaways from this? Marketing overtook science as the engine growth of pharma. I'm going to talk in the next lecture about some specific examples that you'll see, and you'll understand how and why this happens in such a persistent way throughout the industry. But I really think that the innovation in terms of discovery of new drugs takes not a back seat, but becomes aligned fully with the idea that it's one thing to come up with the discovery inside the lab, it's one thing to get the patent on it. They have a whole legal department now inside most of these companies to go ahead and make sure that they get the patent and they protect it, and they sue anybody who's trying to step on it, and then they lose those suits sometimes. But that's not as good; you then have to sell it. You have to make yourself stand out. You have to have the relationships with the doctors to go forward, and you need to create all this ancillary business in the medical tribunes, the magazines, and the JAMA's.
- 3.71** Disease definitions and demand are manufactured, and you see this starting in late '50s and early '60s, but without any doubt, it reaches a peak frenzy in the mid '60s, as we'll see in another lecture, when it comes to the benzodiazepines and Librium and Valium and the level of mild anxiety for which that is prescribed, and those become blockbuster drugs.
- 3.72** Targeted persuasion, reputation laundering, as I call it, in engineered markets—I tell you, time and time again, you can see a situation, and I see it on the release of especially the me-too drugs and antibiotics—you will have a situation in which the public is filling a prescription for one drug, this is the number one drug in the country, and then they follow the lead through doctors to the next drug that comes in.
- 3.73** This idea that the latest and the greatest is also the best is part of what a lot of consumer-driven companies go on. I get that. You buy a vacuum, and then they have the new and improved vacuum a year later, and somehow you're supposed to buy that. That's what consumerism is about. But it's a bit different when it comes to drugs, because you would think that there's some idea that you should show that the new products is slightly better and less risky before going to it, but that was not the case.
- 3.74** Patients have to question the sales strategy behind the drugs. Behind the marketing tactics, there's a level at which you say, "Why is this being pushed? Why is this the new wonder drug?" The key takeaways for me: the shift from science to marketing is the primary driver of the pharmaceutical industry. Patients need to be more discerning and what I call more critical consumers of pharmaceutical products.

- 3.75** Now, all of this idea of what this shift is toward aggressive marketing tactics and the fundamentals of it, in some ways, the key takeaway as a consumer is to question the sales strategy. You don't see the sales strategy always. I'm not talking about direct-to-consumer ads, not talking about turning on the TV and seeing the ad for it. I'm talking about when somebody goes to a physician, and they suggest a pill, and they suggest the latest version of that pill. The question is: why are they suggesting that pill versus an older type of pill? We'll talk about that later. Because often, they don't have an answer for that. They do have an answer, but it's a canned answer, and it comes over as tinny because it's not authentic. You can sometimes feel that.
- 3.76** How to be a smarter patient on marketing? Not easy because they're spending billions of dollars to market to fool you. I don't mean that to be depressing. But they are spending a lot of money on marketing to make it smooth and not to have it visible.
- 3.77** You're going to be a pain. You're going to be considered a troublemaker. Doctor is not going to like you. But if you do ask your doctor occasionally what evidence supports that drug, just see what they say. You don't want a lecture. You don't want a scientific treatise. You're not asking the doctor to write an article for you. You're not asking for all the peer-reviewed material. You're just asking in lay English, "Tell me as a patient, what's the science behind this new drug? I mean, is it really good? Are you persuaded by it as a doctor?" Put the doctor on the spot a little bit. You don't have to request all the information on the clinical trials, but I love saying to the doctor, "What is the science behind this, behind just the good marketing?"
- 3.78** Request alternatives—and this isn't medical advice because I'm not giving you medical advice. I'm just giving you my feelings, having studied the pharmaceutical industry for a long time. Ask if there are older generics that might work just as well. One thing you know about the older drugs and the generics is that they could be as effective, so you want to find that out. They certainly will almost always be cheaper if cost is a factor in terms of filling a prescription.
- 3.79** There's one other thing. We know from the generics that have been out 15, 17, 20, 25 years, what the side effects are. By then, we know—sometimes it takes 10 years to find out—but we know what the side effects are, and it might be a laundry list. Now, you can balance that in terms of your own risk assessment as to whether those side effects that the generic has are worth offsetting the condition, symptoms, or whatever else you want to take the drug for. With a new drug, the side effect profile will always be smaller because it hasn't been out long enough for people to report all the side effects. So one of the things the doctors will often tell you on a new drug is it has a lower side effect profile. That's true because it's new. We aren't there yet. Come back in 10 years and tell me that's the case.
- 3.80** So you have to make your own decision as to whether you like the idea that there's a lower side effect profile on the new drug, and maybe it will stay like that, it's possible, or you know all the side effects from the generic drug and you might want to take that.

At least tell the doctor that you're not afraid of a generic option. If the doctor makes a good case as to why the generic is no good, fantastic. The doctor doesn't get any bonus. The doctor doesn't earn a commission. It's not like they're earning something from giving you the newest and latest drug. They're often doing it reflexively because that's how they think it should be done. But if you ask them, they sometimes will take a step back and provide the generic.

- 3.81** Be skeptical—I am anyway— of new, improved, and breakthrough. Sometimes there is new, improved, and breakthrough, but marketing claims about the drugs, changes that are minor, that are oversold—there's major drugs. There's a marketing sophistication that comes in with this. I'm always skeptical of the new major breakthrough drugs. I think that as an informed questioning patient, you help offset the power of the pharmaceutical industry's marketing, which is a good thing.
- 3.82** Preview of Lecture 4: we'll explore the human cost of pharmaceutical innovation, and examine tragedies, loopholes, and the fight for drug safety. Now, I say that, and you think to yourself, "Didn't we just hear that? Or wasn't that what we were doing in the 1910s or the 1900s?" It's a continuing process. So what I mean here is that it's not that it's adversarial, but the drug industry is always trying to push the marketing and increase the profits. And at the same time, there's an impact on patients over the long term, sometimes much worse, like with the lethal opioid painkiller epidemic that ravaged the country, and sometimes a bit less than killing patients. But at the same time, there's this constant fight between the drug industry, the regulators, and how much they can get away with.
- 3.83** In the next chapter, we're going to come to probably the most important point in modern American history of when the federal government had an investigation and the idea for a bill that would have revolutionized the pharmaceutical industry. When I tell you that penicillin changed the industry, as we know today, this next chapter, early 1959 to '62, investigation and proposed bill would have changed the pharmaceutical industry that I'm talking to you about today and has changed it, but in many different ways. That is what we're going to talk about in the next lecture.

Lecture 4: Regulation Revolution

At a glance — The regulation revolution.

- 4.1 One half-step, a very brief half-step back to Lecture 3. Thanks again, Trisha. We were talking before about how the pharmaceutical industry in the 1950s is coming out of this period of penicillin and establishing itself as an international force with its products and doctors. It also eliminated— not "it" eliminated, the AMA had a hand to—competition from an outside source, the possibility that homeopathic medications might compete with a patented pharmaceutical medication. So the idea of that had bothered the AMA for a long time, the fact that doctors who weren't real doctors, who hadn't gone to medical school and followed a standard procedure and ended up instead as what they would call quacks, might be describing some other form of an alternative medication or a homeopathic medication.
- 4.2 Now, in Germany, before World War II, homeopathy was big. Even though that was the birthplace of places like Merck and Bayer and Schering, there was an active part of the homeopathic industry that flourished in Germany. The pharmaceutical companies viewed it as competition, but not necessarily as something they had to eliminate. Here, it was viewed as though homeopathy really should be not just pushed aside, but extinguished.
- 4.3 The best evidence of that is that in 1900, there were 20 schools in the United States for being able to go practice, learn the craft and then practice in terms of homeopathic medicine. In 1950, the last one closed. So, just as they're coming out of World War II, 1950 is the point at which the American drug companies, the top 10 companies, are running 80% of the prescriptions in the world and 95% of the profits, and they've taken over from the decimated German industry. The good news for them in that very year: the last of the homeopathic schools closed. That's another reason why doctors and the drug companies end up as allies, because they are both working one enforcing the other. The drug industry needs the physicians because they are prescribing the drugs. So they encourage you only to think in terms of traditional doctors. And the doctors find that the drug companies are fantastic because they want no competition from alternative or complementary healing of any type.
- 4.4 So here in this chapter, we're talking about the cost of innovation, the fight over prices and drug safety. This particular lecture shows you, in some ways, the way that real patients pay the cost of drug innovation. What I mean by that is new drugs—fantastic. We'll find out what the claims are and how they are. But new drugs also carry risk, and sometimes patients become the test market, in essence, for those drugs. It's only later, when side effects and reports come in of what's wrong with them, that the industry learns the effect of what has taken place.

- 4.5 So innovation: double-edged sword. The same drugs that can heal can also harm, and too often that harm becomes visible only after the profits are banked. So you've made the sales, you've already had the market, now you find out there's a problem. Even in an instance in which you hear about a drug recall, that's great, but they're not giving the money back. Never heard about a drug recall, and they say, "Oh, by the way, we'll refund everything else we've ever sold." That never happens.
- 4.6 So the pharmaceutical industry has grappled with this issue, I think, for a long time. Every major advance—and that's really if it's antibiotics, anesthetics, chemotherapy, hormone therapies, whatever—they carry this extraordinary promise, but at the same time, they carry a risk that patients are sometimes unaware of. This lecture is going to explore those safety failures and how the government came to try to deal with it.
- 4.7 Kefauver's crusade—that's my term for it anyway—Estes Kefauver. His investigation into the drug industry that starts in 1959 is unparalleled because today—I say this with some confidence. I never say never because I've been proven wrong so many times. Once I say never, then the exception always comes up. But I don't believe that today you could have the will in Congress, and I don't care if it's Congress dominated by Democrats or by Republicans, we talked about this before, to have as intensive and as deep an investigation and potential upheaval to the pharmaceutical industry as you had here in this 1959 three-year hearing that takes place and the bills that came out of it.
- 4.8 Kefauver was looking at this, and there's something interesting about him. He really does have what I call star power. It's hard to imagine because he's 6'3". He's a little bit lanky. He doesn't have the greatest posture. If you see pictures of him over a period of time, he's always a little hunched over. He's got glasses. He uses the glasses. They're thick glasses. He has them off. They're dangling at times. He's hiding part of his face. He's not the photogenic star that you might think of.
- 4.9 But he launches in 1950 an investigation into organized crime, and he happens to overlap with the advent of television, television news and cameras. The television channels like CBS and ABC and others said, "Why don't we televise some of these hearings about these mobsters, Valachi and all these people and these guys who came forward straight out of Hollywood casting into American homes?" It turned out to be what I call the True Crime 1950s reality show. It was fantastic. These hearings made Kefauver into a star, a star that people knew across America.
- 4.10 He's got this strong Southern drawl. And if you can listen ever online to any of Kefauver's interviews, or you listen to him—I'm a Northerner. I understand that. I was born in San Francisco, but he's hard to understand. So again, you don't think he's somebody who's going to be a television star, but they liked the fact that he was not necessarily a showman. That played off very well on this investigation. And the mobsters came in. They were colorful. They came in with counsel. They were colorful. They blustered against him. They refused to take—they took the fifth. They refused to talk. Sometimes they did. And he just seemed nonplussed through all of it.

- 4.11 When Kefauver took his show to New York City, which he did for some of the witnesses, for eight days for gripping testimony from top gangsters, Time magazine noted, quote, "It will occupy a special place in history. People had suddenly gone indoors into living rooms, taverns and club rooms, auditoriums and back offices. There, in eerie half-light, looking at millions of small frosty screens, people sat as if charmed. Dishes stood in sinks, babies went unfed, businesses sagged"—pretty dramatic—"and department stores emptied while the hearings were on. Never before had the attention of the nation been riveted so completely on a single matter."
- 4.12 Now, you think to yourself, maybe the writer at Time was a little bit into hyperbole, but it turns out to be that 90% of the people with TVs had turned into those hearings at some point. That's an astonishing figure. You talk about having some influence—30 million Americans. David Halberstam, the historian, described him as a result as a Southern Jimmy Stewart. That's how he came out of that.
- 4.13 So what did he do? He ran for President in 1952 and 1956. Nobody remembers his footnote to history. Kefauver thought, "Oh, yeah." Now he's got star power, runs for President. He loses both times to Adlai Stevenson, who wins the nomination in both of those years, although he becomes Stevenson's vice presidential candidate the second time around, but they lose to Eisenhower. So, there was a moment at which he thought he could be Vice President of the United States, wanted to be President of the United States, and wasn't.
- 4.14 When he lost that second race in '56, he had been building up his brand—I know that's overused. I hate people's brand. But if Kefauver even knew he had a brand, he was building that up because he was chairman of the Antitrust Committee. After finishing with the mob, he went on—I do love this—to steel, auto, and bread industries, looking for evidence of collusion, conspiracy, and everything else. And then, he decided after he had lost the vice presidential run and Eisenhower was in office, "Maybe I should look at the drug industry. That could be as good as the bread industry or the steel industry."
- 4.15 So in 1959, he announces and starts an investigation for antitrust into the practices of the drug industry. This is a time when he says, "It seems to me there's a consolidation in power in terms of antibiotics." And he names them: Parke-Davis, 45% of all your products are just antibiotics. Lederle: 40%, Pfizer: 33%, Lilly: 20%. Even the companies that weren't antibiotic blockbuster companies were really running on one product that they had done a cross-licensing deal with other companies to control it. There was an antihypertensive, Diuril, that made up 40% of all of Merck's sales. So that was their big money maker. A derivative of that drug, Orinase, was making up 20% of all of Upjohn's sales. And Smith Kline & French had the antipsychotic Thorazine, which provided nearly a quarter of all of their sales.
- 4.16 So he, Kefauver, didn't like the fact that there was so much consolidation. He thought these companies had worked together to carve out parts of the market that worked for them. He wanted to look into that. They were a little worried, a little worried, but they

should have been more worried. He launches the investigation in '59 to find out if they're exploiting consumers through pricing practices and patent abuse. You know already, it seems like a rich area to me. He is somebody who's been around now enough that he knows how to do it: focus on competition and pricing.

- 4.17 He doesn't think they're going to come in voluntarily, so he issues subpoenas to 19 companies to get testimony from the top executives and the CEOs on drug costs and potential collusion. That's making a few of them a little bit nervous. The executives come in and they testify in these very confrontational hearings at times. And what do they do? Number one item: "Our prices are high. Sorry about that, Senator, but of course, our research and development costs are very high, too, because we are developing the greatest drugs in the world to be able to cure people." And they claim the patents are necessary to recoup the investments. That's their line, and it's one that they will stick to.
- 4.18 In the course of his preparation for his hearings— and I got the Freedom of Information files on this—his research staff and his investigative staff uncovered and wrote a memo about the Sackler family, who we know was already doing promotions. They have a memo that said, "We think we had never heard this family before, that there's a Sackler Empire, this empire being their influence over the way the drugs are marketed." And they listed a number of things that should be investigated about the Sacklers. As you will see, Arthur Sackler was eventually called before the committee on a very serious matter, but I won't give you this spoiler yet.
- 4.19 Exposing the FDA's conflicts of interest—he wants to know how the top FDA official—remember, before I said that the head of the Antibiotics Division, that's Henry Welch, was receiving lucrative payments from pharmaceutical companies. So, pulling back the curtain on pharma. One of the things Kefauver is doing—and this was the mortal blow. If you were Superman and you're worried just about kryptonite, the drug companies are really worried about their patents because that, more than anything, they have unfettered pricing power in America.
- 4.20 We are the only country, post-World War II, that gave them unfettered pricing power. The other countries in Europe that had national health care systems or government-run health care systems had boards appointed that negotiated the prices with the drug companies. In America, they always ended up being higher because here that didn't happen. Free, full market capitalism. But they had a protection because the patents meant that nobody else could come up with that specific version, and they could say that that was better, and they could charge higher prices to hospitals, clinics, the government, everything else. He wanted to look at the patents and see if really they needed it for that long.
- 4.21 Me-too drugs—he announced that that was part of what he was going to be looking at. Developing the slightly modified versions rather than really going after new treatments, should that be protected by patents?

- 4.22 Aggressive marketing—it really wasn't even known by most Americans how the drug companies were marketing their drugs, because they wanted to keep it hidden that they were so heavily involved. But he had picked up enough information through his investigators to know that it was a problem. So he wanted to talk about the misleading claims that drive the sales of the products.
- 4.23 Weak oversight—I'll tell you something. I've been through these hearings, and they are mind-numbing after a while because Congressional hearings are pretty, pretty bad—watched, confirmed. But he really exposed time and time again, not because it was the goal of his investigation or the hearings, but just because you were saying when you would read this, "Where's the government? Where's the FDA? Where's the FDA missing in action?" They didn't have anything. So what Kefauver thought was, "We've got to pass a law that gives some teeth to the FDA and to the government to bring in these pharmaceutical companies."
- 4.24 Confronting the executives—I'll tell you this is a moment, Francis Brown of Schering, John Connor of Merck, Harry Loynd of Parke-Davis, they came up one after the other to defend their practices during this period. And I will tell you, it made for great television, for the most part, I mean, for the public. The bad news for the pharmaceutical executives was that the public had gotten introduced to Kefauver through the mob hearings and organized crime. So everybody that came forward was a bad guy, essentially. Then you had the steel industry, you had the bread industry, and everybody's doing collusion of some types. There are no good heroes that come out here.
- 4.25 So, by the time the pharma execs come in, they don't look very different than the mobsters except they have a quieter suit and tie on. They don't have a flashy suit and tie on. They're already being cast in the public's mind as something potentially being wrong. And Kefauver is really looking at the price fixing, not just transparency over safety, because that's part of his issue, but why isn't there more transparency over what I call proof of effectiveness, not just safety? He says that in the hearings, and that is very concerning to drug companies because, "You mean we're going to have to prove the drugs work? That's not going to be possible." But that's one of the things he's suggesting.
- 4.26 Transparency in terms of clinical data—show us, show the government or show the FDA what your internal studies show. That might be revolutionary. What about restrictions on misleading promotional materials? Proof of efficacy before treatment is one of the things that he suggests. Informed consent in clinical trials. If we have clinical trials, maybe there should be informed consent. It might be a pretty good idea. And stricter manufacturing standards, because it was pretty much the Wild West in some ways. Some companies were very good, and others were not.
- 4.27 So Senate investigators uncover the Sackler family's industry influence. They discover how they gain significant control over drug development, testing, promotion, and

publication. The Sacklers had, at this time, 24 different companies under different names, which I worked, when I was doing my book on the pharmaceutical industry, to put together, often finding a telephone number they shared or an address they shared: 15 East, 62nd Street, 17 East, 62nd Street. They were all different companies. Some of them would be companies that would do research, consumer research; some would do drug testing. They all were working for Arthur Sackler and the one company that he had been hired to promote as a drug. But they appeared to the public to be independent, and that was important.

- 4.28 Sackler wasn't the only one doing it, but he was the one who was doing it most effectively: a complex, vertically integrated operation in which this is a small, what I call, business empire that's coming out that spans everything from drug research and development to advertising and medical journal publications.
- 4.29 Sackler's Connection to FDA official Henry Welch—it was the Sacklers who had established the relationship with Henry Welch and put him in charge of an antibiotic journal that essentially is promoting the Sackler's combined combination drug from Pfizer. Then, Arthur Sackler starts The Medical Journal—that's its name. The Medical Journal becomes one of the most established medical journals in America for decades. It does a lot of great work in everything else, but he knows he can also stuff it in the early days with articles that are favorable, that are PR-related. It's not enough at this point to go ahead and just pass an article on to a health writer at a local magazine or a newspaper in the hope they will write it. Why not have your own medical journal that everybody subscribes to and has to take a much better idea?
- 4.30 Extensive web of interconnected businesses—over 20 that they own or control through various proxies and hidden interests, the Senate investigators find an effort to uncover the full list of influence. Those investigators acknowledge in their documents that their findings were limited. They wanted to look into the Sacklers more, but every Congressional investigation has the same problem: money. What's your budget? How long will it last? What's your time limit? Some things get cut that you want to look into. The Sacklers on this end, as a family unit, were very, very lucky because they dropped down the list of importance. And as they dropped down the list of importance, they emerged mostly unscathed.
- 4.31 The FDA's conflict of interest— Kefauver's investigation exposes how Henry Welch had been receiving lucrative payments from these pharmaceutical companies. Now, Welch has gone ahead and taken a position together with Arthur Sackler's partner, this fellow called Félix Martí-Ibáñez, on an antibiotic journal. At one point, he, Welch, jokes inside the FDA that his annual salary—it's like \$27,000—couldn't pay for his taxes on the amount of money he's receiving for his consultancy on the antibiotic journal. They thought he was joking, so nobody reported him. They figured, "Why would he say that?" Nobody can quite figure out why he would say that, but he was boasting about that. It turns out he made about \$287,000 from the journals at a time, almost 10 times

his salary. The \$17,500—I'm sorry about that. He made \$287,000. His annual salary was not \$27,000; I gave him a raise. It was \$17,000 a year.

- 4.32** In May of 1960, he resigns after the HEW Secretary demanded it. But in his resignation letter—it's a fiery resignation letter. It says, "I did nothing wrong. I never said or endorsed anything that I would have endorsed because of its scientific integrity. It wasn't just for money." Everybody will say it's not just for money. An internal investigation concluded; they had this big investigation. "Oh, my goodness," they said. "Is the FDA somehow corrupted or influenced or bought, or there's some problem in the FDA in terms of pharmaceutical money affecting the way that the regulators are supposed to operate? We'll do an investigation and find out." And they came back with the investigation and said, "No, everything's fine, hunky-dory, just that guy: Welch. He's the only bad apple of the entire group. No more widespread conflicts of interest. We absolve the entire agency," and so that was the end of it.
- 4.33** Sackler's Medical Tribune—I said before that while they're launched, he launched the Medical Tribune, it becomes, for a while, the most successful medical publication in the country. The covert corporate structure—he controls the Medical Tribune through his corporate structure. At first, people don't realize how Arthur is controlling the Medical Tribune. He secures the backing of this with a renowned Harvard professor to come and endorse the Medical Tribune as the one source you should go to if you're really looking for the latest information. This guy, Georges Doriot, publicly backs it, invests in it, and it makes the Medical Tribune almost overnight the journal that you have to use.
- 4.34** Inside the Medical Tribune, Sackler writes a regular column, and that column, called One Man and Medicine, expresses his views on a whole range of things and serves to further the interests of his clients, from Pfizer to Merck and Hoffmann-La Roche and on. Profiting from Medical [Tribune] is one of the most successful business ventures the Sacklers do until they later hit OxyContin, of course, and expand their influence and financial interests in the pharmaceutical industry.
- 4.35** Sackler's fake ads and conflicts of interest—this becomes a key source of what takes place in the Kefauver hearings. The FDA's first approved fixed-dose antibiotic combination was Sigmamycin. It was a mixture of two parts of tetracycline, one part of oleandomycin, and that's a weak relative of erythromycin—I love the -mycins, by the way, don't you? Can you imagine? These are, by the way, the generic names. These aren't the brand names in that sense. The people are thinking, after they discover it, "What are we going to name it? Oh, erythromycin. No, I like tetracycline."
- 4.36** This is the first one in the group. This particular drug is the one that sets the groundwork for what I call the future of fixed-dose combination antibiotics. They advertise, as you see here, this is something Pfizer is advertising inside of a medical journal, "superior control of infectious disease through superior control of the changing microbial population." Then they go on to have doctors talk about that. Now, that sounds fantastic. So, you would think that they have studies that show their "superior

control of infectious disease through superior control of the changing microbial population." No, that's just PR. That's how they write in. That's how they sell it. But they don't have any tests that actually show that. But they give it to Arthur Sackler to go ahead and try to show that this is a drug that can sell.

- 4.37 He gets Henry Welch to go ahead and give a keynote speech at a major conference. Henry Welch, then the FDA Chief for Antibiotics, says, "I think we"—meaning the public—"are in the third wave of antibiotic discovery." The first wave was penicillin. The second wave was tetracycline and all these others. And now we're in the world of Sigmamycin. Pretty impressive to get from the head of the FDA's Antibiotics Division, but he was doing what he's supposed to do.
- 4.38 Now, he eventually, as you know, has to quit, blames politics. But there's something wonderful that happens inside of this. A writer for the Saturday Evening Post is doing a story about the success of these drugs in 1960. This happens to be the first full year that Kefauver is doing his hearings. He goes to a doctor's office, and he's talking to this doctor about Sigmamycin and why it's so effective. The doctor says, "It's the therapy of choice." He says, "We all end up using it. Here, I'll give you an example." He takes out of a drawer a pamphlet, and he gives it to Lear. On that pamphlet are the endorsements of eight doctors who are full-throated endorsements for Sigmamycin, changing their view of how effective and powerful an antibiotic can be. The first fixed-dose antibiotic is changing their views of it.
- 4.39 These are doctors from around the country with addresses, telephone numbers, and office hours. "I've conducted some experience myself"—this doctor says to Lear—"and I find it's of use, but I haven't seen anything as revolutionary as these eight doctors." So, Lear, the reporter, decides to call the eight doctors up to get their feedback and find out why they're so enthusiastic about it. He telephones them. None of the telephone numbers work. He writes to them, and the addresses come back; it's returned addresses, not deliverable. The doctors, it turns out, don't exist. They're fictitious. They're not real. There are eight doctors listed giving these endorsements for Sigmamycin, and it's done as a novel. But nobody had told that. Even doctors believed it.
- 4.40 So Arthur Sackler is called before Congress because this is his ad campaign, and he blames an over-eager copywriter at Pfizer. He says, "Oh, my God, I didn't realize that they did that. That's amazing. I thought these were real names or that. I'd never do anything like that." And Arthur skates out of there without being held responsible for that. He's the puppet master. He's the one describing every ad. You had to know Arthur Sackler. They didn't know him. They don't know that he's scratching out words that come over from anybody in the company and rearranging them so they're in the way that he can best sell them.
- 4.41 But that ad campaign was the moment in which, if you look back, you think to yourself, "An over-enthusiastic copywriter takes the blame for it." It was very, very close to the

end of Sackler. He and his colleagues had taken advantage of an FTC loophole. So not an FDA problem. We're talking about regulation here: FTC loophole this time. All ads to doctors were exempted from any oversight. I know. It's astonishing. You think, who came up with that brilliant idea? But the idea was behind it, giving that exemption, doctors, they're smart, they're knowledgeable. They won't fall prey to misinformation or what's false. They'll be okay.

- 4.42 If you ever thought that might be right, the idea that doctors were prescribing Sigmamycin because they thought eight other doctors endorsed it shows you that they are as prey to misinformation and bad advertising as anybody in the public. There's no protection just because you had the MD license that you couldn't fall prey to what I call these heavy-handed and, at the same time, somewhat subtle and sophisticated Madison Avenue techniques of selling drugs. So doctors are being targeted to sell drugs. They don't even realize it. The drug makers and the people who are selling them, like Sackler, can do whatever they want. That was exposed here and became an issue for Kefauver as he's moving forward.
- 4.43 Kefauver focuses on drug pricing and patents. Now you start to get the drug industry a little bit nervous because one of the things they've always had going for them is this ability to charge prices of whatever they wanted. Now you have somebody who's actually saying, "Well, maybe I'm going to hold you responsible for a little bit of time on this."
- 4.44 In the hearings—and I will tell you that these numbers are actually pretty low because I can give you examples years later in which there are multiples of this— but they took a US tranquilizer and the same tranquilizer in Argentina. It was costing three and a quarter here, was costing the equivalent of 75 cents there. So, over a four times markup. A UK tranquilizer, the same one, about half the cost of the US.
- 4.45 Antibiotics, one of the cheapest things to provide because they're doing it out of these large fermentation tanks. They have the infrastructure provided from World War II, from the government. They built new infrastructure on top of it from the profits they put in—by the way, from 1950 to 1960, the drug industry put \$600 million back into infrastructure and research and development. It's a lot of money. It shows you how much they were making as well, because we know that that's not even half of their profits. And they have 400 new drugs that they were able to put out, most of them variations of the same type of drugs. So 400 new drugs, \$600 million.
- 4.46 Antibiotics, they were churning out at a very, very low amount— \$18, more than twice the price of what they were selling for across a contingent of European countries. And so Kefauver was asking the question: why the difference?
- 4.47 His legacy—he's exposing pharmaceutical misconduct, and he's sparking some public awareness. The public is not thinking that these guys are all knights in shining armor at

this moment. He's creating some groundswell, at least in the public, for some reforms. They don't know exactly what.

- 4.48 One of the things, by the way, the drug company has always had going for it is that it's somewhat complex. So it's not easy. It's life sciences, it's technicalities. So once you start talking about reform, they just out-science you. They say, "Oh, you're thinking of reform like a traditional industry, like gambling. We're not gambling; we're science. Don't you understand? That reform is going to cripple innovation." So that's their fallback always. And then they give you a little bit of the science talk, and people walk out with their eyes dazed, and they say, "Oh, yeah, lucky we didn't do any of those reforms, or we would have stopped the next great cure for cancer."
- 4.49 Advancing drug regulation—his efforts led to stricter drug regulations—we're going to talk about that—for consumers in the US. Championing consumer protection. I mean, this was really a guy who was about consumer protection way before Ralph Nader was even around.
- 4.50 How they stopped him. Here are the things he proposed. This was his line of studies. He finally comes out with this bill in April of '61, comes out with a Senate Bill, Senate Resolution 1552. It's a sweeping bill. As a matter of fact, it's called—I love this title—the Drug Industry Antitrust Act. I probably like that because when I was on the law review at school, my law review paper was about an antitrust exemption. So I have a fond memory of antitrust law in an odd way. I know that may seem strange, but I love a bill called the Drug Industry Antitrust Act because it makes it sound like you're breaking up some monopoly or price-fixing entity.
- 4.51 Number one key: what does he want to do? He wants to bring patents from their current 17-year protection down to five years because he sent investigators out to Europe, and they talked to all the European countries. European countries allow patents, but they have much shorter periods: some three years, some went out to six, some went to seven. The average was five. Kefauver thought five was enough time for you to make a good return on your money. So he wanted to bring it to five. By the way, the 17 went to 20 later for a series of exemptions that you could do. Forget about gaming it past that, but it never went down; 17 to 5, that they were worried about.
- 4.52 But then, he added something else in as a kicker. After three years, you would be forced to license it. So now you have the patent exclusive, but somebody comes to you after three, they say, "I want to license it. I'll pay you a royalty on top of that," you have to license it. No more cross-collateralizing. They had done this process where if you had a successful drug, you'd cross-collateralize it with another company that had done some of the research, you'd share in some of the royalties, and both would end up blocking off competition on both use patents and on the patent themselves. That was going to be banned under his bill.

- 4.53 Four, if you are trying to obtain a patent by slightly modifying the chemical structure of the drug, you would have to prove that the alteration resulted in, and I'm quoting from the proposed bill here, "a significant therapeutic benefit." Oh, come on. That absolutely caused headaches and consternation across the drug world. "You mean we're going to have to prove before we get a me-too drug approved that there's a significant therapeutic benefit?" It doesn't sound like something they want to go for.
- 4.54 They also—this one hurt Lilly—he also wanted to have the rules changed on the law for extending patents on existing drugs so that to make alterations to continue your patent as it was expiring, you would have to show that you had made an improvement to it that not only had a therapeutic improvement, but that was better than what any generic brand name could do.
- 4.55 Now, adding to Arthur Sackler's consternation was another clause inside the bill that required the drug ads to include, quote, "full and accurate information"—you would think you could imply that full and accurate information should be the standard, but it was now going to be codified in a bill—"and list all reported side effects." That really drove people crazy. They would require advertisers to tell, quote, "the whole truth."
- 4.56 What do you think the opposition was, the general opposition from the drug companies? You will be surprised or not. Science and innovation: "you're going to ruin science and innovation." I get it. "You're going to stop the development of life-saving medications." One of my favorites is "it's unconstitutional and anti-American." You're going to make America into a socialist government, just like all those governments over there, like the National Health Service and everything else in England, and that's not what we're about.
- 4.57 Arthur Sackler argued that it's anti-American more than anybody else. Interesting that he was a card-carrying member of the Communist Party at the time, and that his medical company, to his credit, would give jobs to journalists who were blacklisted in the great Red Scare of the '50s when McCarthy and others were. There were blacklists in Hollywood, and people were losing their careers because of some rumor or aspersion about having been a Communist. And if they lost that and they lost their job, Sackler would give them a job at his company, writing ad copy for drug companies. It's not what they necessarily wanted, but it was at least a job for them. But meanwhile, when it came to this law, he was adamant that this was anti-American as it was.
- 4.58 So they fought—they, meaning the drug companies, fought. How they stopped Kefauver was essentially going after him, going after the bill. By the end of that summer of '61, if I had been an analyst then on the drug industry and on politics, I would have said to you, "It's dead. It's dead in the water. They've killed it. It's not going anywhere." Because it looked like both sides of the aisle had said, "Oh, God, this guy could fall for it. He really went overboard."

- 4.59 And then something happened. Remember when the Elixir Sulfanilamide disaster took place in 1937, 107 people die, and then they act and they pass the 1938 law? Well, what happens next is thalidomide to the rescue. And so the next great drug disaster takes place, which is this drug, prescribed as a sedative and anti-nausea medication, mostly over the counter in Europe to pregnant women. And it caused a health crisis starting in the late '50s and 1960s. Ten thousand babies were born with birth defects worldwide, and really severe limb deformations, organ malformations, thousands of fetal deaths.
- 4.60 The US avoided this tragedy because it had a hidden hero: Frances Kelsey. She was a medical reviewer at the FDA, and she was a relatively new medical reviewer, one of the few women in the FDA at the time. The company from Europe that was trying to get thalidomide approved in the US had presented all its studies. She said, "I don't know if they're rigorous enough." They said, "What are you, crazy? It's being used by tens of thousands, hundreds of thousands of women who are pregnant all over the world. They're the best example that it's safe."
- 4.61 There was a rule then called the 60-day rule that if you, at the FDA, did not respond to a drug application within 60 days, the company got it approved. That's pretty remarkable, considering everything. But she was able to extend that because the only way that you could not have it automatically approved is if you said they were short on their paperwork and what they had to supply. And she just had this feeling, and she kept asking them for more information. So she stands up to the pressure from them and refuses to approve thalidomide without some more safety evidence.
- 4.62 Now, there's no rule in America that requires that a drug company has to show you, at this point, that it's effective, but they do have to show you safety. And she's not convinced that the European studies that they had used are rigorous enough. This is her own decision. Her determination protected tens of thousands of American babies from the same disaster of the side effects of thalidomide. Her single action is what caused Congress to then act because they said, "Oh, my goodness, we just barely avoided this because it happened to land on the desk of the right regulator at the FDA. But maybe we should get back to the Kefauver bill and pass something that will be lasting legislation," and that's what they end up doing.
- 4.63 So the backlash from the pharmaceutical industry, campaign to discredit Kefauver, they're doing a whisper campaign behind the scenes that would do opposition research and political campaigns proud. They're saying everything from the fact that he's lost his touch. He could be a little senile. He was paid for by the Reds and socialists to go ahead and destroy the drug company. Nobody's putting their name to it, but they're doing it. "The trail of the investigation is a precursor to socialized medicine. You let Kefauver pass this bill, you're going to get socialized medicine just a little bit." Lobbying efforts constantly to undermine the hearings and accusations of political motivations, time and time again.

- 4.64 But thalidomide has changed the discussion. It's put safety up at the top of the governance list. The end result is the landmark safety regulation. So what comes out of this period—and I do think that it's important because you end up with legislation that, like Elixir Sulfanilamide made the 1938 law, this ends up making the 1962 law. It's called the Kefauver-Harris Amendments—not quite as exciting as the drug antitrust act, but still, it's pretty good. Here's what it ends up doing, the seven key takeaways, as I say.
- 4.65 First of all, it's the first time, believe it or not, that drug manufacturers had to register with the FDA. Before that, you could be a drug manufacturer, you could set it up, you could be running an operation, you could be manufacturing drugs, and unless the FDA caught sight of you or knew you were doing it or came across it for some other reason, you never had to register that you were a drug company. Now it says, "You're manufacturing drugs, you're selling drugs, register with the FDA." That would seem simple, but it wasn't part of the law until that moment. The 60-day time limit for an FDA examiner to approve a drug application or respond, they got rid of that, scrapped it. So that became history, which was actually pretty good.
- 4.66 In addition to registering with the FDA, by the way— I personally like this next part—all drug companies had to submit to periodic on-site inspections from FDA regulators about safety, cleanliness, everything else, and whether they were really doing it. They had never had to have anybody from the government show up at any facility before to check that.
- 4.67 The FDA got the power for the first time to recall drugs—and I'm going to quote here —"that judged an imminent hazard to the public health or that failed to deliver their therapeutic claims." That meant the agency no longer had to wait for a trail of dead patients to decide why that a drug might not be meeting its therapeutic claims before recalling it. And that becomes a tool that they use like a hammer later on the fixed-dose antibiotic combinations to kill off most of that market. Right away, the drug industry fights it, but that's how they end up using it because it failed to deliver on their therapeutic claims.
- 4.68 The statute also empowered the FDA, much to the consternation of Arthur Sackler and his brothers, who talked about this, to ensure that advertisements to doctors wasn't as much as the first draft—and this is the language from the act—"struck a fair balance." That's the rule now: "a fair balance" between promotional hyperbole and scientific accuracy. The law did not define what fair balance meant. So you know it's going to get pushed to the edge until somebody decides you've gone past fair balance, but at least they had some language in there.
- 4.69 Those are all good, but this is the biggest: before a pharma firm could get its drug approved going forward, it would have to show not just that the medication was reasonably safe, but that it worked for what it was intended. We waited till 1962 to get a requirement that the drug companies actually have to show us that the drug they're

selling us does what they claim it will do. They have to show a therapeutic benefit. And not only that, but the therapeutically useless medications would be taken off the market— that puts at risk a whole bunch of things that they're making big profits on— and that drugs would have to establish "with substantial evidence"—that's the words in the law—"substantial evidence and adequate and well-controlled studies."

4.70 That's the start of clinical studies and trials that drug companies have to do to submit to the government to get their drugs approved. So now they have to submit clinical trials that will show not only that they are safe, but that they actually have a therapeutic benefit. That seems to me—I don't know how you feel about it—to be basic. That's what you want to have as part of the drug industry from the very start. But it took this law in '62 to get it done. It wouldn't have even gotten done to this extent if it had not been for thalidomide, which is really rather amazing.

4.71 I will tell you that internally—and I've seen the documents later produced in other congressional investigations from this period inside the drug industry and the internal memos—they breathed such a sigh of relief that patents were not touched. So they had to deal with all this; they weren't happy about it. But they still had the 17-year patents in place. They still had unfettered pricing power, and that was absolutely critical to them.

4.72 So, the overview, first major overview since 1938, and I will tell you that there have been drug laws passed since—we're going to talk about them in later lectures—but nothing like this. The reason I spent so much time on this, this is the largest single change in drug laws ever. There have been modifications over time, but this is the one that set the standards that said you're going to prove that you're actually delivering what you said you did.

4.73 It ended the industry self-regulation. If you think about it like this, that's really what the industry was doing. It was regulating itself beforehand. It was saying, "We'll find out. Oh, people are dying. We're sorry. We'll pull that drug. We'll recall it." No, no, no. Federal regulators will now recall it. "We'll tell you if it's safe." No, you're going to have to prove it's safe in clinical trials. "Trust us. This is what it does." No, you're going to have to prove what it does in clinical trials. You're not going to be able to regulate yourself with this. Now you have regulatory oversight.

4.74 Now, it's a different problem. The FDA gets this, and let me tell you, they are suddenly swamped. The FDA is trying to expand. They're trying to deal with this. The drug companies are preparing these clinical trials. So it's not an easy transition. It's not like the FDA is prepared for it and says, "Oh, yeah, we can run all of this. No problem. We'll be on top of it." There's a real rocky period for a period in the '60s in which this is taking place. But the framework is here. It empowered the FDA to ensure safety and efficacy. Before that, it could just ensure safety to a certain degree, but not really doing much about it. So the Kefauver Amendments are a pivotal moment in the history of drug regulation. I really can't overemphasize that.

- 4.75 I did forget to say that in addition to inspections, they had to meet quality control standards. They didn't exist before. It was the "trust me," the honor system. It might be all right if you have the honor system and what you're manufacturing are widgets or something, but if you're manufacturing drugs that people are consuming, you might want to have actual quality control standards.
- 4.76 The FDA sets regular inspections. Part of that problem becomes later on. The FDA is supposed to also inspect the facilities of foreign drug manufacturers who are making pills that they're sending back to the United States. Antibiotics, most of them are made abroad now. And the FDA spread thin those regular—they're supposed to be unplanned, but you really can't plan to go into China and do a drug inspection on the spur of the moment with four different FDA inspectors. So the foreign drug manufacturers—and we can talk about this later—always know when the FDA inspectors are coming, and when they arrive, everything is spick-and-span, wonderful. It's just fantastic. Never seen such a clean plant in all my life. Every standard is met. I wish American plants look like this. So foreign plants very seldom have any problem at all.
- 4.77 The FDA Recall Authority for imminent hazard to public health— it's not used very often, but when it is used, drug companies will fight it. The question is: what's an imminent hazard? Accurate pharmaceutical advertising— they did hate that. But once the threat to pricing power was defeated, the industry was happy to actually return to these questions of what I call oversight and safety.
- 4.78 So what's the legacy of this era? The pharmaceutical industry hates the extra cost of the steps for the safety proposals and the clinical trials. No question about that. But if you said—I believe no one offered this as an alternative to the CEOs of the drug companies at the time—"I'll give you a choice: you want to spend a lot of money and slow up the process for drug approval by having to do clinical studies and submit safety and therapeutic test to the FDA, or do you want to lose your unfettered pricing power and your patents go down to five years?" They would say, "Neither one is a great choice, but I'll take the safety regulations," and that's what they ended up doing on this.
- 4.79 Pressure for faster market entry—this is something that happens all the time. Pharmaceutical companies will come and they will say, "We need expedited handling. We've got to have this one fast because we've got a drug out here that's so important. If we don't get to the market, people are going to die. People are going to suffer needlessly. You've got to expedite your approval process." In the hope that it makes things a little bit faster, maybe they don't have to do as much work.
- 4.80 Eventually, you'll see this, the FDA goes ahead years later and starts a program in which if you're a drug company, and you want an expedited approval, you can pay for it. You can say, "This is an important drug"—and not just a few dollars; it's a big fee. And then, they will rush it. They'll put more people on and everything else and get it through

faster. Well, guess what? A lot of pharmaceutical companies assume that's the price of doing business. They add that on to the cost of the drug. Almost three quarters of the FDA's budget comes from the expedited fees. That's bad. When you are the regulator, the oversight of an industry, and the industry is paying you money to expedite your reviews, and that's over half your budget, sorry, bad sign, but I'll talk about that later.

- 4.81** Approval does not guarantee safety. I think that one of the biggest errors made by the way the press covered the Kefauver hearing—I look back at the contemporaneous press—was that their view was, "We did it. We finally have a guarantee of safety for drugs going forward because they're going to be shown to be safe in a clinical trial, and they'll show to be therapeutic." We learn drug after drug after drug that the times when we have to have drugs that are having lethal side effects, causing cancers, causing blood clots, causing breast cancers with hormone replacement in early versions, the pill, ten times the amount of hormones that had to be in the pill when it was approved in 1960, the first lifestyle drug, those are all drugs that were approved later on through a process of clinical trials, almost every case. Yet we still had side effects that were terrible.
- 4.82** So one of the takeaways from that is this process is good. It helps. There are drugs that don't make it out because they aren't safe enough or they don't have a therapeutic value. But it's not a guarantee of safety. You still don't know until it's out in what I call the big clinical trial, which is the public, when it's prescribed in big numbers for more people.
- 4.83** Ongoing need for vigilance—the history of these tragedies that take place occasionally, and the industry's tendency to prioritize profits, underscores the continued need. You can't just pass a law like the Kefauver Act and then hope that everything is going to go away. This error did not eliminate the danger. It just put some of it into a statute from Congress.
- 4.84** So summaries and takeaways: safety failures shape drug regulation. We get that. Cycle of tragedy, reform, and then industry resistance—I'll tell you you could take those three, put it down, and the same thing is going to take place when you have something else that happens. Vigilance is essential for patients, and the industry opts for tighter oversight rather than cutting profits. Profits still stay big.
- 4.85** How to be a smarter patient: ask if the drug is new. Again, it goes back a little bit to what I was talking about in the last lecture. The safety record is not as long. You just want to know if it's new. It doesn't mean it's bad, but they prescribe a drug to you, something for hypertension or whatever else, if the doctor doesn't tell you it's new, you want to know if it's new. How long has it been out? Has it really been studied with a wide patient population? Sometimes—again, not medical advice— you might prefer older, well-studied medications. They have the safety record that's there on the record.

- 4.86 Treat marketing claims with skepticism. This has never been truer in this day and age, from 1997 and on, from the time that we have had direct-to-consumer ads when drug companies really are selling to us these remote possibilities on television. It's odd to think about because why are they running ads to us, the consumer? Because we can't write the prescriptions to get the drugs, even if we want it. They're doing it because Arthur Sackler had a memo in which he said to Pfizer, back in the 1950s, "I think that we"— meaning the industry—"can create consumer demand."
- 4.87 They couldn't do it back then. They did it by having articles written in a newspapers and publications that they thought the consumer would read. They targeted publications like National Geographic, Time and Newsweek because those magazines in an era before— most people never remember going to a doctor's office. You used to go to a doctor's office and sit in a waiting room, and those were the magazines that were most likely to be on the site for patients to read before the doctor saw them. And so they'd want the articles in there. That's how they targeted it.
- 4.88 Now they targeted the consumer directly because they want us to go ahead, read the article or listen to the ad and say to our doctor, "Hey, what about that new medication I just saw the other day? What about that great one that's supposed to do this? Why aren't you talking to me about that?" So you drive the demand by doing that.
- 4.89 Monitor symptoms and report side effects. I'll tell you: you get a new medication, you get it, tell your doctor if you have side effects. If you really have unusual side effects, you should report them because I think that one of the things that happens is we only get safety profiles of these drugs as we find out how they react. You have to filter out for people that over-report every adverse effect. There are people that will Google a drug and see what the adverse effects are, and then they think they have all of those. So I understand that. But on the whole, I'm saying that if you do have a side effect, you want to try to report it as part of being a vigilant patient. Never assume that regulators caught everything because it's not a guarantee of long-term safety. Approval is a snapshot in time of that approval, but it doesn't tell you what's going to happen afterwards. And we'll see an example of that later.
- 4.90 In the next lecture, I'm going to cover a bit of America's addiction crisis. And here I don't mean opioids, believe it or not. We start with something not more mundane, but back in the day, it's going to end up talking to you about mother's little helpers, the Rolling Stones, called it, Valium, Librium, the drugs, the benzodiazepines that became massive successes and how they were created to. This prescription drug evolution from tranquilizers to today's opioid epidemic, it sets the DNA and the groundwork of it as you go forward. It also shows you how the public perception of drug safety changes over a period of time, the influence of the pharmaceutical companies coming in and prescribing the drugs, and the impact that addiction has on society as a whole.
- 4.91 By the way, just as we leave this, yes, I'm going to talk about amphetamines also tomorrow. Amphetamines became big drugs in the '60s in terms of diet centers. There

were thousands of diet centers around the country prescribing essentially pure amphetamines and speed, and then they got abused and abused. But Valium and Librium, its predecessor, become these massive drugs. Valium becomes the first \$100 million drug in a year in sales. That was the big news. And they become the first billion-dollar drugs in drug history. And who was the person Hoffmann-LaRoche chose to market them? Arthur Sackler.

- 4.92 The drugs that were starting to get prescribed as off-label, as they learn different effects of the original use of a drug, for example, like insulin, and we see it with sildenafil and tadalafil, originally made for vasodilation. And then, we saw other effects on people. And so they marketed almost in a vanity way, like losing weight and for ED and such. So these companies that discover a new use for these drugs, I would be curious to see how much more they make or sell based on the new suggestion, the new thing they're trying to—they saw use for it. Sorry.
- 4.93 We'll talk about this in another lecture, but off-label prescribing is a massive industry. Think of this: when the doctors were given the authority to do that in 1938, a small number of drugs. And then, over time, as that's growing, it's still an ability of the doctors, and there are any number of drugs. The hypertension drug that when people were taking, they noticed that they thought it gave them a little bit more hair growth, that becomes minoxidil, eventually. But when that was being prescribed, not for hypertension, but instead for hair growth. That was all off-label, and the companies like Upjohn love that. They sit back. Botox was the same thing. Botox was approved under a thing called orphan drugs—we'll talk about—for a problem in the throat in which you had a stricture, and it would soften that, and you'd be able to deal with this condition. But then people noticed that it softened wrinkles and that, and it was being prescribed for cosmetic reasons from Beverly Hills to New York to wherever else. And the company Allergan made far more money from the unauthorized prescriptions of Botox than it did from the actual prescribed use.
- 4.94 Now, in the annual reports in public companies, they will list what the sales were for a drug. But you don't know as the public, whether those sales were for prescriptions for the actual purpose for which it was approved by the FDA or for an off-use label that was being used because the drug had become popular for that. It's the same with Viagra, which was an offshoot of, again, a hypertension drug. They realized, "Oh, my God, it's for erectile dysfunction." They came up with that name. But that was all off-label.
- 4.95 Those companies eventually, and I mean, years later, in some instances, go to the FDA, they run a clinical study, and they get an approval officially for that off-label use because it's so large. But meanwhile, they have essentially done the clinical testing by having, at times, millions of people try the drug for an off-label purpose. If there aren't a series of deaths and people that are walking around unable to raise their right arm and their neck is to the side, they're able to say this is relatively safe.

- 4.96 What I find, and I rail against, and I personally find so disturbing, is that we have this safety regulation in that you have to submit clinical trials to show that the drug is approved for a therapeutic purpose, and it's safe. And then, you have doctors prescribing it for something for which it wasn't approved, and you don't know if it works for that or not. Yet, the FDA allows that to take place because that's the way the rule is, and nobody has closed that off. Drug companies don't want it closed off because it makes it possible that when you're out in the market—
- 4.97 And this goes to what I said before about the possibility of bad side effects. The drug's approved, we think we know what it does, and then there could be bad side effects, but there are also side effects that companies can cash in on. It has another use. But why shouldn't they be required at that point, if doctors are starting to prescribe it, to go back to the FDA to get approval for it? And they don't.
- 4.98 So time and time again, it's one, to me, of the greatest loopholes in the business. AbbVie has a drug called Lupron, for instance. Lupron is used to chemically castrate sex offenders and is approved in a precocious puberty when hormones are too high for children sometimes to balance it up. But it's used off-label as a puberty blocker for children who identify as another identity. That's off-label use. It's never been approved for that. And yet, AbbVie is making a lot of money from the use of that drug as a puberty blocker for which it's not approved. So my point is these companies—and they know it, they don't mind it—they allow it to flourish because there's nothing technically wrong with it. But it's a great gamble every time it takes place.
- 4.99 With the government subsidizing and apparently endorsing so many drug companies, the true innovation and competition and pricing, we stifle that with the government being so involved. And even with the FDA approval of so many companies and not approving homeopathy or other methods of taking care of ailments, what's the solution or the balance between the government being expected to take care of public health and safety and not over-endorsing or subsidizing certain companies and offsetting the competition?
- 4.100 I think you hit the core question. It's a balancing act, and often isn't balanced very well. There are some times in which it seems the regulation is too heavy-handed, but not often, because I tell you, drug companies really figured out a way around it. We'll talk about some of these with pharmacy benefit managers and orphan drugs later. So law is passed for good intent, or to have some regulation in place. And then, the drug company looks at it and says, "Okay, that's interesting. That's a roadblock. How do we get around it?" And they're always faster than the regulators and the government. So it's hard for the government to overregulate, but they can.
- 4.101 The biggest factor that has moved some of the drug manufacturing abroad has been cost. The drugs are just manufactured cheaper abroad. No question about that. If you're selling the same drug and you can have it made in India or China, two places that have been enormous in terms of the pharmaceutical industries, and you can have it

made at 20% of the cost of what it might cost you to make it in Iowa, the drug companies move over there internationally without any doubt.

- 4.102** But I think that not all innovation has been stifled. There are still efforts. We are at the forefront, the beginning era of biotechnology, biologics, what could be done in the future. The FDA doesn't even get power to the 1970s to look at biologics, which is pretty amazing. They're always behind the eight ball. But I do think that in the future, there is some company out there, and I don't know which one. They're all chasing the same thing. They all want to have the next penicillin. They want to have the first biologic. They want to have the first ability to go into one of these technologies that might be able to turn off a cancer cell, as opposed to turning it on. So they're looking for that because they think—there are scientists working on it because they think it's for the public good, but they also know that's a multi-billion dollar hit, and everybody else is going to have to license it. I do think there is innovation, and we'll see it in the future. It hasn't been totally stifled away. I wish there was a lot less of the me-too drugs.
- 4.103** In your research into the drug administration and the FDA and all of the history, I'm wondering, has there been any collaboration between the food and beverage industry and the drug industry, the pharma industry, as far as pushback on FDA regulation or any collaboration or anything like that?
- 4.104** Yeah, so Big Food and Big Pharma sometimes are allies, right? It depends on the issue, without any doubt. And there's a fight that's gone on for years in the state in which I live, in Florida, about streptomycin used on citrus crops. So citrus crop is very important in Florida, big part of the market. It's bringing this old-school, very cheap because it's so old, antibiotic on the crops to keep them away from pests and everything else—maybe not the greatest idea in an era of rising antibiotic resistance.
- 4.105** The food industry and the pharmaceutical industry is lockstep on that as to why that's good. It's not too costly. It's effective. Don't worry about the antibiotic resistance claims; they're overblown. They have the same thing. So yes, not always, but there are those instances in which the drug industry is applying its wares to the food industry, and to that extent, the food industry is moving together with them. They both have muscle, but they're independent of each other. They do sometimes work as allies.

Lecture 5: Mother's Little Helper

At a glance — 'Mother's little helper' — psychiatric drugs.

- 5.1 We're going into Lecture 5 of The Secrets of Big Pharma: How To Be a Better Patient. I looked last night, just out of interest, at the audiobook for Pharma, my book about the pharmaceutical industry, where a lot of this research and material came out of after years of working on it. The audiobook is 23 hours for the narration. I didn't realize that coming into an eight-hour course, I'm being more selective.
- 5.2 So what you are getting is not all the secrets. The eight hours of this lecture, I've tried to cover and put together for this course, I've created just what I call the highlights and throughlines. So sometimes, yesterday, we ended up in 1962 on what I thought was this major legislation passed by the government from the Kefauver hearings. Today, I'm going to go back just for a second into the '50s, into the early part of what I call this chemical calm that's being sold and then come forward.
- 5.3 Because this ends up being—you keep finding that if you go through a straight line on the pharmaceutical industry, you get a series of events of what happened. But if you're doing it in terms of themes of how the industry operates, and you tie together different products—we have the addictive products, we have the products that are sold for a whole series of anxiety and other issues, you have the products that become the big blockbuster sellers—there's a theme to them that sometimes has to tie back, and you just need to step back for half a second.
- 5.4 So I call this Lecture 5. We're going to explore how they do Mother's Little Helper to the addiction crisis of opioid. But this really becomes marketing by the pharmaceutical companies, but also in a particular sense, in how they target women. This becomes a feature. I have a separate chapter on this. That's not to say they don't want to sell to men. They want us taking the pharmaceutical drugs as well. But they have a particular way that they're selling in the '50s and '60s, and it becomes a throughline after a while.
- 5.5 So it's really the middle of the 20th century when America discovers what I call this "chemical calm." Millions of Americans are exhausted, anxious, overwhelmed by the pressures of modern life, and it creates the market for pharmaceutical marketing.
- 5.6 Now, you think about this, this is post-World War II. So the drug industry is just coming into its own after penicillin and that. But post-World War II is also the time when America is in its golden age, having won World War II, you would tend to think the 1950s are fantastic, but there are some technological developments coming in. There's a whole series of new inventions from TV rolling out at the end of the '50s, everything else is taking place.
- 5.7 And there's a sense of anxiety with all of the progress and change. Leonard Bernstein in 1960 did a symphony called The Age of Anxiety, based off of W. H. Auden's poem of the

very same name, The Age of Anxiety. There are Time magazine articles and Newsweek articles in the '50s talking about this being the age of anxiety. They should have fast-forwarded to today to see the age of anxiety, but at least in the '50s, it felt like that for a while. So pharma sees this as a marketing opportunity.

- 5.8 If emotional discomfort, if that anxiety could possibly be medicalized, then maybe you could end up having drugs and create products that could be successful. This is when really the tranquilizer boom comes in. So there's this cultural shift. It lays the groundwork for what's coming later, but it's not yet what I call the blockbusters that come into the 1950s. This is the time when the drug industry recognizes this profit potential, as I say, in these conditions, if they can come up with the chemical solutions for it.
- 5.9 Targeting women in a broad sense—we'll get into this in more detail. We're going to explore how the pharma companies targeted women in particular, looking for a profitable market, and how the tranquilizers and opioid booms, Miltown, Librium, Valium, and then the transition to opioids has this gender throughline through it, leading to the creation of the modern addiction crisis, the opioid crisis, which has ravaged the country for the past 25 years. This lays the groundwork.
- 5.10 We started with addiction back in the days of the Civil War and morphine and Charles Pfizer and all of those companies. They got their starting addictive products. Then they had to go cold turkey after that was all banned. But they end up finding ways to find new addictive products. And when they put out products that they don't know are addictive at first—we'll see this when it comes to Valium later—when they realize that there's a physical addiction to it and dependence, they don't pull it off the market right away. They just are quiet about that, and they bury the information until they can't ignore it anymore.
- 5.11 This is really about medicalizing stress in some ways. It's pathologizing normal emotion and seeing if a pill can restore the balance to you. This is from the book I had: "Miltown created the psychological template for the blockbuster that followed." Barbiturates, which we talked about before, which had been around in the '40s and in the '50s, were very, very tough. They were given to people, but they caused you to be out of it most of the time. They caused a dependence. They were heavy downers mixed with alcohol. People would occasionally overdose. Lilly had the biggest seller into the early '50s called Secanol. Secanol was a powerful barbiturate.
- 5.12 So pharmaceutical companies are looking for something a little bit better than that. This French surgeon comes up in 1953 by accident—he's looking for a cure for malaria and ends up finding Thorazine, which he thinks nobody's going to be able to take. But Smith Kline says, "Okay, we'll try that." They buy the rights to Thorazine, and they put it out, and it becomes a hit in '54 and '55. Three million prescriptions go out, 3 million prescriptions in the very first year, which is an extraordinary number for such a heavy drug.

- 5.13** If you look under the 3 million, you start to look under the numbers, 80% of those prescriptions are written by psychiatrists. Most of the psychiatrists writing the Thorazine prescriptions back in '54, '55 are working in mental asylums. So it became a way to calm patients, so you didn't have to worry about patients as much. That wasn't bad for Smith Kline; they were making real money from that. But that's where the market was.
- 5.14** Arthur Sackler, who at that time was advising a number of drug companies, was telling them—because he's a psychiatrist, his two brothers are psychiatrists— he says in these notes, it's quite fascinating, "If you really want to hit a drug, you've got to come up with a drug that isn't just prescribed by a psychiatrist. You need to come up with a drug that ordinary doctors will prescribe." You need—Sackler's words for this: "you need to have Thorazine light," which is interesting, if there could be such a thing, an emotional aspirin. The idea is that you could almost have a feel-better pill.
- 5.15** So, is it possible that you could have a lifestyle pill in which you weren't actually given a prescription because you were ill or had a disease, but you were given a pill because it made you feel better about doing everything that you normally do in everyday life? That was the ultimate dream. No side effects, a feel-better pill, and everybody takes it. So he's thinking big at that point.
- 5.16** Now, there's a researcher, a scientist who comes up, he's working for a company, and he comes up with a version of what he thinks might be Thorazine light. He works at Carter's. Now, Carter is interesting. Carter Products, they made a thing called Carter's Little Liver Pills. They made a laxative that was selling for \$450 million a year. And their other two big hit products: they had Nair, which was a cream they used to take hair off, and they had Arrid, that's their deodorant.
- 5.17** They really weren't in the pharmaceutical industry. So, they weren't an Arthur Sackler client because they were not thought of as a drug company. But the scientist working for them who's supposed to be coming up with consumer products, comes up with the idea inside this lab for what he thinks is a better version of an anti-anxiety medication. The company names it Miltown, after the lab in which they were doing the original testing in Jersey— not the greatest name in the world, I will say.
- 5.18** At first, Carter decides, "You know what? It's not for us." They hold off from going to the FDA for nearly a year because they think, "There's not going to be any market for this. Who's going to prescribe it? We're not sure there's a need for it." Eventually, they go to the FDA. American Home Products comes in, likes what they're doing. It says, "By the way, we'd like to license it at the same time. Give it to our subsidiary, Wyeth." They'll put it out under a different name called Equanil. So they'll be chemical twins.
- 5.19** So Carter says okay on that, and they say to the FDA: how are we going to get this approved? They go to the FDA, and they say, "What are our options?" The FDA says,

"Well, you could have it as a tranquilizer." And Carter says, "No, we don't like a tranquilizer. Tranquilizer is Thorazine, and people think of that, so we're not looking into that." "You could put it up as something along the line of a barbiturate." No, they don't like that. So Carter said, "Maybe you could create a category for us called a mild tranquilizer." The FDA said, "Okay, that could work." So they called it a mild tranquilizer.

- 5.20** The minute it's called a mild tranquilizer, there's nothing else on the market like that. So the idea that doctors might think it's safe really takes off. Carter comes out and puts that pill out. Let me tell you how successful it was: they were caught completely by surprise. Inside the company, they're talking to each other, saying, "Can you believe this thing is selling like that?" It took off. So in 1954 and 1955, Thorazine accounted for 99.6% of all tranquilizers sold in America. I mean, talk about having the market. That was the market.
- 5.21** One year later, Miltown and its chemical cousin, Equanil, sold by Wyeth, had gone from under 1% of the market to 70% of all tranquilizer sales. It essentially had taken over. Carter held a symposium in New York, took all the doctors who were speaking enthusiastically about that, took a page out of Arthur Sackler's marketing, and sent out bound copies of that symposium to 153,000 doctors in the United States and said, "By the way, this is the next wave of good pill." Thirty-six million prescriptions, one-third of all medications provided in 1957 were Miltown prescriptions, which is really amazing from nothing. One in 20 Americans had tried it after two years on the market.
- 5.22** It was the type of thing where there would be stories about it. So there's a famous drug store in Hollywood called Schwab's. It has this long history, and it ran out of stock in four months, 250,000 pills, which meant people wanted even more because you couldn't get it. There's a comedian at the time, Milton Berle, who said, "I'm going to change my name to Miltown Berle because it's so big." The writer Norman Mailer, Tallulah Bankhead, the actress, Tennessee Williams, the playwright, talked about how it had changed their lives.
- 5.23** One of the things interesting about that is this was a period in which people weren't ashamed of saying that they were taking a mild tranquilizer, so they talked about it. And then this very strange thing happened across the society of ours in America, where some people started to brag about taking it because they thought that meant that they were really productive and successful. They needed Miltown because they were doing so much and had so much on the plate. You were doing nothing at all over there. It doesn't matter. You don't need Miltown, but I need Miltown because I'm really doing and achieving a lot. That was a very unusual take.
- 5.24** Then what happens is, there's this wonderful moment in 1958— I happen to like this. I don't mean wonderful in terms of drug history, but in terms of just a side light. Carter says, "The AMA's having a convention in San Francisco." And it turns out, they said, Salvador Dali, the artist, his wife, Gala, was a Miltown user. They talked to her, and she

said, "It changed my life." And so they said, "Well, will your husband design a piece of art for Miltown at this AMA convention?" Salvador Dali ended up creating—\$100,000 they spent on this; \$60,000 was Dali's fee—this 60-foot long, undulating, parachute-looking, it looks like an intestine of sorts, but it's supposed to be a dream state that you could walk through, and it was moving a little bit, and that was the feeling of being on Miltown.

- 5.25** I can't show you that because all the pictures of that are all locked up by Getty and these companies that have bought the rights to those photos, unfortunately. If you go on to do a Google search, you'll be able to find it. It's rather fascinating. The amazing thing is that, of course, was destroyed after the awards. It would have been today a multimillion-dollar piece of art if it had been, but it was very, very clever.
- 5.26** Now, what happens is this penicillin of psychiatry, Sackler is a little jealous that it happened in a company that he wasn't representing, but he likes the idea of what Miltown was doing. Miltown sets the basis. Afterwards, what happens, by the way, fortunately for them, it doesn't happen until the early '60s, but the groundwork for the replacements— this is the DNA and template of the pharmaceutical industry: a drug comes out, it's a major hit, and then we realize afterwards there are problems with it. The more problems that come out, that drug sales go down.
- 5.27** What happened with Miltown, it was high-dependence. Oh, my God. The possibility of overdosing together with alcohol is fairly high. All the problems that you would have had with barbiturates, it was barbiturates light, but it was still barbiturates in that sense. It becomes a problem that then creates an opportunity for other drug companies to come in with a new and improved drug, and that's what happens. So Miltown is the hit. The question is what will come up eventually to replace it.
- 5.28** And by the way, these are some of the ads on Miltown: "Stop Emotional Suffering in Chronic Disease." I love this. "Believe the Anxiety Behind the Tension. Relieve It." "Proven, Scientific, Selective." This is before any of the proven, scientific, and selective was required at any time by the government. "The Battered Parent Syndrome." If you were a housewife, you needed this. "Pregnancy Can Be Made a Happier Experience." Think of this. We always say, "If you're pregnant, don't take this, don't take that. Be careful. You're pregnant." It's on everything. Here's actually being suggested that this is a way to make it better. So these are the ads that are being run to doctors. "Reliefs Seriousness In Allergic Patients." I mean, you're talking about every possible thing. Miltown is that fix for everything. And for a moment, it does hold the idea of the market.
- 5.29** The executive monkey experiments, bad science—two of you know about this a little bit: Walter Reed Hospital, Army Hospital in the 1950s, starts to do studies because they really are testing stress. They're not doing this for pharmaceutical companies. They're taking monkeys, hundreds of them over time, a bad series of experiments.

- 5.30** There's one set of experiments in which they lock two monkeys into a contraption that looks as though they're locked in. They have very little movement except for the ability to move their arms a short amount of space, but their heads are inflexible and so are their legs. Then, the device is attached to their feet delivers an electric shock to both monkeys. They shock every 15 or 20 seconds. One of the monkeys has a small lever which is in reach. If it reaches out and touches that, presses it, it stops the shock from coming in. Monkeys, pretty smart, They learn very quickly, and over a period of time, that monkey operating the lever keeps pressing it to stop the shocks coming in. That monkey is dubbed the executive monkey because it's making the executive decision to stop the shocks from coming in.
- 5.31** Those experiments would go on and on. When the monkeys died, if one died, the other one would then be euthanized. They would do an autopsy on the monkeys. And what they found in each instance was that the monkey who operated the lever, the so-called executive monkey, had ulcers. Basically, that was the worst situation that they had. Some coronary thickening of the arteries, but the primary problem was the development of ulcers. The non-executive monkey, the one who just was there without operating the lever, didn't have the development of those ulcers. So there was a level of stress that obviously came to be—what they said was the executive monkey had a level of stress that caused the ulcers to be made.
- 5.32** What happens with that is Arthur Sackler says, "You know what? We could use that for marketing drugs. We could definitely use that for marketing anti-anxiety drugs." Because think about this: this is the 1950s, early 1960s. The breadwinners, the people going to work in the traditional family, were men. They're the ones going on in society, bringing the money, and women are at home with the children. So men are the executive monkey. They develop ulcers because they're under such stress. So we should be able to prescribe things like Miltown or other drugs to men so that they don't develop ulcers. It'll make them better at work.
- 5.33** And women, meanwhile, who are at home, they're going to need all the help you can possibly get because, of course, their medical conditions, if they are complaining about them, they really take exhaustion or frustrations or what I call any type of stress disorder. It's correctable with pills, but they pathologize that into women being somewhat overwrought, hysterical. They need the assistance of being calm so they can be better mothers and better at housework and better for the family and better for the husband when he comes home.
- 5.34** That is the most remarkable. If you would think today, if somebody suggested that, you would say, "You're the most sexist, ridiculous, out-of-sync person I could possibly imagine, and that will never work." But at a time when 95% of doctors were men, believe it or not, that heavy-handed approach ended up working. So, they used the executive monkey experiment, this bad science, for what I call blockbuster marketing coming down the road.

- 5.35 The drug ads, in which Sackler plays a role in some of these as they come out, you'll see them here—Ritalin. One of the interesting things about this, I love this, so Ritalin is a stimulant, we know it. It's used for attention deficit disorder and is over-prescribed. But I love the first of the Ritalin ads: "You Can Do Better Housework If You Just Have Some Ritalin." Use a little bit of speed, and you'll clean the house in no time at all.
- 5.36 "Now She Can Cope"—Butisol. Just doing housework again; give her a little bit of an anti-anxiety. Serax, the same thing: "You Can't Get Her Free, But You Can Help Her Be Less Anxious." "How World Orbits Around Doctors With Psychic Tension Rules Her Universe." "How Mother And Baby 'Picked Up'" —Blatz, another stimulant. Mornidine — "Now She Can Cook Breakfast Again." You need a prescription to be able to do that. Mellaril—"She Can Be Reached At Home," again with the broom in the hand. And Nervine—"Helps You Feel Calm."
- 5.37 Now, you will see no ads—and this are just a handful that I've taken out. Serax for anxiety, Butisol for everyday situational stress, Ritalin for fatigue and apathy, Mellaril for reaching her at home, Mornidine for morning sickness. My God, back in the days, almost you're thinking of the worst possible days with thalidomide, but it wasn't as bad. And even Nervine for nerves. So these ads reinforced the idea that women's emotions and stresses required chemical solutions. You don't see these ads for men. You didn't see them there. Nobody was showing them, somebody on a golf course, waving a golf stick, "Help Him Play a Better Round," he needs Nervine or that. It just didn't happen.
- 5.38 When you eventually get to the marketing of Valium, and we'll talk about that in a little bit, Sackler runs a whole series of ads for Hoffmann-La Roche. This is one of my personal favorites because I find it so objectionable, but it was also one of the biggest ads they had. Women were repeatedly cast to responding to stress an emotional outburst and taking a gloomy toll on their own mental health, so much so that sometimes their children or their husbands would be affected. You didn't want to have that, so therefore, medicate the wives. So Valium is for exhausted moms and moody ones.
- 5.39 In this case, a still-hoping-to-be-married, single woman, whom they have pictures of—she's fictional; it didn't exist, of course. They created this. She's with her father initially, and then she's with friends as she goes along. And then there are pictures of her with boyfriends This is her life album. She comes all along until the last one. She's 35. She is single. And obviously, because she's not married, she is psychoneurotic. There's no such condition as psychoneurotic, but it sounds very impressive.
- 5.40 So, if you turn up at a doctor's office and you say, "I'm a little anxious about life." "Oh, I have just the thing for you." Is it surprising that into the 1960s, women got 63% of barbiturates through that period, 68% of all the anti-anxiety meds, and 71% of antidepressants, and 80% of the amphetamines? Remarkable because these were

blockbuster drugs, with women getting most of the prescriptions, because this was the easy marketing that went forward.

- 5.41** Here's Valium and Librium. Librium and Valium are about to replace Miltown. But in between comes something else: amphetamines. Now, you thought we were past all these drugs, right? They were gone. Cocaine inside of Vin Mariani and Sears and Roebuck selling cocaine, and then we had morphine. All of that was gone, and now they were back to real drugs. They were treating diabetes with insulin. But you know what? That's when I said to you early on that the origins, the DNA of the industry, is in addictive origins, and that's how they end up coming back to time and time again.
- 5.42** So, in the 1960s, [amphetamines were] marketed as diet pills, heavily marketed to women to lose weight and maintain a slim figure, promoted also as pep pills, energy boosters that could help stay alert, active, and productive throughout the day, and sold as mood elevators as well: pleasant, upbeat mood. Fantastic, with no downside, right? They aggressively marketed these, and when I say aggressively marketed them, diet clinics started to open up around the United States. We ended up with thousands of so-called diet clinics that would then have these pills.
- 5.43** Now, where's the money? Energy boosters, pep pills, diet pills, they go down the line. The money—most of the diet clinics, by the way, were subsidiaries of existing drug companies. So they opened them up; they knew how to do it. One hundred thousand 10-milligram amphetamine pills cost the diet center \$71. That was the price they paid for it. And when they sold them through the diet center, they sold them for \$12,000. So \$71 in cost, \$12,000 on the other end. This was big business.
- 5.44** It's not surprising that by the mid-1960s, the government estimates that about half of all of those diet pills, the amphetamine being manufactured by drug companies and being sold through diet centers, was being diverted, the same as happens with opioids to the street. So people were getting addicted to their amphetamines, and then they were filling prescriptions at multiple diet centers, and then they were going out and selling the pills themselves for more money, and so it becomes a street problem.
- 5.45** That's what happens with addictive drugs that are wildly popular. The same thing happens with OxyContin after it gets diverted. People develop an addiction. They want to keep the addiction going. They move it over to that market.
- 5.46** Where's the FDA? We have the FDA. They're empowered. They have all this authority after 1962 now to do things. The FDA is the regulator. So they're going to come in, and they're going to close up these diet centers. They're going to put these drug companies in their place and everything else. Not quite. The FDA also set its own template, which is: it's behind, it's slow, it's slow to react. Even though it was receiving reports that came in and filtered in that—I'm going to quote these—"that talked about amphetamine-induced psychosis, antisocial behavior, and crippling paranoia," the FDA

was told by the drug companies that those were really the exception. They hardly ever happen. But they continue to add up.

- 5.47** Then, in 1963, the FDA had such a benign view of amphetamines that the drug company, CIBA, said, "We want to get our prescription amphetamine, Dexamyl, approved for over-the-counter use." The FDA almost did that. They stopped at the last minute from making an OTC. So think of this: we have this boom taking place for amphetamines, the drug regulator almost puts it on the market, so you could go and get it. Again, back to the period we were with patent drugs in the early days of the drug industry: anybody could get it. Fortunately, at the last moment, in their own vote, they decided not to do that.
- 5.48** But amphetamines opened the market. They exaggerated the market that Miltown had created. They opened the market for the next big wave of addictive products. So this is the rise of what I call Mother's Little Helper, tranquilizers capturing the female market. So Librium and Valium become blockbusters, and they are targeted toward women.
- 5.49** Miltown is starting to lose its favor. There's much higher rates of the overdose with alcohol, dependence, sedation. And one of the things that really is knocking the hell out of the Miltown market is psychomotor dysfunction. People are saying, "I take enough Miltown, and instead of feeling really good, I'm stumbling, or I'm having trouble doing things, or I go to pick something up, and I'm dropping it." It's because it's a stronger pill than people think. Mild tranquilizer description from the FDA, not quite right.
- 5.50** So Hoffmann-La Roche comes up with Librium, which is a new class of drugs, really, a breakthrough: benzodiazepines. It has a better, without any doubt, side-effect profile than Miltown. So you're comparing, remember, it's always fantastic, you're comparing to the last drug that was ground. Barbiturates, Secanol were really tough. Thorazine was slightly better, but a little bit better than barbiturates, but not by much. You didn't have the same addiction to Thorazine; people weren't getting addicted to Thorazine and selling it on the street. Barbiturates, they were getting addicted to. Then you get Miltown, better than Thorazine by quite a bit, but still in the danger zone.
- 5.51** Then you get benzodiazepines, which, in lab tests, look as though—and they did not have the same dependence level, it appeared. They did not have the same tendency to overdose with alcohol in the same lower prescribed doses. There's a whole series of things that appear to be better with benzodiazepines. This was what Arthur Sackler was waiting for, as this emotional aspirin.
- 5.52** Hoffmann-La Roche goes to him. I think that one of the things that they are able to do, the message that Sackler says to Hoffmann-La Roche, and they published a small book about this. The message to women is, "Feel anxious, feel overwhelmed, feel sad, lonely, frustrated, or unheard? These are not social problems. They are not cultural pressures. They are chemical imbalances." Once you start to think about that, the solution? Chemical fix.

- 5.53 So the sales idea, the ads: pale, worried mothers, stressed homemakers, elegant women, glowing with relief after, the husband's marvel at them for—and these are words from the ads—"renewed calm." How's that for an adverb? The pills can give you a renewed calm. References made to "female nerves"—you didn't know you had nerves that men didn't have, female nerves. Irritability, of course, so that gets to the issue. Get rid of irritability for women. Make them less irritable by doing this. "A Tranquil Woman Is a Good Woman"—hard to imagine that anybody could say that and not be locked up in jail, but that actually is what they're marketing. A medicated woman is a manageable woman. Oh, my God. It's enough.
- 5.54 I do work, as you know, on my books with my wife, Trisha, who also is an author. This enraged her. She wanted to go out and take out the marketing people who had put this out and to get a court order to take their bodies out just so she could throttle them again.
- 5.55 The marketing emphasizes women's needs: emotionally unstable, in need of medication to achieve this calm. Doctors prescribe benzos overwhelmingly to women; 70-80% are given to women. That's that male-dominated medical profession. It really is the success. Valium becomes a cultural symbol of female compliance. I really believe this to be case. At one point, by 1970 or that, 20% of adult American women are taking the drug regularly. Pretty remarkable.
- 5.56 For Hoffmann-La Roche and for Sackler, that was an unbelievable success. Remember, this becomes the first \$100 million drug. This is at a time when drug prices were high, but it was tough to get to \$100 million. It becomes the first billion-dollar drug. Librium becomes the number one selling drug in America after it's released in 1960, replaced only by Valium, which is released in '63.
- 5.57 There's a moment at which Sackler, the puppet master, Arthur Sackler and Hoffmann-La Roche have a discussion about, "Hey, should we release Valium?" Because Valium is just a slightly modified, same type of chemical structure. It's a me-to drug. It's a me-to drug of Librium. The thought was, Librium is number one selling drug right now. It's replacing Miltown. We're doing so well with that. Maybe we shouldn't cannibalize ourselves. They said, "No, it won't cannibalize it. It'll be additive." There are people who are taking Librium. They'll market Valium as the true equivalent of an emotional aspirin with very, very little downside. Everything he's been waiting for in some ways comes into this period.
- 5.58 We know this prescribing bias. They give the scripts to women. The diagnosis, literally, the diagnosis from some of the doctors that end up coming out as to why it's being done: hysteria, nerves, domestic fatigue, the male-dominated field in terms of the doctors doing it, and a lack of representation. I think that because women were underrepresented in medicine and in these fields, it was easier for this type of absolute propaganda to take hold and move forward.

- 5.59 So this is the strategy that becomes, moving forward, the idea for pharma: identify a brand, a poorly defined condition, but you've created here an emotional calm. You medicalize normal human experience. The stress, anxiety that we all have every day about something, discomfort, you now tell people, "No, you know what? That's not normal, and you should take a pill to correct it." You position the drug—this is key—as universally safe. Safe, non-addictive, no downside, and it'll manage the emotional pressures. You don't have to worry about the side effects.
- 5.60 Promote heavily to physicians—let me tell you: they did sponsorships, conferences, everything else. Doctors were flooded with information about Valium time and time again. They created that cultural demand for doctors and for the public as well: ghost-written articles, tailored advertising, doctors getting hit big on that.
- 5.61 Cross-cultural demand—leverage the insights, what I call these cultural icons of tranquilizers, the need for their products. Try to get public figures or others talking about it. Make it part of what Sackler wanted to do: make your drug part of the cultural discussion. If you could do that, it would make it more acceptable, not just to doctors, but to patients as well.
- 5.62 Twenty percent of American women are regular users. The materials promote celebrities at times. Elizabeth Taylor, there were addictions and overdose cases, Elvis Presley. You would think that when those started to take place in the '70s, and celebrities who first said, "By the way, I think Valium is a game changer. I feel so much better in my life," then developed dependency to it and had trouble getting off, that people would say, "Oh, wow, maybe that pill has some downside. I don't want to do it." But instead, the reaction was, and what the advertisement kept saying is, "Oh, some people overdo it. They just take too many pills. If you take what your doctor prescribed, you'll be at the happy equilibrium."
- 5.63 By the '70s, by the way, the FDA was not doing anything because Valium stays the number one drug in the world for 15 years. That's astonishing. That's an unbroken record. There are other drugs that have made more money since, billion-dollar drugs that have come in and everything else, but Valium was unmatched in that. It was in the early '70s, only nine years after it had been released, that the FDA was starting to, what I call, pile up the list of addiction withdrawal, cognitive impairment. Those are starting to pile up, but nothing is being done to bring the drug down.
- 5.64 By the late '70s, those reports were at the FDA at a moment where they were thinking: what can we do to solve it? So Valium becomes psychic. Aspirin is a cultural phenomenon, and it really is this cultural phenomenon. Mother's Little Helper, the Rolling Stones are doing a song in which it's mentioned. And by the way, The Stones are doing a song that is not really good about it. They're talking about the fact that, if I have this here, "Mother needs something today to calm her down. And though she's not really ill, there's a little yellow pill." That was pretty much it. That was the idea. Maybe The Stones had a more cynical view of that than a lot of people did in the United States.

But that same year, novelist Jacqueline Susann put out, she named Valium "dolls" in her best-selling Valley of the Dolls. It became part of that cultural milieu.

- 5.65** When the calm wore off, the percentage of Valium users that had side effects or complications 70%. One national survey of doctors came back with patients up to 70% of physiological addiction— benzos can create a physical addiction that people just didn't think was possible. Sixty-five percent of those who are regular long-term users of Valium trying to get off have problems with withdrawal. Rebound anxiety— you took a pill because you were nervous and anxious. And then, when you stop taking it, the rebound anxiety is worse. That's why some people go back to taking Valium because that rebound builds in. Cognitive impairment—thesame test was given in one study to people, on Valium and off, some cognitive impairment in over half the cases. And misuse with alcohol and then later with opioids— big.
- 5.66** One of the things that happens later with opioids, with OxyContin and the Sackler family in Purdue, is when the government first starts to look into the question of whether OxyContin is killing people—this is five years after it was out. It was released in '96. In 2001, they're getting some reports. Then they send in people, and they get medical examiners to send in the reports of those who tested positive with OxyContin in their blood from the toxicology report and were dead. Purdue would say, "But it's not just OxyContin. There's other stuff in there. Look at it. They've got benzos, they've got alcohol. It's a combination. If they were just doing OxyContin on its own, they probably would still be alive." That put the FDA off.
- 5.67** People use multiple items. I get it. They have a couple of drinks, and then they've taken some Valium or Xanax, and then they take some OxyContin on top of it, and eventually, you depress the heart rate enough, and some people overdose. But the drug companies use the combination of drugs to argue against their drug being the culprit. And benzos is very, very tough. Despite this risk, the industry, Hoffmann-La Roche, continued to promote the benzos as safe and effective.
- 5.68** At the same time that they have created this market, they've decided to create another market. Roche is targeting women with psychotropics, and Wyeth and others have decided that there's a different market in which they can target women. This relates to menopause. They've at least this time come up with a passage of life that affects only women. It's not as though they've decided we're going to create something that women are hysterical and more anxious than men are in any case. They said, "No, there's this natural part of life: menopause. What we're going to do, though, is medicalize it." We medicalize it, then we make it something that you don't pass through naturally.
- 5.69** You can't just say, "I have hot flashes, my estrogen levels are dropping, I feel a little bit different, maybe my hair is a little bit thinner over the long term," all those things that might happen through menopause. We say that's not a normal part of life. It's what happens to you, but we have the fix for it. We're going to replace the estrogen that

you're losing with hormone replacement therapy, taken from the urine of horsemares, by the way, that was the original process as they developed it.

- 5.70** The marketing is anti-aging as it comes out, the original idea for this. Pharmaceutical companies say they know how to do this very well. The same reason some people take GLP-1 drugs; they appeal to vanity. So if you don't take it, you're going to get old, shriveled, wrinkled, terrible. If you take it, you'll restore your youth and vitality. Beautifying effects on skin, hair, overall physical appearance, all of the cultural norms of femininity, restoring natural femininity—recapture your natural femininity and hormonal balance.
- 5.71** These pharmaceutical companies are pretty clever at how they go about doing it and how they make it into a big revenue stream and make HRT one of the biggest sellers in the 1960s because they end up having a book that sells the market. A star gynecologist in New York City, Robert Wilson, ends up putting out a book called *Feminine Forever*. That book really changed the way that American women thought about menopause and HRT. He says in there—and it is one of the most sexist books you could imagine if you read it today—that if you go through menopause and you don't treat it, you'll lose not only your femininity, you'll lose your sexuality, you'll lose your ability to think clearly, you will become just a shadow of yourself.
- 5.72** His example of it was his own mother, who was this vibrant woman. He was married as well, but he didn't talk about his wife having gone through this. His own mother, who was vibrant, intellectually sharp, had done all of these things, and then she went through menopause, and she was this discrepant woman whose bones were weak, and she looked as though she had aged 20 years, and she had lost her sexuality and her vigor, and she was a shadow of herself. Then, he gave her Wyeth's HRT hormone replacement, and she blossomed. She came back. It was like that time machine. She came back to being the person that he knew beforehand. So it's never too late. You've passed through menopause already, don't worry about it, because if you take HRT, you can come back.
- 5.73** It turns out—nobody knows it at the time—Wyeth was subsidizing him. They paid for the publication of his book. They paid for his New York office. They paid for him on speaking tours. They had worked with him to make sure that the product was pushed, and other HRT makers helped him get the word out.
- 5.74** Trisha did a book years ago, in 2000, 26 years ago, called *This Is Not Your Mother's Menopause*. She has on her side of the family, all the women had had breast cancer, and so she was worried about taking HRT. As a result, she went through menopause naturally and wrote about that. She found, when she was doing that book, something that I later ended up writing about in *Pharma*; she talked to the son of Dr. Wilson. Dr. Wilson was dead. And the son told us a remarkable story in which his mother got breast cancer twice. The second time, she died. The first time she got breast cancer, his father and his mother and the Wyeth people talked about: what do we do? Because we

don't want to think that she's on HRT, and it might be a cause of her breast cancer. So they kept it secret. They didn't talk about it. The family decided, "We don't talk about the breast cancer because that could hurt sales if anyone thinks it could be related."

- 5.75** Well, what happens here? Barbara Seaman, a writer, wrote a book 10 years later. Congress investigated this. The FDA was getting adverse impact reports, adverse reaction reports, about women taking the early versions of HRT. The same with the pill, the contraceptive pill, by the way, which had been approved in 1960, a big decision by the FDA. They approved it, the first lifestyle drug.
- 5.76** We talked before about a feel-good pill that maybe you could take if you really weren't sick or didn't have an illness. Well, the birth control pill is a lifestyle pill. It allows you to make choices about whether you're having children or not, but it's not for anybody who has an illness. So that was very controversial. It's also filled with, of course, lots of estrogen. It's affecting the cycles and affecting the way.
- 5.77** Both HRT and hormone replacement therapy and the original pill end up having levels of estrogen 10 to 15 times what would be considered safe by today's standards. It took until the 1970s when there were so many reports of blood clots, which was the biggest problem; there were people actually dying because of the blood clots. And in the longer term, if you take enough of an amount of estrogen and you are susceptible and have the family genes that you might develop a tumor that is estrogen-receptive, not everybody does, but if you do, then those cases of cancer in breast cancer, those cases weren't coming out for four or five years after people started to take the pill. So they started to come in later.
- 5.78** Once Congress did the investigation and once they looked into it, then all of these companies said, "Oh, wow, we didn't realize these levels of estrogen were so damaging," and they developed new products that came down lower than the original Premarin. But this particular story of how it was marketed to women for a condition of life that every woman—
- 5.79** The amazing thing is, all women, if they live long enough, you think all women would share childbirth, but that's not the case. We don't have children, so not all women have childbirth. But all women, if they live long enough, will go through menopause. That's it. Only those who pass away early. So menopause is one of those things that everybody goes through. That's why, to people like Sackler and the drug companies, that is a real opportunity, because the entire female population is going to come to this change of life, as it used to be called. And then, how do we tell them, "By the way, you don't want to go there"?
- 5.80** One of the things is so fantastic, do you remember before how I said that Miltown, when it first came out, there were people who bragged about taking Miltown because they thought it made them tougher and that? Well, one of the things that Trisha used to say, when she had her book, she said, "Menopause is a room emptier. You mention

menopause, and people walk away from you. Nobody wants to really talk about it." So, here's a situation which—it used to be called, I remember what my mother said: the change of life. They wouldn't even say the word. It's like cancer used to be the big C. They wouldn't say the word. Menopause was the same type of thing. Nobody was happy about going through menopause, and they would feel off kilter.

- 5.81** Remember, we said the other day: do you take a drug to treat the underlying condition, like you take an antibiotic to get rid of the infection, or are you taking something as a pain reliever to get rid of the symptoms? The symptoms sometimes in menopause can be bothersome, with hot flashes and items like this people will feel. You don't feel yourself. Therefore, the idea of that is what pharmaceutical companies are able to medicalize.
- 5.82** If you control a society's definition of normal emotion, you can market the drug. These are related along with Miltown and Librium and Valium and the legacy of the tranquilizer era. To me, it lays the groundwork for the opioid crisis. It really does. It's different. When we get to the opioid crisis, in a summary, I'll tell you that at the end of this, before I talk about how the Sacklers had set the blueprint for this. But they've medicalized discomfort. They've taken the normal human experiences, and they make them into lifelong pharmaceutical dependencies.
- 5.83** By the way, today, even with all the revised versions of biologic hormones and that for the equivalent of HRT, doctors, even if you don't have a history of breast cancer, will try to say often, "Try to limit it to five years." The symptoms sometimes might come back, but five years is the most. In the '60s, they wanted you on it forever. So once you hit it, once you started it, you were going to take it for the rest of your life.
- 5.84** Overstating safety, minimizing risk—I'll tell you: Valium was not—they didn't just say it in a whisper; they marketed it as safe, just like OxyContin was downplayed. In 1996, when OxyContin goes on the market, the Sacklers are able to convince—by the way, Arthur Sackler, the chief guy who redid—he gets an award from the Advertising Association for having revolutionized, being a pioneer in marketing and advertisement for pharmaceuticals. I don't view that as a badge of honor, but that was the award that he got. He dies in 1987. So when OxyContin gets approved in the US in 1996, he's no longer around. But his two brothers, Raymond and Mortimer, had been with him long enough to understand how to do it. They applied that same ability to OxyContin marketing.
- 5.85** The same error that was made by the FDA, different people, when they said, "Oh, Miltown. Oh, mild tranquilizer." I get it. "Oh, amphetamines,"—CIBA— "maybe we should do over the counter," always underestimating the risk. Purdue went to the FDA, and they said, "We have something new here. We have a pain reliever that's wrapped in this invisible polymer that we have a patent on that we have marketed in England for end-of-life, so that people who are terminal cancer patients can go home to hospice. Now we're going to market it here in the US with oxycodone inside instead of

morphine." And because it's only released every 12 hours, even though their test showed really it lasted about eight hours, but they were able to get approval for 12, that means people will be less likely to abuse it.

- 5.86** The FDA said, "That sounds reasonable." Of course, the person who agrees with that later goes to work for Purdue a few years later, but that seemed to make some sense, even though there were no studies that showed it. And they put out pills at 160 milligrams at one point, which is an amazing amount of oxycodone. Even if you were an addict or you were diverting it or wanted to abuse it, guess what? You take the pill, and you smash it, you take the powder out and take it right away. The 12-hour release meant nothing. It could still be abused as easily as a Percocet or any of the other pills, the instant release, opioids that were then out. So from '96 to 2001, they have this period in which they're just selling, and they are starting to make real money.
- 5.87** Then in 2001, the FDA starts to get some reports about diversion of the drug. That's interesting. The same thing happened with amphetamines. Then the New York Times and other newspapers are running stories because the original marketing had been in Appalachia, the hard, blue-collar working areas. Some towns that had 3,500 people were getting 5 million pills. It didn't seem quite right. There are stories about how this was wrong.
- 5.88** Eventually, over the next few years, the FDA continues to investigate it. In the end, the government brings in action against Purdue for over-marketing because its sales reps were going out. They couldn't give out free pills; they couldn't do that part of the Sackler routine. But they would say things. They got bonuses. The most lavish of all the bonuses in the drug industry went to the OxyContin reps if they were able to resell. They would say to doctors, "By the way, this is less addictive, less damaging. People never developed a dependence."
- 5.89** And then, they would whisper things that the FDA hadn't approved it for. They had tested—they, Purdue, had tested OxyContin—to see if it might help on osteoarthritis. It didn't. They couldn't use it for that. But the reps would say, "By the way, it's not officially approved for osteoarthritis, but we're hearing from so many doctors about how their patients say it's a miracle for osteoarthritis." The doctor would think about that. One of their patients would come in with terrible osteoarthritis, and they would say, "Maybe you want to try some of this OxyContin." The same thing happened with back pain. It's not just supposed to be for regular back pain. It's supposed to be for chronic serious pain after surgery and items like that. Now you've got a back pain. Instead of going to the chiropractor, maybe you can try one of these Oxis.
- 5.90** In 2007, Purdue ends up signing a consent agreement. "We over-marketed, we pushed it too much, we broke all the rules, we're so sorry, we won't do it again." Then, the government looked away, they went back, and they did the same thing. So much so that we built that opioid crisis up to 100,000 dead over the years. I tell you, emotionally, the toll. We dealt with so many of the parents who lost children to

OxyContin. One mother lost a child to OxyContin, who partied with it one night and didn't realize how strong it was and ends up dying. You hear those stories about fentanyl all the time. But this human cost and tragedy—

- 5.91** And what did it do for the Sacklers? Arthur wasn't around to see it. He would have been in heaven. It would have been for him the dream. His idea came true, not through emotional aspirin. But in 2015, the Sackler family entered Forbes: the richest family exists at a \$14 billion net worth. Wow. They were called the Oxy Clan by Forbes. And guess why? They were a one-hit wonder. They didn't have systems like Merck or Pfizer or some company that was putting out instant release, Percocets, and opioid release pills, which were a small part of the sales of the overall company; they had a whole list of other things, from hypertension medications to diabetes medications. Purdue had nothing but one hit, had OxyContin. One single drug had made them a \$14 billion family. That is a sad commentary on the pharmaceutical industry and the legacy of what we're talking about here.
- 5.92** Capturing physician prescribing behavior—aggressive marketing tactics, tailored advertising, sponsored research, shaped how doctors approach it. And with Purdue and Oxy, they created a speakers bureau. The speakers bureau says to influence leaders, some of the best doctors around, "Hey, by the way, why don't you come over? We'll send you out to talk to other people and other doctors." "Where are you a doctor? You're a doctor in Minnesota? Wow, that's amazing because we're having a conference in February in San Juan, Puerto Rico. Maybe you want to get away from the snow and the ice. How would that be?" They always had them at some great location, some resort and whatever else, and these doctors would push it.
- 5.93** They got very lucky because, remember, yesterday we talked briefly about how medical doctors and pharma are definitely allies, often on the same path. We talked a little bit about how Big Food can sometimes be allies. Big Food can be so bad that it can send patients to Big Pharma, but that's another issue. But in this case, in the 1980s, there were a group of doctors who on their own, started to reevaluate opioids because they said, these guys like Dr. Portenoy out of New York and others, they said, "You know what? We think that opioids have been tarred and feathered for too long as being addictive and abusive. Therefore, people have pain, we should treat pain first and not worry about the addiction."
- 5.94** Then, they came into this wild idea. There's a footnote in the Journal of the American Medical Association that said that of 38,000 people that had been examined in a hospital, those who used—they were all given opioids, while they were in the hospital—and only two developed addictive problems. My God, that's minuscule. So, it's not that addictive. Well, of course, first of all, when you publish something in a journal, it's peer-reviewed. But when a letter is published, it's not peer-reviewed. So that got in. It didn't mention that they only measured the time that they were in the hospital. Nobody followed them afterwards. Nobody knew if they were prescribed opioids

afterwards. But that was cited constantly by those doctors and later by Purdue and others as evidence that there was very little risk.

- 5.95** Then there was a maxim that came in that said that if you prescribe opioids for pain, truly for pain, you take it for pain, you won't get addicted. That was brilliant marketing as they thought of that. Some of these doctors believed it. They became the key influence-leaders for Purdue and others. They sold their soul. They went on, essentially, and were the ones who helped Purdue and the other makers of prescription opioids expand the market.
- 5.96** Now, of course, after all of this lethal crisis, if you talk to people in medical schools or people who have just become doctors, it's gone all the way back to what it was in the '70s: opioids are viewed as terrible. They're banned. You can't get a prescription for an opioid unless you're literally dying. Trisha, who had breast cancer surgery four years ago, coming on five years ago, couldn't get a painkiller after the surgery. They gave her some Tylenol. So we've gone back to the point where now everybody is petrified about them, but it takes a long time for that to happen.
- 5.97** A profitable blueprint for addiction—the Sackler family and other industry leaders created this model that's used, and it's used time and time again. The fact that the Sackler family itself, the ones that were called the Sackler Empire in 1960 by investigators for Estes Kefauver investigation, could 60 years later, 65, become billionaires from the idea of the template they had is to me, one of the most infuriating parts of the story of pharma. So the transition of opioids, new drugs, old playbook, it's the same pattern: engineering dependency, this time with more devastating consequences.
- 5.98** I do mean to say: doctors are amplifiers on this, but they know that with opioids, they are dealing with a real condition. They're dealing with pain. And pain is one of the most common ailments reported by people time and time again. So you're talking about a cure for pain that's not addictive, you have a major drug.
- 5.99** Summaries and takeaways—pharma engineered the dependence by targeting women. The tranquilizer and opioid crisis are linked by marketing, not just chemistry. There's no question about that in the addiction crisis. Patients must recognize this history for what I call selling by stereotype because it's part of that. This is the history they have of lucrative markets, creating these markets. It's crucial for patients to understand it.
- 5.100** So, how to be a smarter patient? It makes you want to throw your hands up. Gendered marketing—question. A medication is offered to you for mood, anxiety, sleep, weight, or menopause, be cautious of it. Remember, I'm not a doctor. I don't even play one on TV, as Marcus Welby used to say in another generation. If you're going to go with your doctor's advice, just be questioning them a little bit.
- 5.101** Ask about non-drug alternatives if they are available. For instance, I know that in Germany, the German Medical Association has approved as an authorized use for

doctors to prescribe a standard amount of black cohosh, which is an herb for women who are having hot flashes from menopause. In the United States, it's considered an alternative theory that may or may not work. Black cohosh doesn't help everybody, but no single drug helps everybody, but it might help. If that's a symptom that somebody's having through menopause, they want to try black cohosh, maybe that is an alternative short of actually doing HRT. But again, I'll leave it to you and your doctor.

- 5.102** Beware of drugs marketed as restoring balance, calm, or energy. If we could, in a pill, encapsulate the meditative state of what it would be like to be in nirvana, everyone would be taking it. But we haven't yet been able to do that.
- 5.103** Understand your doctor's prescribing bias, particularly benzos to women. I don't want just the women in here to be particularly suspicious of the doctors that they go to. Remember, the fact that today you go to a woman as your physician, an OB/GYN or whatever else, doesn't mean that she is not also susceptible to the same types of marketing that men are. So it doesn't mean you're automatically going to get away without being hit by pharma.
- 5.104** So we've looked at drug companies and dependency. What's coming next, we'll delve into what I call the industry's pursuit of profits, leading to political capture and the erosion of some public safeguards. And you think to yourself: erosion of public safeguards, how is that possible? What safeguards are in place now? But it gets worse.

Lecture 6: Pills, Power & Political Capture

At a glance — Pills, power, and political capture.

- 6.1 Pills, Power, and Political Capture. By the late 20th century, the pharmaceutical industry was one of the most profitable of all industries in America. That's not surprising, considering the story where they're selling a lot of pills and they're pushing it aggressively, but it doesn't necessarily explain their dominance, why they stayed as dominant as they have over a period of time.
- 6.2 I think when you examine it, they've shaped legislation at times, they've influenced regulators, they've neutralized threats, and they do it very effectively. They've also worked to make sure that reforms, when they do take place, stop just short of really being a major reform. Sort of what they did, same thing. Think of this. Even back in 1906, with the Pure Food and Drug Act, they were able to make sure that that regulation was just—they were one of the voices in their early infancy to say, "Hey, just put the ingredients on the front." Nothing about safety or anything else. That was it. Then, even in 1938, it's just about safety, but don't make it about therapeutic value, to prove we're doing what we say we're doing. And then in '62, we have to prove what we say we're doing, but don't take away our patent protection and our pricing power. So there's always something that they pull out as a win that stops short of really turning the industry around. We're going to look at some of that today.
- 6.3 The postwar seeds of influence—I'll tell you: unlike tobacco and alcohol, it's hard to believe, pharma entered the 1950s and the '60s with public trust and as innovators, both. Today, a lot of people, if you ask a list of industries that you don't trust, the pharmaceutical industry would be pretty high in the ranks—used car salesman, Congress, pharmaceutical industry as a list of people. But that wasn't the case back then. Forget about that.
- 6.4 It's hard to imagine a time in which Americans overwhelmingly had a good impression of the drug industry. Why? Because of penicillin. Penicillin had changed people's lives, and it did seem as though it had made lives better. In addition, not only did that happen, but following penicillin with all the different antibiotics that are out, antibiotics were never that expensive. They made a lot of money for the drug industry because they sold so many prescriptions, but it wasn't as though people were saying, "You're overpricing your drugs."
- 6.5 And then comes the polio vaccine, and they see a change. People's lives are changed. There's a real scientific endeavor taking place. Even with the early parts of Miltown and Valium and Librium, they think that they're treating something we didn't even know we had as an illness. So there's trust in the doctors. There's trust in pharma itself. And that works to pharma's benefit for a long time.

- 6.6** The access to policymakers—I'll tell you: I think in some ways, they have invested in Washington as stakeholders in some ways. They back initiatives. And if you look at pharma's lobbying, it will end up on both sides of the aisle. They do things that the Democrats are pushing as their favorite new policy agenda item, and they'll do things the Republicans want as well. They want to make sure they're trying to keep people happy on both sides of the aisle.
- 6.7** Influencing regulation—the FDA has this enormous power over them. They can recall drugs. They can have safety standards, marketing claims. They can look into all of this. That meant that the drug industry had a single incentive: how do we influence policy before it reduces profits?
- 6.8** PhRMA—you've seen that? Some of you think I have a typo there? I left out an A. No, that's the name of the pharmaceutical lobbying group, believe it or not. It was formed in 1958. Interesting time. They knew they needed a unified group as they were really starting to sell real things. There's this explosive postwar growth, and antibiotics are driving profits. Even though the individual companies are doing their own lobbying, they say, "We should have a lobbying group," and that will be it.
- 6.9** They have a very, very good sales pitch here. What they're doing is they're coming in not as tobacco or alcohol or the arms industry or the auto industry or anything else; they come in as a healing profession. You've got to deal with them somehow different. They're providing cures to save people's lives, extend their lives, and make lives better.
- 6.10** Dependence on patents—that's the underlying thing. Long market exclusivity. And remember to recoup R&D cost and drive shareholder returns. Now, I understand. I said yesterday I was a capitalist, and I get it that public companies have to meet every quarter's expectations of what analysts think profits will be like. So they're on this churn. But again, you are dealing with public health, and I hate that cycle that they are on where they're always looking for bigger and better profits and everything else gets pushed aside.
- 6.11** Vulnerability to regulation—the government has the power. The question is how they will use it. Singular corporate mission from pharma, if you could summarize it in a single word, to me, it's: influence that policy before it influences profits. And the rise of pharma becomes this dominant political donor and lobbying force in Washington. More people are working for it than work for Congress in just a few years. They are applying their efforts on behalf of the industry. They are hiring former Congressional staffers and FDA officials at times for drug companies. They're funding patient advocacy groups—we talked about that yesterday. Groups come out and say, "I want a drug developed because I have a condition or my child does."
- 6.12** They shape debates on pricing, patents, and innovation, and they push narratives that regulation has in other countries stifled innovation, and that's why America is the place where all the great innovation comes out of. They keep pushing that narrative very

effectively over a period of time. By the '90s, they are spending more money on lobbying than any other single group in America. In the 2000s and 2010s, they were replaced by tech. Tech is a broad group, but it's social media, it's Facebook, it's Instagram, it's Twitter, it's TikTok, it's everything else. Tech now spends more than pharma. Maybe one day we'll talk about tech's effects on the adolescent brain and profit-driven motives, but that's for another lecture.

- 6.13** The loopholes and monopolies, the art of extending patents, and there is an art to this. First of all, I mentioned before that Lilly is one example, and I don't mean to say Lilly is the only one. Oh, my God, there are a lot of examples, but Lilly was very good on insulin. They do evergreening. They do these little minor tweaks. They come up with a way, often with insulin, because the product isn't changing, the delivery is better, come up with an easier delivery format. That means that you're able then to extend the patent a little bit because now you have a new use patent. That doesn't make the therapy any different, but it does allow the company to continue to use it. They're very good at extending the patent portfolios around the time.
- 6.14** Clinical trials and outsourcing the global risk—this is quite interesting to me. To the extent that there was regulatory scrutiny here in the United States, some of the pharma companies starting in the '80s, more in the '90s, started to outsource their clinical trials. You have to do clinical trials to get your drugs approved. Maybe it will be cheaper somewhere else. Maybe you can put it together, and then you come back and tell the FDA what happened. I mean, Purdue did all of its results in Puerto Rico. They decided to do their work on OxyContin, they had done. Some go further abroad to do that.
- 6.15** Now, what happens when you're doing clinical trials abroad? You're coming back with the results, but oversight is weaker. Consent rules are often lax. They don't have the same consent rules they have here for participating in some of the trials. Costs are lower. I believe from my own study on this that adverse events were less visible, not because they hid them, but because the standards for the way the clinical trials were sometimes conducted did not meet a level at which those adverse effects would have been easier [to detect].
- 6.16** Patients, more importantly, who were harmed overseas with a drug that turned out to be really bad, never ended up suing in an American court, never became part of a class action. Here, the lawyers are waiting behind a clinical trial to find out if there's some way that they might be able to take a pharmaceutical company to court because they settle a lot of cases. That never happens abroad. I do think that political influence at home enables this, what I call risk exportation as it takes place.
- 6.17** The patient advocacy mirage—they're legitimate, most of the patient advocacy groups are absolutely legitimate. They are really patients who have gotten together who think that they have a condition, illness, a genetic problem that's not being treated. They want pharmaceutical companies and the government to pay attention to it and to put

some funding and research into it. The drug companies saw that those patient advocacy groups existed. They were in their infancy in some ways, in the '50s and '60s, and said, "We can amplify it. We can make it better. We're working on a product that might fit that market. Let's make sure that the government's on top of it."

- 6.18** So by the time you get, for instance, to OxyContin, Purdue has patient advocacy groups on pain. Some of the most prominent ones are asking the government to make the rules easier about using prescription painkillers, and that there's more need for it, but they're being funded by the drug company. So there is a point—and the shame of that is that it tar and feathers the legitimate mom-and-pop patient advocacy groups that have already started, who don't take any money from the pharmaceutical companies and want to do it. It's not easy to always know, but I really detest—there are a lot of things I dislike about pharma, but I detest their use and misuse of the patient advocacy program to push this idea of what they want.
- 6.19** The revolving door—I say the regulators become regulated. So in the '80s and '90s, pharma ramps up. You see the amount of money that they're spending on lobbying gets ramped up to influence Congressional races, the 90s and 2000s, more than any time. It happened beforehand, but it really picks up in terms of this revolving door between industry and FDA leadership. Let me tell you, that, for me, is a loophole. I mentioned this yesterday. It should be closed not just for pharma, but for defense contractors and across the board. You want to do public service? You want to work for the government? Fantastic. You want to go to work for an administration? Great. Getting your paycheck from them, good. You want to leave government service? Okay. But you know what? For five years, whatever it is, seven years—we'll come up with a number—you're not going to be able to go to work for somebody in that industry.
- 6.20** What would Arthur Sackler have done? If he had been banned, if it said, "You cannot leave the FDA to go to work for a drug company," he would have had them go to work for his medical advertising firm or maybe a company that did clinical studies or something like that. I don't mean to say, I'm not naive enough to think that if we pass that rule, that'll stop everybody from leaving government to go in. They're going to go into some subsidiary that doesn't quite meet the definition of the company that's banned, but will make it more difficult. I like the idea of making it more difficult because these FDA reviewers, like in the case of OxyContin, joined the companies after having approved key products. Well, that always looks like a conflict of interest. We were told in law school: avoid the appearance of impropriety. That looks like impropriety to me.
- 6.21** Pharma executives obtained advisory roles inside the agency. I forgot to tell you that. Oh, yeah, sometimes you're a pharma executive, and they would give you an advisory role at the FDA because the FDA would say, "You know all about the drugs, and so therefore, maybe you can help advise us." The advisory panels that would be put up by the FDA to advise on whether a drug should be approved, I'll tell you: there are

advocates that are put on these panels that, at times, you know where the answer is going to be. So recently they had a panel look at HRT and hormone replacement therapies that had a black box warning on them, and all of them were advocates for HRT, and they recommended taking the black box warning off. That wasn't a surprise; I could have predicted that beforehand.

6.22 The regulatory decisions increasingly reflect corporate priorities, and the revolving doors are ensured because regulators view the companies as partners, not adversaries. Now, the FDA had a philosophy from its creation, and through most of the FDA chiefs, that I think is right, maybe is right. They said, "We don't want to be adversarial with the drug industry. We want to not be partners with them, but we want to have a friendly relationship with them." I think that's actually wrong, even as I'm saying it. You want to almost be the regulator and have a somewhat—you don't want to be fighting them on each and every issue; you have to pick your battles—you want a somewhat adversarial relationship. You want the companies to complain about the regulator because the regulator is being strict. You don't want to be liked. Just like, as a parent, you don't want to be liked necessarily. You don't want to be a friend to your children. You want to be strict at times. That's what was missing here. The boundary between oversight and collaboration at times blurred, and I think safety regulations dropped down.

6.23 In the 2000s, not only did the pharma execs, as I say, obtain the roles inside the FDA, regulators viewed companies as partners, not adversaries. In the 2000s, there were instances in which the industry—I'm using that as a broad term, some companies more effective than others—literally wrote sections of the laws that were proposed and issued in Congress. That's not surprising. And so that incremental capture is not one dramatic moment. This did not happen when I can say to you that it's like the release of a drug. "In 1963, everything changed." This is an incremental buildup over time. It's relentless and invisible for the most part.

6.24 A law to promote generic drug competition, 1984. Big moment. The drug industry is bristling at it. They're really worried about it. This is the law that encourages generic drug competition. Congress is responding to somebody that is more important than drug company executives: voters. Voters are complaining by the '80s about the price of drugs. Now it's become a complaint. No longer is the drug industry considered so harmless and always to be saints along the way. They're making a lot of money, and their drugs are really expensive. So as a result, there's public pushback.

6.25 Congress feels it on both sides. They passed this rule: the Waxman-Hatch Act. It made generic manufacturers a force. Literally, before Waxman-Hatch, the year before, generics accounted for 1.5 billion dollars of \$21 billion in sales. A year and a half after, they accounted for \$5 billion of sales. So it hadn't taken away the entire market, but it took away much more of the market, making it easier for generics to be approved and streamlining the process. Unintended consequences—pharma, of course, what does

it do? Looks for loopholes to extend their own drug patents and block generics. That becomes important for them. They start to look for the ways.

- 6.26 Patent evergreening—minor reformulations for drugs goes on steroids. You want to make sure you can keep the patent longer.
- 6.27 Pay-for-delay agreements—I love this one. This is a case in which you've done everything. You've tried to do evergreening to extend your patent. You've tried to be able to hold on to it for a longer time. You can't. And there's some company that's coming out with a generic, and they're going to be competing with you. Instead, you say, "Okay, I give up. How much do you want to delay your generic? You don't put it on the market now: 20 million, 30 million, 100 million, 120 million dollars?" And the company agrees to delay the introduction of that drug for a year or two.
- 6.28 All the pharmaceutical company is doing is making a calculation based upon what it expects its sales and profits to be. If its sales and profits by keeping the patent will be X amount, and if the generic comes out, they have to lower their price, so it drops, and they cut into it by 30% or 40%, they are now saying, "We agree to share some of that profit that we will make with you." You still get to put your generic out; it's just delayed. What a deal. The only people who suffer are the public, of course. They don't get the benefit of the lower price. That's legal. That's done. Nobody steps in.
- 6.29 Aggressive patent portfolios—I'll tell you: the delivery systems, and this happens on a lot of the drugs, ends up getting refined, and they are able, in instances, to keep these monopolies going. And then, of course, they acquire competitors. So some of the big companies buy the small little companies that are specializing in generics and envelop them into the big company, and everybody's happy on that.
- 6.30 How did Arthur Sackler deal with Waxman-Hatch? Brand-name drug promotion—it gained advertising momentum after generics were really introduced, and came in. So in some ways, it helps the business for the drug advertisers because the drug companies who have patented items want to reinforce to doctors the idea that we have quality. There's a variance allowed on a generic drug. It has to be almost as good as what? It might be just the same therapeutic value, but there's a little bit of room given for it. And so doctors say, "You know what? The clinical trials were done on 80 milligrams once a day, and that's where the benefit is. We're the quality manufacturer, so you have to stay with us." Generics are a bad, across-the-board type of idea, and they're advertising more for that.
- 6.31 Skepticism toward generics—Sackler teaches the drug companies, the big ones, Pfizer and others that he's dealing with, to underwrite studies and get friendly articles into Medical Tribune that highlight alleged safety and therapeutic issues, so it casts a public doubt. How many times I remember my parents, bless their souls, would say, "I'm not so sure that generics is as good as the real drug," the real drug, as if the generic wasn't as good. There was a generation that questioned that. Not as driven as heavily on price,

but thinking they were paying more meant a better quality. That wasn't just the fact that they came up with that idea, but that was out in the ether. I know that they had a doctor at the time who very much believed that.

- 6.32** Advertising imbalance—the generic drug manufacturers invest almost nothing in advertising, and branded drugs have the promotional advantage. It's another reason that the promotional empires that are built around the drug business, like the Sacklers and other drug advertisers, hate generics because generics aren't advertising. You're just going in and getting a generic. You never usually see the ads for them. This is one of the periods.
- 6.33** Now, in September of 1985, Arthur Sackler ran an article in Medical Tribune, his magazine. He talked about the psychiatric intensive care unit of the Charlie Norwood Veterans Administration, Athens, Georgia. Here's what the article said. You'll like this: "They had switched"—the providers at that Veterans Administration—"had switched from the brand name tranquilizer, Thorazine,"—still brand name—"to a generic. Eleven previously stabilized patients ran amok and needed increased dosages until the hospital went back to Thorazine a month later. At that point, the patients returned to their previous calm state as if a switch had been flipped."
- 6.34** That's pretty effective. That's amazing. So, you take them off of Thorazine, you put them on the generic, they all go crazy, and then you put them back on Thorazine, and everything is hunky-dory again, except it wasn't quite true. Why are we not surprised at that? It turns out that half of the patients didn't receive Thorazine or a generic that they were talking about. They didn't exhibit those behavioral problems described like that, nothing remotely as dramatic. There's only one person quoted by Sackler in the story that he does about this, and it's the chief psychiatrist at the clinic, this guy, Dr. Richard Borison.
- 6.35** The FDA decided nothing wrong was done. Not even Arthur Sackler could have imagined what Borison went on to. He's eventually charged with his colleagues on 172 count indictment for gaming drugs at the Veterans Administration that were being given out and pocketing \$10 million on it. He also turns out to be the doctor who has conducted a study that says, AstraZeneca's antipsychotic Seroquel is the most effective treatment ever developed for post-traumatic stress syndrome. It turns out the information in it was fake. The study was wrong. AstraZeneca didn't know it. They relied on him, and that eventually blew up. He was convicted, sentenced to 15 years in prison, fined and stripped of his medical license. But he was the only source for the article that Arthur Sackler ran on the dangers of generics. So, if you read that article and you didn't know the follow-up, you might think, "God, if they can't get it right on something like Thorazine, how can they get it right on anything?"
- 6.36** So, case example: Ayerst. They have a high blood pressure medication, a hypertension drug at the time called Inderal. Some of you may have heard of it. Inderal was a big moneymaker for Ayerst. In 1984, 20 generic versions had gotten an FDA approval

within a year. Some were priced at the original cost. They thought, "God, there's so much competition here. We might be able to price it higher than a real generic." But others were undercutting it. At that time, you could be 30% away. Now, it's much less as 10%. At that time, you could have more of a variance on a generic.

- 6.37** Ayerst took that and ran with it. They said, "Hey, we're not talking about an acne cream. We're not talking about something that might take away your headache. We're talking about your hypertension, your blood pressure. If it's too high, you could have a stroke. This is serious stuff. You give up Inderal because you want to save a couple of dollars? That's going to hurt." So they funded these academic articles that highlighted the risk of generics without disclosing—of course, they didn't disclose it—company sponsorship. Doctors went ahead and took the check from Ayerst and then wrote the articles.
- 6.38** They minimized the cost savings. They always took the generic company with the highest cost and said, "You're not saving that much anyway," even though there were other generic companies that were offering pills that would be far less expensive. And they cautioned pharmacists— I love this big bully routine about legal risk. "If you dispense a generic to Inderal to one of the patients that comes in because it was written for that, if you don't tell them that, maybe they should talk to their doctor and think about taking the name brand, gee, if it turns out that the generic is not good enough, anything happens to them, I wonder, you know the way lawyers are in America? They're terrible. Lawyers sue everybody up and down the list. They're going to probably name you as the pharmacist as well." Pharmacists would think about that and wonder if there was any downside.
- 6.39** Doctor incentives for branding—this is great. Free airline tickets. You prescribe their product, you can fly off somewhere. They had only two years left on the patent, and they were intent on making sure that they got as much of that as they could have.
- 6.40** One of the things that comes in to help the pharmaceutical company, and we talked about this the other day if you have a drug that's approved for a given reason, you have a patent on it, and you might have some generic competition coming up down the road, there's always a great way to be able to cash in. It's widespread. This first line here to me is—I say I'm not surprised by anything, but then I am surprised by things time and again. One of the things that surprises me is that nearly half of the prescriptions in the US, today we're talking about, not 1950s, 1960s, currently, are off-label. That's remarkable. It's an enormous market. We're talking hundreds of millions of pills, billions of pills around the world. Half of those are for indications for which the drugs weren't approved. We're not just talking an occasional time.
- 6.41** There is no doubt that if some very clever researcher doctor at Johns Hopkins thinks that this drug that's been used for hypertension might have an effect and help somebody on a lethal disease, and you can make a breakthrough like happened, it would be fantastic. But in other cases, you're using Lupron, as we talked about

yesterday, AbbVie's big hit drug as a puberty blocker, not quite life-saving. The companies encourage off-label use. I'm going to talk about a few of these in just a bit more detail.

- 6.42** The FDA had approved Upjohn's Loniten; that's what the brand name was. It's minoxidil is the underlying name in 1979 to control high blood pressure. I love some of these discovery stories of the off-label use. At the University of Colorado, a first-year resident asked the chief of dermatology to check on a patient in her 40s who had a skin reaction and had complained about a skin reaction. When this first-year resident went to check on the patient who had the skin reaction and had been using Loniten, she had hair growth on her legs, arms, and along the hairline. Now, people take minoxidil; some people respond to it with a considerable amount of hair growth, and some not at all. It's the nature of how the physiology works. This is obviously a woman who responded to it. He said to her in jest, "It would really be something we could figure out a way to put that on the top of our head." Little did he know that he was making a joke about what would become a thing.
- 6.43** The word started to spread among doctors in the early '80s. It was, remember, we said before that there was a time when Thorazine was put out, 80% of the prescriptions were written by psychiatrists. Well, who do you think is writing most of the prescriptions off-label for minoxidil in the beginning? It's dermatologists. They turn out to be the area. Dermatologists have a new thing that might have hair growth help. It's fantastic. Dermatologists wrote, on average, more than two times the number of prescriptions that cardiologists wrote for high blood pressure. That's a significant use.
- 6.44** The FDA never moves, never finds out. People who have normal blood pressure, even with the new restricted numbers of 120 over 70, are now taking minoxidil in a pill form, a prescription form, because they're hoping that it might grow hair. Well, what does that do to blood pressure? Does it make it too low? Does it have an effect? No one knows because no one has done a study on it. It was not long before, in the mid '80s, before they went ahead and got it approved, that you had hair restoration clinics popping up in the US with compounded forms of minoxidil, where you would be able to go and get it done.
- 6.45** When Upjohn realized it had a problem—here's the problem they had. The problem was manufacturing enough minoxidil to meet the demand. They had to go to foreign suppliers and ask them to create in their own manufacturing facilities enough minoxidil to send back here. When did that happen? In 1979, it got approved. It's found out in the early '80s. Finally, Upjohn says, "We'd better go to the FDA and get it approved." So, in 1989, they do that. For nine years, they get a free run on off-label prescribing. They make twice as much money as they would if this was just hypertension. They finally get it approved, reformulated as a prescription liquid as well, and they call it Rogaine. Better than Miltown as a name, but Rogaine becomes part of the way we view drugs.

- 6.46** A year after Rogaine hit the market, a British Pfizer scientist discovered Sildenafil. Revatio was the brand name. Nobody really knows that by the brand name today. It's a condition also for—it treats high blood pressure, but more of an effect of blood pressure in the lungs that causes chest pain and shortness of breath. Then, the same type of thing: a nurse later recalls she went to check on a patient's blood pressure in a clinic, and the men were laying on their stomachs, and she wasn't sure why. It turned out, she found out that they had erections and reported that as a side effect that might be something to do with the drug. That was thought to be a real problem. Even at Pfizer, they said, "Oh, that's an uncomfortable side effect. We don't know what to do about that." Arthur Sackler was not advising them then; he would have said, "You're onto a multi-billion dollar drug here. My God, you just aren't thinking of it that way."
- 6.47** There was even, I know this is hard to believe, but it's on the record— one of the things that made it go viral in a pre-social media age—oh my God, imagine if there was Twitter and Facebook and TikTok and everything now, it would be even crazier—but then, a rumor went around that there was this drug that did that and that it actually made penises grow. Well, that made men crazy. God, they were probably lining up for that. That, of course, was false, but it was one of the factors that led to this frenzy.
- 6.48** So what happens? The same type of thing that happened before on Loniten. Pfizer could not ignore the off-label success of the drug, and it took off. 1989 is when that drug was first approved. In the early '90s, it's become a rage given by doctors to men because it's one of the most filled prescriptions. It's not until '98 that Pfizer goes before the FDA to say, "Let's get it approved now to find out if there might be any downsides or anything else, " which there are, risks all the way along, but they get it approved, and they mint the condition erectile dysfunction. That is how they have it; off-label use leads to it.
- 6.49** Botox, I'm going to talk to you about in a bit more detail because Allergan is so wonderfully diabolical. I don't mean diabolical, illegally diabolical, but clever in terms of how they were able to master that.
- 6.50** Lyrica is the last one, just for a second. Lyrica is a drug— Pfizer has a cousin to it in gabapentin, I'm going to talk about that as well. But Pfizer has a drug, Lyrica, and it's really meant to treat epilepsy and severe anxiety. But there were some doctors that said, "We think that people who were taking it for severe anxiety and had nerve pain, they seem to have less nerve pain. So maybe Lyrica helps nerve pain." Nerve pain is one of the hardest types of pain to treat. People who have nerve pain know that. You can't just take a couple of Advil, and it's going to go away. It's really ingrained, chronic pain.
- 6.51** The idea that Lyrica could help nerve pain made it widely used. I will tell you that Lyrica keeps getting expanded over a period of time for additional uses, but its main thing is, doctors would tell you, if you went into an office at Pfizer's encouragement, "There's this wonderful side effect of Lyrica: it's going to take away your pain." Millions of prescriptions are sold.

- 6.52** It took until 2019 for a major report to come out in which a doctor and researchers have—Christopher Goodman studied Lyrica, did his own study. FDA is not doing it, Pfizer is not pushing forward to see: is this true? If it's true, it's fantastic. Maybe something should be done. They give a placebo to some patients and others. Here's the result: "patients and physicians should understand the drugs have limited evidence to support their use for many conditions, including for pain. There can be some harmful side effects like somnolence, dizziness, and difficulty walking." That, you think, is the end of it. People aren't going to keep prescribing it as doctors, right? That comes out.
- 6.53** Since 2011, off-label dispensing of Lyrica provided half of all of its annual revenues and continued to do so. During an eight-year period, Lyrica was at or near the top of the 10 best-selling drugs worldwide. Lyrica alone brought in 37.5 billion dollars to Pfizer. Remember when we said that Valium reached \$100 million a year in sale? That was big news. Those days are long gone. A \$100 million for a drug sale would be considered a failure—37.5 billion, half of it off-label. So this becomes a way in which the companies who have a patent are still looking at other ways to be able to use it.
- 6.54** Medicare prescription drug plan—Congress gets together and says, "You know what? Under Medicare, you pay for drugs out of pocket. They're expensive. Let's put in a prescription drug plan." Is it perfect? No, but it's pretty good.
- 6.55** This is key: the noninterference clause. The pharmaceutical companies said, "Okay, you're going to put in a drug plan. Can we get something in called the noninterference clause?" That is really fantastic. It says essentially that Medicare, the government, cannot negotiate drug prices with pharmaceutical companies because Medicare is really set up to take care of health concerns for those over 65. So drug companies say, "They know nothing about the pharmaceutical industry. Those bureaucrats up in Washington have no idea about the cutting work we're doing in research and development for cures to save people. So if they negotiate prices with us, they're going to cut into our profits, which will cut research and development, which means less innovative drugs, more people dead." Congress says, "Okay, that sounds reasonable. So we'll make sure that Medicare doesn't."
- 6.56** We know that's changed. From the first Trump administration, the Biden administration, finally, they're allowing Medicare to negotiate a group of 10 drugs, the most expensive drugs, based upon prices in the rest of the world, and it is bringing down the cost of some of those drugs. But it has been by tooth and nail to even have 10 drugs negotiated, much less than that. And that's the non-interference clause that gets put there in 2003. That's the result of pharma's lobbying efforts. There's no question, the federal government doing that.
- 6.57** That single clause guaranteed massive income streams. It protected the high cost of drugs to consumers, and it cemented pharma's pricing power. That's really the key. Without government intervention, they continue to dominate. And it shifted the cost to taxpayers, so the prohibition on government price negotiations really works out to be a

detriment for the government who is now stepping in to pick up some of these costs. Same as on Medicaid, where the government is covering the cost of the medications. If pharma continues to have the medication cost high, they continue to pocket money.

- 6.58** Now, the ACA comes in under the Obama administration and expands drug coverage, which is good. It expanded Medicaid eligibility by raising the income level to 133% of the poverty line, and that translated into another 10 million people eligible for Medicaid. All developments in which, if you're looking at a social safety net, it's fantastic. What we don't realize is that the ACA made prescription drug coverage a mandatory benefit of health insurance plans. So once you went on to the marketplace to look for a health insurance plan, those companies that were providing insurance had to include a prescription plan. It wasn't always great, had high copays, maxed out a certain rate, but they had to have it for the first time. So the ACA makes that happen.
- 6.59** Medicaid increases in terms of its size, and has 10 million new people; there's a wrinkle on that. The ACA also improves, of course, coverage for 2 million children in the Children's Health Insurance Program for necessary medicines. The ACA reduced the Medicare "donut hole," as it's called. Trust me, you could have an hour lecture on the donut hole, and you would walk out dazed and confused because the donut hole is one of those things of how you create—it's a period in which you have Medicare, and you have to pay for it after not paying for it. Then, after that hole is covered with more payments, they go back to paying for it. Low-income subsidy provides coverage for 13 million individuals who misqualify for Medicaid. So we increase the Medicaid rolls: 10 million new people on. And then we have this buffer zone in which millions more who are almost at Medicaid but not quite qualifying get this benefit.
- 6.60** Here is the kicker. Why is all this important? Why am I talking about expansion of people on a Medicaid and everything else? Look at the last point: "The federal government is legally required in its drug benefit"—in its own legal language—"to cover every FDA-approved drug, whether a prescription is written for an inexpensive, generic, or a really expensive biologic." If you're covered under Medicaid and you have a condition and a doctor prescribes a pill that is \$500,000, \$600,000, I'm going to talk about some of those in a minute, there's no option; it has to be covered. It's not like private insurance where it may not be covered.
- 6.61** Now, essentially, the drug companies realize, "Hey, the poorest people, those most at risk, the people on the government support program, we have to really target doctors who are in those clinics helping people on Medicaid. We have to tell those doctors, 'You are supercharged to be able to prescribe the best biologic and the best and the latest drugs because your patient is not going to have to worry about paying for it.'" It's a focus at which immediately the drug company thinks: how can we maximize our profit from it?
- 6.62** ACA's impact on drug spending—prescription drug spending after the ACA with Medicaid expenses rose 25% in the first two years to \$40 billion. In 2024, last year, I put

it together: \$100 billion from zero. So ACA, I understand exactly why it's trying to do what it did. It's a good overall social net, but I guarantee you that nobody writing in, including probably President Obama, thought they'd be spending \$100 billion a year on drugs. So that's going into drug companies, but patients aren't paying for it. We all are paying for it. It's taxpayers paying for it.

- 6.63** Brand name dominance—by 2015, now think of this: the change comes in 2003. Then, the ACA makes the change. By 2015, brand-name drugs make up 90% of Medicaid's drug spending. How is that possible? Generics are on the market. They're competing. But inside of Medicaid, 90% of the prescriptions are for name brands only. Why? Because they know somebody's going to pay for it because that's what the government has to do.
- 6.64** Medicare's growing role—Medicare's prescription costs share, so the Part D costs the government, jumped to 30% after the ACA, making it the second largest payer nationally. So now you have Medicare paying more for drugs than just about anybody other than civilians.
- 6.65** The federal government's substantial contribution—when Medicaid and other drug benefits are taken into effect, the federal government covers 45% of retail prescription costs in the US each year. Remember all that talk we used to have at one point about socialized medicine? "We don't want to go there. Don't do that." That's how they were fighting the Kefauver and everything else. This is what I call the unintended consequence of having a social program, a health program, in which you decide that you're appropriately going to cover people who can't otherwise afford it, the lowest end of the totem pole financially, for their drug costs in a system in which drug prices are high. But you don't put any limits on the drug companies for what they charge. That's madness. That's absolutely crazy. "We're going to pay for it now as the government. Go ahead and charge whatever you want."
- 6.66** Then, finally, how does it take until the last two presidential administrations to be able to say, "We're going to at least have 10 of the most expensive drugs. We're going to try to work those prices down." And pharma fought them tooth and nail on every part of that. They should be negotiating all of the drugs across the line for lower prices, and they aren't.
- 6.67** They continue, by the way, I don't want you to think they've stopped targeting women. They don't give it up that easily. So how are women depicted in psychotropic drugs? This is current material shown in a professional role: one out of five times is somebody shown in a professional role, shown in a family role on drug ads that run, including those direct-to-consumer ads. Ninety percent of the time, still in a family role. In a leisure role: 77% of the time, or sleeping: 77% time. So in a professional role, less likely. They continue to have those medical problems and target them. It's a reliable tool for pharma. So gender targeting political influence is quiet, but what I call powerful feedback. The industry frames women's everyday experiences, we know that, as

medical defects. It's built up this massive market around weight, stress, menopause, et cetera, and continue to bring in huge profits.

- 6.68** How reform was neutered? Mastery of marketing—there's no question that marketing continues to be key. Plus, in 1997, they get the wonder of direct-to-consumer ads. Mastery of lobbying—they're excellent at lobbying. Reform efforts consistently fell short, and meaningful change keeps getting blocked time and time again.
- 6.69** A case study, a quick case study: the rise and risk of gabapentin. Some of you may have seen recently that there was a story in the Wall Street Journal, an investigative story that talked about this very issue. Others have written about it. Now, gabapentin is this Pfizer cousin to Lyrica, the same type of drug. It's merging concerns about its effect and how it's been applied. So gabapentin was first approved for seizures and nerve pain from shingles. The word pain appears, so you know they're looking at that. And it does help if you have an outbreak of shingles, which are extremely painful—never had it myself, but I know people who had. So, decades ago: seizures, nerve pain associated with shingles.
- 6.70** Now, today, as I'm talking to you, it's the seventh most widely prescribed drug in America. That's pretty amazing. You mean that many people have seizures and nerve pain from shingles? I don't think so. You look for what the drug is approved for, and then you see its sales, and you say, "Okay, something's happening here." You have to find out what's happening. But over time, it's a dramatic increase in how it's being prescribed. Its journey, as I say it, from specialized drug to one of the most widely prescribed, is a perfect example of what happens here in the drug industry.
- 6.71** So first, to prove seizures and nerve endings. Then, over time, it has a wider spread acceptance and prescription growth. Research reveals that gabapentin is associated with a greater risk and side effects. There's a list of these studies that come up, but nobody's paying attention to them. It's a similar pattern as we've seen with the pill, HRT, and opioid painkillers.
- 6.72** I'm going to give you a stare. 2010 prescriptions: 24 million prescriptions; that's what it had in 2010. You think to yourself, "There was a point at which I was telling you that Miltown had 36 million prescriptions at one time in a year, so 24 million, not that big." In 2015, it keeps moving up off-label now, off-label for fibromyalgia, migraine, more generalized pain—if it helps for nerve pain for shingles, maybe it helps for nerve pain for others. 2020 prescriptions, 2024: 60 million. It goes from 79 per 1,000 people, we're getting it to 178. It's doubled essentially in terms of its dosage and its use.
- 6.73** At the same time, it is marketed as an absolutely safe drug. There's no side effect. The side effect profile was really small for gabapentin in the beginning. But research shows that it has a list of—possibly contributing to dementia, certainly having some effect on cognitive behavior. There are some instances in which it increases suicidal ideation or may contribute to that. For people who have a lung disease, didn't check on it before, it

causes additional breathing problems and can, in fact, be fatal. Edema is a side effect that people don't like but can be bad for heart.

- 6.74** Now, drug is tough. It has 12 separate pathways in the body, so it's a complex drug. And you know what? Pharma companies love complex drugs because they get to say, "Oh, that's a side The product must have come from the interaction of the 11th pathway with something else they were taking. They were taking cranberry juice, and they should never take cranberry juice with that 11th pathway." And they try to explain it away some other ways. Varied effects in individuals. No question about this. So it acts differently in different people. Some people get benefits from it; some people get harm from it. That's the nature of this particular drug.
- 6.75** For conditions like neuropathic pain, that type of nerve pain, gabapentin, I could line up a group of patients here who will tell you that it helped change their life for the better. They had crippling neuropathic pain beforehand. They went on gabapentin, and it helped them. Other people, there's a range of serious adverse effects that they learn about, those people have to stop taking it even if the return is to pain. The most important part is that the results are unpredictable, and doctors are prescribing it often at a high rate without worrying about how it affects that particular patient.
- 6.76** It's hard to imagine that this pill, developed not as an opioid painkiller, still results in withdrawal syndromes. So if you've been on gabapentin for a while, not all patients, but some have really debilitating withdrawal symptoms. Doctors are examining still whether it's a physiological dependence or just a psychological dependence, but there's no question that people have trouble giving up gabapentin at times. The medical establishment is 100% behind Pfizer and saying, "Those patient reports are the anomalies. Gabapentin is not habit-forming."
- 6.77** Why isn't the FDA on top of this? Why isn't the FDA requiring gabapentin to have clinical trials for the alternative uses which made it the best, seventh-selling, best-selling drug in America? With all of these questions and reports of adverse effects, why isn't gabapentin being put on that microscope of clinical trials? That's the real question. Why isn't the FDA enforcing a stricter rule on its prescribing for pain that results from shingles and on seizures? Because there's no rule that stops the off-label prescribing. Before, we were talking about off-label prescribing for Rogaine, the side effect is maybe growing more hair. Minoxidil, maybe somebody's sex life is enhanced or that. We're not talking about the potentials here with gabapentin and other drugs that are prescribed off-label.
- 6.78** Here's the combination with opioids. This is the rate of opioid deaths involving gabapentin and opioids. Now, remember what we said, we do the autopsy, the toxicology report afterwards, and there's benzos involved and alcohol, and somebody has smoked some weed, and they've taken an Oxy, and they're dead. Now, we're finding in the new reports, guess what, 2018? Not that much. It increased 22% from 2018. It's only 5,400. I understand that. And I love that because that's what the industry

says: "It's only 5,400 cases. A hundred thousand people a year are dying of lethal prescription opioids and fentanyl, and this is less than 5% of it. And we don't know if gabapentin is to blame. Maybe they were taking gabapentin to offset their pain, and they were taking opioids just to party, and that combination turned out to be bad because they took too much of the opium."

- 6.79** It's the same thing that Purdue and OxyContin did when the government first called them in and went through the reports of overdoses that led to deaths from different medical examiners around the country. Purdue—and I have the slide show that Purdue presented to the government. It's so rudimentary. You wonder how it ever could convince anybody. It just showed all the mix of drugs that were in all of these different deaths, saying it can't just be OxyContin that's killing them. It's got to be the combination. It's their fault. Always blame the victim if you can, and that turns out to be very good.
- 6.80** So we're seeing this increase year by year, in which doctors and medical examiners are finding that in overdose deaths, gabapentin is there. It's not yet larger. Will it go up again when the next group of tests come in in 2024 and 2025? I don't know, but it should be an alert to the FDA. It hasn't been yet.
- 6.81** How to be a smarter patient. Besides not mixing your drugs on your own, if you have four prescriptions from a doctor, ask them if there's any interaction between them before you take them. Don't assume high drug prices reflect innovation. I tell you: it's not like buying a television or a car. I don't even know if high prices mean that a television is better than another TV. It just has some development that they think is the flavor of the month to be able to try to sell it. But don't assume here, just because the price is high, that it necessarily means better.
- 6.82** Be skeptical of claims that regulation follows progress. If you get into a discussion with a friend of yours and they tell you once, "I think the drug industry is overregulated," you should say to them, "So what cures would we have today if we didn't have an FDA at all? Would there be something? They've cured cancer, no heart disease, no more strokes? Would dementia be a thing of the past? You mean we've held them back from doing all of those things because we regulate them?" I don't think so. I know that regulation can be stifling at times and increase cost, but it's not the reason for the lack of innovation.
- 6.83** Know that patient advocacy campaigns may be industry-funded, and that's why—it's not difficult to find this out, but it's work, and so I don't expect everybody to do it. I'm always taking these patient advocacy groups that are often set up as 501(C)(3), they're nonprofits, your contribution to them may sometimes be tax deductible. Then, I'm looking through the 990 reports, which are the filings that they have to make with the government to show the sources of income that they have. Then, I'm going through the names of the people, not the individuals with a \$100 contribution, but the large contributions at the top. It doesn't say Pfizer, Merck, or anything like that. It's got some

name for consumers hoping for health reform or that. Then, you've got to go into the corporate record check and find out who it is.

- 6.84 Sometimes the patient advocacy group is clean, which is fantastic. It's grassroots, and it's the real McCoy. Then sometimes it is pharma-funded. That does not mean that the issue they're talking about isn't real. It just means that it's been wrapped in a pharma envelope, and I'm always worried that that means that something's going to get approved at a very high price.
- 6.85 Recognize the limit of the FDA's power—I'll tell you: people assume, not people here today, because you'll be savvy enough not to make that assumption, but many people assume that if a drug is FDA-approved, that means it's safe, good to go. "Oh, FDA approval, I'll buy that." So be careful. It doesn't always mean that.
- 6.86 Understand that the system is not designed around patients. I'll tell you: how is that possible that we have a healthcare system not designed around patients, but we do in terms of at least the drug industry. This strategy that they've used for decades just means to me that you can make informed decisions. You just have to always be cognizant of the fact that it's a product. It's a product that may help you. It's a product that, at times, you will need or help heal you if you're really, really sick. But it is a product, and you have to have that sense of skepticism as you go forward. That's the end of regulatory capture for this hour.

Lecture 7: Price of Monopoly & Orphan Drugs

At a glance — The price of monopoly and orphan drugs.

- 7.1 One thing, right before I get into the price of monopoly and orphan drugs, PBMs, and the billion-dollar drugs—we were talking before about the side effects of drugs, and every drug has them. Pharmaceutical companies know that. They expect that.
- 7.2 One of the amazing things is that in the '60s, I came across a company that was viewing a drug that had so many problems as a gateway drug, meaning not that you would get addicted to that drug, but that it was a drug that would cause, they understood, so many side effect problems in treating the condition that they might be able to develop drugs to treat those side effects. So that happens. What was the one drug that Purdue Pharma had on the books that they were racing to try to get so that they weren't just a one-hit wonder was a drug that would overcome the constipation from opioid use. So they get back the feedback on what the problems are, and then it's in their wheelhouse. They say, "Why shouldn't we be making money from the side effect as well?"
- 7.3 You think sometimes the drug companies are looking at and thinking, "Too bad those side effects exist." Occasionally, they say, "The side effects, maybe we can actually profit from that as well." Now, this idea of what I call—this is new models for profitability, in some ways. The drug industry, as we've come to know it, they really are targeting large populations with common illnesses, and they're trying to have drugs treat those illnesses, even if they're illnesses or medicalized conditions that they have created.
- 7.4 Now, as we get into a later period to today, the industry has realized that it's possible to turn that on its head and to make as much or more money from treating a much tinier, and I would say tiny, slice of the patient population. So, instead of having a hypertension medication for the tens of millions of people who might have high blood pressure or a medication for diabetics, you're now targeting, in some cases, 500 people, 400 people, a much smaller number at a price that can be extraordinary. Once they realize that they might be able to make as much money from treating small patient populations, and it's based upon a law that we're going to examine, you'll see how this takes place.
- 7.5 The inversion of the market: from mass market to micro markets. The patient populations for rare diseases with a small number of patients become a gold mine over a period of time after the passage of an act. We'll talk about that in a second. The profit here comes from scarcity, not volume. So they're not selling 20 million prescriptions at \$40 each. They're selling just a handful at an extraordinarily high price. The highest price they can charge, the profits they can still generate. And they exploit incentives time and time again for rare diseases.

- 7.6 This inversion of the market doesn't happen just by accident. It's a discovery that they come across, what I call, from a well-intentioned legislative act that has unintended consequences. That's all from nature. Things are done meaning well, and then it's the consequences that come afterwards that make the difference. It really comes from this realization that they are able to use the Orphan Drug Act.
- 7.7 So let me tell you what the Orphan Drug Act is. It's passed in 1983. It's a noble concept. It's a federal law that provides incentives to pharmaceutical companies to develop and market drugs for rare diseases and conditions that affect—and they weren't sure what the number should be. When they were drafting this law and coming up with it, they said, "So what's a small patient population?" And they initially settled on 100,000. They thought 100,000 would be the number. And then, they expanded that to 200,000. There was a big debate about that. So, 200,000 or less would be the patient population.
- 7.8 The act was passed in 1983. The reason it was passed was because, of course, there had been a lack of interest by drug companies to focus on these small patient populations who had hereditary and genetic diseases. It makes perfect sense. If you're Pfizer or Merck or someone else, you're going to go for the largest possible market you can. Why would you spend your time developing a drug for a smaller group of patients? They're not thinking yet of what you could charge for that, but there's no incentive to do that.
- 7.9 So what the government decided to do, government decided it would come in and create an incentive for the companies to do that. So noble intentions, but with loopholes. In '83, they passed the law to encourage treatments for rare diseases. Fantastic. Some of the incentives—and I'll go over a list here with you: seven years of market exclusivity, major tax credits, FDA fee waivers, and research subsidies. I'll give you the details on that in a second.
- 7.10 From the 1980s to 2000s, from the time this was passed, orphan drugs become the fastest-growing sector in the pharmaceutical industry, priced between \$300,000 and \$3 million a year. It's remarkable because we think of a pill that we might go and fill a prescription for an antibiotic, and I'm upset if the copay is \$25 or something, and I'm complaining the entire way home. I'm not thinking \$300,000 to \$3 million for some of these drugs.
- 7.11 Exploitations of loopholes—I'll tell you, indication stacking is a process where they put what they think are effects, so they can expand the market beyond just that small little market that it was first approved for, diseases into smaller subtypes sometimes, so they can extend the length of a patent drug. You get it approved for a specific condition, then later you cut it out maybe for a pediatric version of that, and it allows you to have the drug again. And drug repurposing—changing the rules so they can extend it.
- 7.12 Here's what happens in terms of the Orphan Drug Act. Here are the inducements. I want to have you understand these, as I call them, eight, because they're interesting.

You're saying to drug companies, "We want you to create drugs for this small patient population with genetic diseases," for the most part, because those are the ones being affected by it, in most cases, not all. "So one of the first things we're going to do is we're going to give you taxpayer grants to underwrite early research up to half a million dollars a year." That's pretty good. So you can get a half-million-dollar grant to underwrite the research. So the initial money you're spending isn't necessarily pharmaceutical money; it's taxpayer money.

- 7.13** Then you get a 50% tax credit on whatever you spend on R&D, which is much higher than the tax credit you normally get from R&D. The remaining 50% of the costs that you don't get as a tax credit, you get as a deductible business expense as you do it in that year. That meant about 70% of the R&D cost for these drugs were essentially paid for by the federal government. That helps bring in the drug companies. The law also simplified applying for academic and NIH co-sponsorships. It waived the processing fees before the FDA— everybody loves a deal, even pharma companies love a deal, not having to pay the processing fees.
- 7.14** It expedited the approval when the drug went to the FDA. You have to pay for expedited approval. We talked about that yesterday. Now, you don't have to pay for it. They'll put it on the fastest track possible. And the FDA provided grants on a competitive basis for clinical trials that it deemed important. The gauntlet of three or more rigorous trials, dramatically abbreviated.
- 7.15** All of that is inside the industry. What they liked the most was the seven years of exclusive selling. Now, you think to yourself: didn't I tell you that it was 17 years on a patent, and sometimes it had been extended to 20? So, why would they get so excited about a seven-year exclusive selling, what they called—I love this—a platinum mini-me patent? That's what it was called.
- 7.16** The reason it was considered such a good patent is because in the real world, the non-orphan drug world, if you come out with tetracycline or you come out with streptomycin, or your own patented antibiotic, another company can do a me-too drug and come out with Terramycin or Aureomycin and compete with you. This seven-year period meant nobody else could come up with a drug to treat your condition. So even if they had innovation inside the lab, once you get the approval, every other drug company stops looking at that particular condition. You've got seven years on your own without even remotely having anybody else come in to compete with it. So that's why it was considered such a golden thing.
- 7.17** Now, there's absolutely no question that the creators of the bill who passed this bill, and when Congress passed it, they expected it to be used as it was designed to help these small patient populations. And one of the very first orphan drugs proved what could be done. It set what I call the template for the future, an AIDS drug that came in as AZT. So in the early days of AIDS, there were all types of misconceptions. Nobody really understands how the virus is passed. They're worried about it. There was

tremendous misinformation. The disease really was eventually thought of as only a disease that affected the gay community, primarily the male gay community.

- 7.18** At that time, the government is looking for some type of medication that might be able to stop this disease, which in many cases was proving to be lethal. Samuel Broder was head of an NIH team, and he did work—when I say these NIH scientists, he's got this communicable disease, it's affecting primarily, it's also affecting, of course, people who are receiving plasma and blood products. So you have hemophiliacs who are suddenly coming down with HIV and then developing a full-blown AIDS. So it's an additional problem across the board. It's not just in the gay community.
- 7.19** Samuel Broder is running an NIH team, looking to find a drug that might, in the laboratory, slow up the replication of the AIDS virus, which they've isolated. His team of researchers is working on this, and they think, at one point, that maybe they could use a drug that would slow the process up. They focus on retroviral drugs. One of the retroviral drugs that they focus on is AZT. It turns out that AZT was an unused drug that was sitting at Burroughs. Burroughs was a drug company. They had the patent on AZT, but they weren't using it. They had made it.
- 7.20** Broder goes to the scientist at Burroughs, Burroughs Wellcome at the point, and discusses whether AZT might be a possible treatment. Burroughs said, "All right, go ahead, knock yourself out, and go ahead and test it." When he decides to do that, he can't get a sample of AZT from Burroughs because they have not used it for so long that they weren't able to reconstitute it in their lab correctly. They had to do that at the NIH. Burroughs is not only not using this drug, it's on their shelf, they still have a patent on it, but it's just sitting there as an antiviral they've never been able to use for any other type of drug.
- 7.21** So, Broder and his team began the first human clinical trial test in July of 1985. Approval of an antiviral, because antivirals are very complicated and they have long-term bad side effects sometimes, can take eight years, nine years before the FDA sometimes. There was a tremendous pressure, particularly from the gay community, to get this done. They got this approved in 20 months, which 20 months might seem like a long time when people are dying. I understand that. Activists inside the gay community were absolutely furious that it was taking so long, but 20 months is a speed record in terms of the FDA and the antiviral.
- 7.22** The FDA gets it approved in 20 months, the fastest ever for an antiviral. They then say to Burroughs, "All right, so now we're going to distribute this drug, and we'll get it out as soon as we can to everybody." It turns out that Burroughs says, "Thank you very much, but we have the patent to that, and we're going to price it."
- 7.23** Burroughs Wellcome, the pricing of AZT—NIH spent \$50 million in developing this drug. That's what they spent, that otherwise would have been spent by the drug company; \$10,000 a patient is what Burroughs Wellcome set the price at. So there was an outcry

from AIDS activists who marched in front of the headquarters of Burroughs Wellcome and said, "You're killing people with this price because so many people are uninsured." And Burroughs Wellcome said, "Okay, you're right. After the public backlash, we'll lower the price to \$8,000." Still rather unbelievable. The highest price at that point for a drug, for a treatment, was the Burroughs Wellcome. Percentage of AIDS patients without insurance: 35%.

7.24 Now, what happens? A copy of AZT was made by a generic drug company, and they said, "We're going to provide this at a couple of hundred dollars." What does Burroughs Wellcome do? Takes them to court. They say, "You're not allowed to do that. It's under the Orphan Drug Act. There can be no competition whatsoever." They go to court, they go to federal court, and the federal court sides with Burroughs Wellcome and says, "No, you can't put out a generic." Pretty amazing in the middle of what you could consider a lethal epidemic.

7.25 That litigation ends up going to the Supreme Court because now the government is trying to get its hands on the patent, saying, "We did all the work. You guys provided the drug, but you didn't even really provide the drug because we had it created here at the NIH, because you didn't even know how to make the drug anymore. You just technically held the patent." And in their legal papers— I always love this line—the lawyers for Burroughs Wellcome described the NIH as only, quote, "a pair of hands that helped the company, but the product belonged to the company." The court agreed and kept it there.

7.26 Now, it was the most successful launch ever, straight out of the box for Wellcome, the British-based Wellcome. Remember, by the way, when we were talking a few lectures ago about Silas Burroughs, the American traveling salesman, and Henry Wellcome, who was a British, and they started Wellcome & Burroughs back in England in 1880? This is the stepchild of it all the way down the road. So this is Wellcome. And they had only launched the parent company, British-based Wellcome, a year before AZT came out, just gone public. Their stock, after they announced the \$10,000 stock, and then went down to \$8,000, went up 500% that year. Investors liked the idea of a drug that was going to be able to treat the AIDS epidemic as an exclusive drug in the beginning and be priced at that amount. In terms of business, big success. In terms of the Orphan Drug Act, not so great. The impact of the pricing.

7.27 The percentage of AIDS patients without insurance: 35%. The average annual cost for a patient: \$10,000. The government spent about \$300 million a year on subsidizing the uninsured. And as the price went down to \$8,000, you still had a percentage of AIDS patients without insurance. It was still costing the government \$400 million. You'll see here, even as the number of AIDS patients go down, the price changes over time. They raise the price of AZT. The government is spending serious money on subsidizing this drug.

- 7.28** Not only did the share price, by the way, double; it eventually went up 500%, then doubled again, but AZT was bringing a third of the company's revenue, about \$450 million a year. It took until 1994 for it to pass \$2 billion a year in sales. Nobody—oh, sorry, one great wrinkle, and I don't mean great as in good. I mentioned before the orphan drugs, 200,000. That's the max; that's the biggest size of the patient population. So nobody had ever asked the question in Congress when you're passing the law, what happens if you have a small patient population, you design a drug for it, and then it's some type of illness or disease that keeps growing and surpasses 200,000? Do they then lose all of their orphan drug benefits? Do they have to give back all the tax credits and everything else? Can there then be competition?
- 7.29** No, no, no. One of the very first examples, they have to decide what to do with it, and they do decide. They say that if you come into the market when it's under 200,000 and you get orphan drug classification, you're protected, even if it then grows to 5 million. So even though the number of people infected with HIV grows far past 200,000 in the US and millions worldwide, they are able to sell AZT under the protection of the Orphan Drug Act.
- 7.30** So you think this has to be an exception. It can't be something that's repeated. But 30 years later, I'll bring you forward how little things have changed. Some of you may have heard of a drug called Truvada. It's called a HIV preventative drug of sorts. It helps reduce the possibility of becoming infected or passing on the transmission. This drug comes in, approved in 2004 for treating HIV; that's the original purpose of it. Then, they repurpose it as this ground-breaking pre-exposure prophylaxis, PrEP, as it's called, to prevent HIV infection in healthy individuals. They are looking at a very small end of the market.
- 7.31** Here's the research journey: CDC begins looking into whether they can have medications that might be able to slow the spread of AIDS in 2006. That's when the CDC is spending some money and effort on this. In 2008, Gilead, who owns Truvada, sends free doses of Truvada to the CDC for work in the lab and with primates. It says, "See if our Truvada might be of any help to you." In 2010, the results of CDC and San Francisco AIDS Researcher's work are published in the Journal of the American Medical Association, in the New England Journal of Medicine. In 2015, the US government patents Truvada as a new prophylactic use for a small part of the AIDS population that might benefit from it.
- 7.32** What's interesting is that this time the government didn't wait around. It said, "We've done the work on this. We've created it. We were the ones. The Bill and Melinda Gates Foundation had given \$17 million to the research. The NIH had paid for the human clinical trials. A San Francisco AIDS researcher had used \$50 million in federal grants in doing this work." So the federal government said, "You, Gilead, you were uninvolved. You send the medication over. Thanks very much. We're going to patent it and make it inexpensively available." And Gilead said, "I don't think so."

- 7.33** Gilead fought for the patent. It went to court for it. This court battle between the US government and Gilead for the right to the patent is, to me, one of the seminal battles in modern drug history because Gilead prevails. It really is the inviolability of the patent. Once you own it, you have the exclusive right to it, and nothing can upset that. So if somebody takes your drug, to which you have the patent, and they knock themselves out with clinical trials and tests and everything else, even if it's the government to provide for public service, no court is willing to take away your right to that ownership. They gave it back to Gilead.
- 7.34** Who is manufacturing Truvada at \$6 a cost for a patient—that's what a dosage would end up costing—they priced it at \$1,800 a patient, and the markup percentage is 25,000%. So what does it mean? It ranks every year as the top-selling prophylactic in the top 20 drugs in the country. It reaches \$3 billion a year in income for Gilead, \$30 billion in income for Gilead from the time of its inception until its patent expires, by the way, in 2021. They finally lose their patent. Don't feel sorry for them because they have a new formulation called Descovy, and they're trying to compete with that. But the generic is out there, and it's being widely distributed. But this is the example of, again, talking orphan drugs and how it works. It happens first with AZT, then it happens here with Truvada.
- 7.35** How did this become a blueprint for the rest of the industry? Or you think of it, "Oh, maybe it's just an exception in the way it applied to HIV and AIDS." Within 20 years, from the time of the passage of that act in 1983 until 2004, '05, '06, in that 20-year period, orphan drugs become the fastest-growing sector in the pharmaceutical industry and the most lucrative. I guarantee you that if you go to 10 people on the street, well-informed friends of yours who read about the drug industry, who follow the news and know everything else and say, "By the way, you want to tell me about orphan drugs?" They've never heard of it. For the most part, orphan drugs are this unusual part of the market that, unless you've had some connection to it, unless you've had a disease treated, unless you've had a family member, somebody else who's had it, you don't even realize that it's an orphan drug or what it means in terms of it.
- 7.36** So, they're priced between \$300,000 and \$3 million a patient, astronomical amounts. And one of the benefits here is, you'll see this later on a specific drug we'll talk about, they're very smart. Drug companies know they can't just go ahead and pick a price out of the air. So what they often do is they end up coming in, and they say, "We're treating this rare genetic illness for a small patient population. Now we're going to tell you what the hospital cost and clinical cost is for that patient over a five-year period, seven-year period. Our drug may seem really expensive every year, but it's cheaper than you paying for that medical cost."
- 7.37** The European countries that negotiate lower drug prices make the drug companies do that very thing. "Show us that the price of that drug is less than what we would pay if

we had to take care of it in a national health care system." So the drug companies are calculating this all the time to be able to justify the high prices.

- 7.38** Orphan drugs extended to broaden the market. We'll go over that in a second. You start as an orphan drug for a rare disease, and then you end up with a drug that's treating a lot of people for something else.
- 7.39** Orphan drugs used to claim monopoly control even when the underlying science is not novel. I will tell you this time and again when something is not unusual, and we'll see an example of that coming up, in which the drug itself isn't necessarily groundbreaking. All that's groundbreaking is the price that comes up.
- 7.40** Allergan's Botox, exploiting the Orphan Drug Act—I don't want you to think that Allergan is the only one doing this. They're not; they just happen to be the example today, in part because it goes back to what I said the other day: most people are surprised when they hear what orphan drugs are about for rare genetic diseases or unusual illnesses, for patient populations under 200,000 people that Botox cosmetic could have started out as an orphan drug. But this is one of the best examples of how you can make a lot of money starting out on a small thing.
- 7.41** Botox, originally developed in 1990, approved for two orphan illnesses related to involuntary eye muscle spasm. Allergan—so it was an ophthalmologist, as a matter of fact, who had a drug called Oculinum, and that was approved for two orphan processes. Allergan, a year after that approval was made, bought his company for \$8 million. They were basically an ocular care company at the time, Allergan. So this was in their field, in their wheelhouse. They paid \$8 million to buy this company out. So it's to treat these two different disorders. They acquired the rights in '91. They bought the rights to Botox.
- 7.42** By the way, Botox, you know what it's short for? Botulinum toxin. It is because it is a poisonous toxin. Botox, the beginning of each of the words. That's all. So they have that as a thing. Doctors are using it to treat these spasms on the eye. Later, they have it approved for also a throat stricture problem. And they notice, around the eyes, that there appear to be less wrinkles, less formation of wrinkles; they appear a little bit softer. Once that word gets out, what is that? That's off-label dispensing. We know that. So dermatologists start to do off-label dispensing of Botox approved for this problem of these two small orphan drug approvals for spasms of the eye, and it starts to take off. Dermatologists use it.
- 7.43** At this point, Arthur Sackler died in 1987. He would have been proud of the way Allergan made Botox part of the cultural milieu. *Sex and the City* was an HBO show. There's a line from Kim Cattrall in it where she says, "Husbands, they come and go. But Botox, that's forever," or something like this. It's used the same way that people used to talk about Valium.

- 7.44 By the way, one step back, Valium became for a while to be used across the board by people to refer to anti-anxiety medications. They would say, "You are taking a Valium," the same way people used to say, "Are you making a Xerox?" when you were making a photocopy. Some words become so ingrained in the public mind. Botox is one of those that becomes that in terms of cosmetics.
- 7.45 So here's what Allergan does. It acquires the rights to Botox by buying the company for \$8 million, without, at that point—by the way, I will say, they did not know, they had not done any secret studies to know about the side effect regarding the relaxing of wrinkles or muscles. So it wasn't like they bought it and said, "Oh, we have a golden goose here." They found out about that afterwards, and then what they did is figured out how to make money on it.
- 7.46 Ten years later, they get it approved for a disease for cervical dystonia. From 2001 to 2010, they launched a secret campaign to essentially promote it for headaches and pain. Oh, God, we're back to pain. It's so amazing. It's fantastic. Pain is such a moneymaker. People are looking for a solution from pain, whether it's Lyrica, gabapentin, OxyContin. It goes back to morphine in the days when we were first getting the drug industry going. It's that solution for pain. So, could Botox help for pain? That's what they're positioning as.
- 7.47 By the way, they get at one point a black box warning. They get a black box warning because they're doing studies, and they find out at a certain point that there are some serious potential side effects. Two children died during trials, whether it would treat cerebral palsy or not. Two children died. They realized, "Oh, yeah, it could have some effects here that aren't necessarily very good." Black box warning comes, FDA orders it, and the black box warning doesn't even make a dent in the off-label sales or the sales that they get because of the fact that they finally got it approved from the FDA for cosmetic use.
- 7.48 But the reason the black box warning made no difference, and Allergan knew it, and they talked about it in their internal memos, is because you did not go to a pharmacist and fill the prescription. You didn't walk away with a bottle of pills. You didn't walk away with a prescription cream. You went to a clinic, a doctor, somebody, dentists nowadays, whoever is injecting it. And so you don't actually see the vial. You don't see the little black box warning that comes on the vial, only the person dispensing it to you.
- 7.49 Do you think that any dermatologist or anyone else who's running in, turn back the aging clock, is saying, "Oh, by the way, before I inject this Botox in you, I want to tell you about the black box warning the FDA put on it and what it means," and then you walk out of the office. Of course, no one was talking about that. So this might have been the black box warning with the least effect of any that was had in all of the pharmaceutical industry. The revenue for Allergan, for about 20% of the company's annual revenue, it's about \$3 billion a year, over \$3 billion a year that it's taking in.

- 7.50** Tactics—they aren't just sitting back. One of the things I don't want you to think of. I don't want you to think that these companies, there's off-label prescribing, and they just sit back and they say, "Let's go on vacation. It's fantastic. It's spreading on its own." They want to set the fire larger. How do we make sure it spreads? How do we add some gasoline to the fire as it's going around? So they dispatch their sales representatives, and they want nationwide to convince doctors to prescribe Botox off-label, and that's the headache and the generalized pain.
- 7.51** They hold workshops to demonstrate—this is one of my favorites. I have a list of favorites, so this is one of my favorites: Allergan, the drug company, is demonstrating to doctors how to disguise their off-label injections so they'll get covered by insurance. I will tell you, it doesn't sound like that's kosher to me. Providing financial incentives: they paid for weekend getaways. This is the old hat. They went back to the Sackler playbook, as I call it. Physician conferences, workshops, dinners, all on expanding the off-label therapies. They create patient information outlets. Patient organization that said, "We'll give you advice whether Botox might be able to be used for off-label." If you went through that patient portal to find out, you were doing a search and you end up there, essentially it cured everything, according to them. So you wouldn't know necessarily that that's being funded and controlled by Allergan.
- 7.52** The legal fallout—Allergan pleads guilty in 2010—a lot of people don't know this. They think of Botox as one of these drugs that's just out there, and that's the end of it—to a single criminal misdemeanor of misbranding Botox with off-label promotion. Pretty remarkable.
- 7.53** Criminal fines and several—they pay a fine of \$375 million. I don't mean that's a pittance; that's not a pittance. I don't mean to diminish a \$375 million fine. But in the pharmaceutical industry, I've discovered this over time, and I can give you example after example. Those fines, when they finally get cost and they have to pay it, it's the cost of doing business. They don't want to pay it, but if they do get caught for overbranding, for misbranding, for overselling, for doing things they shouldn't do, and then they get hit with the fine, that cuts off their profits: \$375 million less in terms of profits, but they're still making more than that. And then, they did a separate civil settlement, the federal government in the States, which cost another \$225 million, so \$600 million in fines and civil settlements.
- 7.54** False claims to government programs—the settlement was because Allergan had helped doctors file those false claims they talked about before for Medicare, Medicaid, and Veterans Affairs. So the government was able to say, "Hey, false claims to us, the government, because we're paying for it with taxpayer money." So that was part of the reason for the False Claims Act.
- 7.55** They signed a Corporate Integrity Agreement. Every pharmaceutical company signs a Corporate Integrity Agreement, the same type of Corporate Integrity Agreement that Purdue Pharma signed, and they went on doing the very same type of things. And if

they don't comply, in that Corporate Integrity Agreement, it says, "We'll exclude you from federal health care programs in the future if you have a material breach of it."

- 7.56** Meanwhile, they are looking at additional uses for the drug that started out as two small orphan drugs. They're looking at it for Meige syndrome, which is this lip spasticity. They're looking at it for axillary hyperhidrosis, which is excessive sweating under the armpits. Neurologic conditions, it might be a spinal cord injury. They're doing what every drugmaker does and push it out.
- 7.57** Why do they have a competitive advantage? Why isn't Botox, after 2010, hurt? There are competitors in terms of the cosmetic field to Botox. Why are they able to still keep carving out these major areas that are so big, these additional seven-year selling monopolies on other orphan drugs?
- 7.58** That's because they really have been like Heinz 57, the ketchup, or Coca-Cola, the secret ingredients. They have kept the basis of their drug pretty much a secret. People know what's in it, but when I say it's like Coca-Cola's secret formula, believe it or not—and you'll believe it because I know by this time you trust me—fewer than a dozen people at Allergan know, and I'll make sure I have this right: "all of the reagents utilizing the exact settings of its anaerobic fermentation process." So even inside the company, they don't want people leaving to give it away to somebody else. So they have a way of creating that drug that's a little bit long. As long as they are the only ones who know that recipe, then they can keep using it for new disease subsets of orphan drugs.
- 7.59** They are testing Botox today for nearly a dozen potential other orphan drug designations, which will give them a seven-year exclusive selling period for the variation of that drug they've developed for that specific. Now, they are adjusting Botox for each of those small markets, but if they get them, they have revenue streams going forward that are absolutely protected.
- 7.60** Exploiting the Orphan Drug Act—I don't mean here exploiting it in a way that they would think this was savvy. Most drug executives would think that this is savvy. After its criminal plea and settlement, Allergan continued to submit Botox for additional orphan approvals straight down the line of the Orphan Drug Act. With each approval, they get new tax credits, government credits, and subsidies to offset the research. Even though Botox itself is a multibillion dollar drug, they are now being treated as though they are a new company coming in to do an orphan drug because the Orphan Drug Act was passed without saying, "By the way, if you've taken a drug that was approved here and made it into a general purpose drug for a mass market, it's no longer considered an orphan drug in the future for other purposes." That's never it. You can continue to apply that drug just for orphan purposes, and they do that.
- 7.61** They've extended the exclusivity periods. For some of these orphan drugs, they've prevented competition, and they've allowed themselves to keep a monopoly on specific drug indications for a longer period than you would have imagined.

- 7.62 Botox's expansion into mass markets—overactive bladder: 10%. This is a current rough breakdown of the revenue that comes in from Botox for the general mass market: 10% used for overactive bladder. Migraines: 20%, fairly large end of the market, and 70% just for cosmetics. So they've expanded even the mass market out into migraines, now approved, and overactive bladder. So they're always looking at that.
- 7.63 The Orphan Drug Act, exploiting loopholes for profit— this is one of those areas in which you think to yourself, "How much more can they do? They did AZT, they did Truvada, Botox gained this system. That's probably the end of it. So how did Medicare end up paying \$2 billion for one drug while the manufacturer paid doctors millions in promotion? Not the best story around.
- 7.64 Acthar was a drug approved in 1952 by the FDA to treat West syndrome, an uncommon fatal epileptic disorder that strikes infants, usually younger than one. That's what it was approved for in 1952. So Acthar is sitting around. It's made from the pituitary glands of slaughtered pigs. And you know which company had first gotten it approved for drug approval? It was discovered by Armour, the meatpacking giant. So, food industry, you did ask the other day, and I forgot about that example at that time.
- 7.65 So Armour gets the original work. They're the ones who are able to pull it from the pituitary glands of pigs. And then it's bought, finally, in the 1990s. Sanofi-Aventis takes Acthar, and they said, "You know what? We're going to be able to use this as a treatment here. It's a very small end of the market. We understand that." They believe that the market affected for this rare fatal epileptic disease affecting children under the age of one is somewhere between 700 to 900 patients a year. That's the idea. So Sanofi says, "You know what? We They think there's still money to be made." They take it in the 1990s, and they price it at \$40 a vial. So \$40 a vial, and they're not making any money. They're losing money, not enough on that.
- 7.66 And then in 2001, Questcor, this small unprofitable drug maker, comes in and buys the rights from Sanofi, who says, "We're not doing anything with it. Sanofi is doing all types of other drugs." And Questcor says, "Fine. We know how to make money. It's just a process of increasing the price." Same product, Acthar gel, and they increase it from \$40 to \$1,650. That's a big jump. And guess what? They're still losing money. They still don't have enough patience to make real money on that.
- 7.67 That is until in 2007, they get a new CEO, a guy called Don Bailey. He comes in and he's a business executive with no background in the drug industry. So that means he's looking at just numbers. He knows, they tell him, the patient population is maybe 800. That's the number he ends up working off of. So, what do we need to make a profit on this? That's all he's looking at is what's the number that we need to have. He raises the price from \$1,650 a vial to \$23,000 a vial. That's a big jump.
- 7.68 He reformulated it to get Acthar gel. Now it's a little bit different. They never had it as a gel before. It's been a solution. Now it's a gel. It goes back to the FDA and says, "Hey,

my patient population is tiny. Can you give me orphan drug status on this?" The FDA says, "It looks good to us." So Acthar gel now has a seven-year run as a monopoly on this particular drug. Nobody else can touch it. In addition to what it was originally approved for, they are now testing—they had pushed the FDA on items like nephrotic syndrome, kidney disorders, rheumatoid arthritis. They were looking at whether it could treat lupus. It's the same thing we always see.

- 7.69** Questcor's stock went from \$0.60 to \$50. So shareholders think Don Bailey is a genius. He's fantastic because he's performing from them. The Insider sold more than \$100 million of stock over a two-year period—that's the side light; nothing against selling stock, but I'm just saying they sold at the highs, not at the lows. And it turns out that the US government, us, all of us who paid taxes, from 2011 until 2016, after this orphan drug status is given to Acthar gel, had paid more than \$2 billion to buy the drugs under Medicaid and under health care programs because it was being prescribed and it was being used.
- 7.70** It's the head of Oregon Health and Science School, this Dr. Dennis Bourdette, who had studied the use of Acthar gel afterwards, said, "It's a mystery to me why someone would be prescribing the drug." It's hard to imagine. One of the reasons why it would be a mystery, you would think, is because there was a Canadian drug called Synacthen, which sold for \$33 a vial. That's pretty good. It's a chemical cousin of Acthar. But what happened? Questcor became Mallinckrodt, another company which bought Questcor. And then Mallinckrodt bought the Canadian company that was creating Synacthen and essentially took it off the shelf. So it didn't exist anymore.
- 7.71** So your generic competitor was bought and disappeared. Sixty percent of the spike in Acthar's sales went to seniors in Medicare. That's great. They're the only ones who need it over the age of 65. What for? For one of these spinal problems that it was approved for? Oh, no, no, no. For off-label use of ailments, including rheumatoid arthritis. There is no evidence, clinically, that Acthar helps rheumatoid arthritis. So you have an orphan drug approved for a specific, narrow purpose, and then you have doctors prescribing it at the suggestion, possibly, of drug reps off-label for things for which it wasn't approved at \$23,000 a vial.
- 7.72** It's being paid for under Part D prescription programs from Medicare. It's being paid for in Medicaid, where they have to cover it. That's how Medicare ends up paying \$2 billion for a drug while the manufacturer paid the doctors literally millions in promotion. This is one of those stories for me that absolutely is enraging. If it doesn't get you angry, nothing else will.
- 7.73** Profits above care—some of you may recognize that name or that face, I should say, Martin Shkreli, a former hedge fund manager and pharmaceutical executive. I use that loosely. In 2015, he gained notoriety because he was looking around, and this becomes an investment. Investors like Shkreli are looking for companies that have a patent drug that might get an orphan status or has an orphan status and isn't being used

aggressively, like Acthar had been. That shows how you can take a drug that was \$40 a vial, make it \$16.50, make it \$23,000, and then make it into a real business.

7.74 So he ends up buying the rights to Daraprim, used to treat a very rare parasitic infection. And he raises the price from \$13.50 to \$750 a pill. A big impact on the patients who couldn't afford it, and there's a real outcry that takes place. So he spends \$55 million to buy Daraprim. You know that this is a straight business transaction for people who come in like this. They aren't paying \$55 million because of altruistic reasons, because they want to get a drug that they can then distribute to people who are suffering from the immune disease. They're buying it because they figured that \$55 million is a worthwhile investment for what they can make on, even though the patient population is small.

7.75 That allows him to control the pricing of the drug. He moves it from \$13.50 to \$750 a pill, a 5,000% increase. Daraprim patients, for the most part, suffering from rare parasitic infections that come from their lowered immune systems, many of them can't afford it. The annual health care costs for a patient now becomes \$35,000 a year. The government paid \$35,556.48 for a month's supply of Daraprim. You know what they were paying beforehand? \$608 a month. It's almost as though you feel like saying to the federal government, "You're suckers. You've fallen for this. You created the law. You created the Orphan Drug Act to do a good thing. Now it's being gained by investors who are coming in, raising the price, and there are some uninsured people who may not get anything, but the ones who are covered under government health, you're going to continue to pay that amount."

7.76 What was the industry response? It was interesting. They pushed back on that particular drug, and Shkreli in particular, because he was aggressive, and he went before Congress, and he was arguing for the price of drugs being high, and he was very unlikable and all of those things. So pharma and the rest of the industry realized he's bad for business. We don't want it to make it appear that we are just bringing investors who are buying drugs and then ratcheting up the prices. It's better that the people who ratchet up the prices are the inventors or the doctors or people like that. We don't want these outsiders from Wall Street doing it.

7.77 So they launched this \$60 million campaign to improve the industry's image—I love that. And this is one of their lines that they're using inside when they're talking about launching this campaign to win over public approval again. They said, "We want more lab coat, less hoodie." The idea that he was just a young guy with a hoodie, and he was at one point, when he walked into one of the conferences. This bad actor narrative, they wanted to say, "This is not who we are. This is just this guy." So here are the problems.

7.78 By the way, something for you to note, he ends up going to prison. People think he went to prison because of the drug pricing. Oh, no, not at all. It was a securities problem. It was completely different than that. It had nothing to do with raising the

price of the drug. And one of my favorite things—again, long list of favorites—is that when he went to prison, people, the average person, even me, initially thought, "Oh, that's the end of this outrageous gouging of the price on Daraprim." No, he went to prison. The spotlight went off of him and his company because of the fact that he was no longer there. People thought he was paying a cost. The company kept the price at \$750. They didn't lower it. They didn't change it. So that stayed the same, what he had done. So lack of price regulation, it's a problem. You can charge whatever you want for orphan drugs.

- 7.79** Patent extensions and exclusivity continued to be a problem. Broad eligibility criteria—I'll tell you: this is a real problem. Affecting fewer than 200,000, then they expanded beyond that, and they get a much larger market. Inadequate oversight—there's no doubt the FDA is missing in action on some of these.
- 7.80** Lack of transparency—the financial incentives and benefits behind the scenes, that the drug companies are using to push the Orphan Drug Act dilutes its entire effectiveness. On top of this comes direct-to-consumer ads. We talked about this a little bit yesterday. In 1997, the FDA says, "Great idea. Let's go ahead and improve direct-to-consumer ads." The first one is Claritin, the one for the allergy medication. I'll read you the text for that. It's a 30-second spot. It has the two people skipping through the field of flowers. Then a man and a woman talk over each other to this upbeat soundtrack. They spent big money for this on Madison Avenue. "Claritin. It's time. It's time. Don't wait another minute for Claritin. Claritin. I'll ask my doctor. It's time to see your doctor. At last, a clear day is here. I want to know more about Claritin. Ask your doctor for a trial of Claritin and for free information and a \$5 coupon toward a prescription for Claritin. Call 1-800."
- 7.81** That sharing plow ad had everybody talking. Once Claritin takes off, a year after the rule change, there was a time in which they thought nobody would advertise for this, and a year after the rule changed, they were spending a billion dollars a year on television ads, which is really remarkable. They covered everything. I mean, some of the early ads were terrible: Abe Lincoln at a playground with beavers, talking stomachs, Andrew Jackson, praising a wonder ointment, even eerie animated characters that morphed into toenail fungus. They were unleashed. In 2025, they're spending \$7 billion, almost \$8 billion on those television ads, \$10 billion overall, a lot of money going into direct-to-consumer ads.
- 7.82** What are some of the solutions for pushing back on this? Tied in eligibility criteria for orphan drug designation—there are a number of proposals for this. It just has no traction in Congress. Implement stricter price controls and oversight—oh, my God, that's going to be socialized medicine. They're going to fight that every chance they can, especially on orphan drugs. Increase transparency in drug pricing and development costs—that would be fantastic. Tell us what it actually costs you to develop the drug. They put in their overhead and everything else on top of it. They

make the cost of the drug look much higher in the development than it actually is. And incentivize generic competition for orphan drugs. Take away that seven-year exclusive period if a generic can show that it has the same quality to be able to treat the disease for which it's been approved.

- 7.83** The economics of the new ultra-expensive drugs— the key is that we're now entering this era of what I call gene therapies, enzyme replacement therapies, biologics, cancer drugs for small genetic subtypes. The opportunities for these ultra-expensive drugs for small patient populations is expanding at a pretty fast rate, faster than the regulators are able to keep up with. The patients can't shop around, especially if you have that rare genetic disease or that unusual disease. Doctors don't look beyond the single medication that's been prescribed for it. There is political backlash if you try to do something to regulate that drug because the small patient population that is affected by it will scream at the top of their lungs that you're doing something to hurt them. It's a very interesting thing.
- 7.84** So, now what we're in, we're in the world, what I say are single-dose therapies. Sometimes in biologics, they're able to talk about giving you a single-dose therapy that may be able to last for years, and that single-dose therapy really makes a difference. So, here's what we have. We have in this area a high price is justified as innovation, while research relies, again, time and time again, on public funding. It happens consistently through this period, and it runs into something else that's taking place: PBMs.
- 7.85** Again, many people have never heard of orphan drugs. Many people have never heard of pharmacy benefit managers. But pharmacy benefit managers are the middle ground between the drug companies and pharmacists and the public that have been inserted since the 1970s and 80s, and they're responsible for part of the reason the drug prices are so high.
- 7.86** So what are PBMs? They're companies that administer the drug benefits for private insurers, the Medicare Part D plans, employer plans, Medicaid managed care organizations. The first PBM started here, where we are taping this lecture in Scottsdale, Arizona. It was a result of private insurance called PCS; that's the name of the company. PCS really wasn't thinking about the extent to what PBMs do today. They went to the large, like IBM and other companies that had large insurance companies for employees and said, "By the way, we will administer the paperwork. We'll do the medical claims. We'll make the applications to insurance." They're essentially like a payroll processor, but doing it for medical claims, which made sense. PCS has a bit of a business in that, so it's creating it.
- 7.87** Congress had passed this HMO Act, a small company, virgin. As HMOs grew, they saw an opportunity in PCS. They're the first firm to process prescriptions, formulary lists, and reimburse pharmacies. They get bought out a year later, essentially, by McKesson, one of the big monsters, McKesson, one of the big drug distributors. McKesson, they're

inside the business; they have a whole different idea of this. They said, "We don't want to just be paperwork processors. We want to influence drug coverage." So now they have, over a period of time, grown to have an independent power as drug distributors.

- 7.88** The companies that have used PBMs, including the drug companies, have allowed them at times to actually say which drugs insurers will cover. You ever have a drug prescription plan and you look to see if your drug is covered or not under the formulary or whether it's not or it's non-formular and whether you have to pay a lot for it? Guess what? That differs insurance plan by insurance plan. You think it's the insurance plan that's making those decisions? It's the pharmacy benefit managers hired by the insurance companies to make the decisions because the PBMs claim we have a view over the entire industry so we can tell you what's best for business and for coverage.
- 7.89** Which drugs the pharmacies can dispense? Pharmacists, more than anybody else, the community pharmacists hate PBMs because they're the last ones to know. They're told what they can dispense, they can't change the drug, and their profits are cut down on that end. How much patients pay? The copay is set by the PBMs. Which manufacturer gains share? Set by them. PBMs administer drug benefits for Medicare private Part D plans, employer plans, Medicaid managed plans, and they get a rebate from drug companies, sometimes, if they emphasize, push, or put at the top of their list a company's drug, the rebate comes back to the PBM, but no one knows how much that rebate is because it's not closed. It's not transparent. It's one of the most frustrating parts.
- 7.90** So, profiting from rebates, the rebates from drug makers spread pricing from what they pay to the pharmacies and bill insurance. It makes another tier of billing. This creates price pressure across the board. Driving up prices? No question about it. It leads to an upward spiral of those prices. The lack of transparency, the fact they're so opaque is one of the biggest problems in dealing with PBMs.
- 7.91** What does it lead to? Higher list prices, larger rebate pool for PBMs. So they love high list prices. PBMs prefer drugs with larger rebates, not lower net prices. They're not looking for the lowest cost for the consumer or the patient. Drug makers inflate list prices sometimes to satisfy rebate. So you don't want to know what the list price is. You want to know what the real price is. Patients pay cost-sharing based on these inflated list prices.
- 7.92** Targeting women in specialty drug economics—that's a continuation of the old book—on autoimmune diseases and thyroid disease and rare genetic disorders and osteoporosis and multiple sclerosis, I could take you down the list here. What had started with Valium and HRT continues in many respects, still targeting women as this unusual group.
- 7.93** Case studies from pharma—the money being spent by the pharmaceutical companies for the regulatory framework, when I go through the money that they spend, that they

report, 80% of them are trying to manipulate what I call the distribution channels. Eighty-five percent are trying to exploit government incentives. They would say that they're just doing the best thing for their shareholders. Ninety percent are outsourcing responsibility. They use the PBMs and others as a buffer to be able to say, "Oh, they're responsible. We're not responsible." Ninety-two percent leverage the loopholes and legislation here.

- 7.94** So what's the result? Lenmeldy: oh, my God, \$4.25 million for an orphan drug. It treats MLD, a progressive nervous system disorder. The patient population: under 4,000, priced at \$4.25 million. Kebilidi: \$3.95 million. It treats a rare genetic brain disease; patient population: under 5,000. Hemgenix, \$3.5 million. It treats hemophilia B; patient population: under 8'000. Elevidys: \$3.2 million. It's an orphan drug for a rare form of muscular dystrophy; patient population: under 13,000.
- 7.95** One of the things that's interesting on this, I think, is for anybody who's looking at it, there was a time, as a matter of fact, when you had other drugs. Novartis had a drug, as a matter of fact, that they recently had a few years ago that they had priced, now it's not so high, at \$2.1 million. They had been thinking of pricing it at \$5 million because that was what they were able to make the argument for against the medical care on a pediatric drug that would have required care for these children. They finally priced it at \$2 million. They were the first company, Novartis, to provide a payment plan, installment plan.
- 7.96** You know layaway? You're going to buy a sofa. You don't have the money for it. You can do it on layaway. So, Novartis said to the insurance companies, "Don't worry about paying the \$2.1 million all up front. You can do it over a four or five year period." And they started to develop that. So you now take the top 10 drugs, the top most expensive 10 drugs in the world, they're all orphan drugs. Nobody knows that. They look at the list, they're drugs nobody's ever heard of, but they just assume they're drugs. They're all orphan drugs. That's how they're able to get these prices at this level for these small groups. Look at the patient population. We're not talking 20 million, 30 million. We're talking only a few thousand. But the profits are big because the prices are big.
- 7.97** The future—genomics, precision, monopolies by design, proliferation of gene editing and cell therapies. You're going to this expand for orphan drug models to target even smaller patient populations. Drug makers will leverage these new treatment modalities for even single-patient treatments. That's the Holy Grail. What's the Holy Grail? Eventually, they'll be able to say, "We have something that will protect a person based upon their DNA structure. That drug is for that person," and you can price that at a high amount, and then you can do that a few billion times.
- 7.98** Microindications in niche markets for smaller genomics subtypes—allowing them to qualify and change the drugs so they can continue the patents. They don't just use it once.

- 7.99** Approaching the cost of a home—I would say it exceeds the cost of a home for most of these therapies that are coming in. Patients bear the consequences of it.
- 7.100** How to be a smarter patient? It's a tough one in this field. Check whether a drug is orphan. If you need it, it's not going to change it. It's just going to be the drug. But sometimes, if you're prescribed a drug and you say to your doctor, "Is that an orphan drug?" The doctor will be shocked that you even know about orphan drugs. If the doctor doesn't know if that's an orphan drug, then you're in trouble. They should at least have an answer to that.
- 7.101** Ask your doctor about non-orphan treatment alternatives. Unless you have a particular genetic illness that can only be treated by that one drug, if you're doing something and an orphan drug is treating your migraines or your overactive bladder or something like that, maybe there is—maybe, remember, I'm not a doctor, it's not medical advice—but maybe there's a generic alternative that might help you be a little bit less aggressively promoted to.
- 7.102** Understand that the PBM shape your drug options. That doesn't mean you can do anything about it. The Biden administration, the Trump administration, they've talked about it. I hear a lot of talk in the current administration about reining in the PBMs. It's good. We're hearing talk about it for the first time. But I've heard talk before about reining in companies, then nothing happens. The way this will eventually happen, I believe, is the drug industry will throw PBMs under the bus. They'll say, "They're the ones to blame for all these years of high drug prices. Not us. The middleman, let's get rid of it." PBMs, by the way, you know what they're like in France, in Germany, in England, and all these other countries, how badly they affect drug prices? They don't because they don't exist. It's a particular American invention.
- 7.103** Beware of specialty pharmacy routing. That is a sign that pricing rebates are driving decision making. So, you get to a specialty pharmacy, unless they have to have that because they're compounding the drug or something else, why are they sending you to the specialty pharmacy as opposed to what I call a chain store pharmacy?
- 7.104** Ask for actual out-of-pocket cost, not the list price. PBM rebates do distort everything. So, forget about what the list price is. Just find out how much they're going to charge you for it. That's all that really matters to you. The list price is going to be some inflated amount. You're always going to feel like you're getting a benefit, even if you're spending a lot of money out of pocket.
- 7.105** And understand the most expensive—again, I repeat, this is not the most effective. Ultra-expensive therapies treat symptoms and slow progression, but none of these drugs we're talking about are cures, and that is rather remarkable.

Lecture 8: Pandemic Profits

At a glance — Pandemic profits.

- 8.1 Pandemic Profits: How Pharma Turned Crisis Into a Cash Cow. Look, these are the same documented patterns that we've talked about before inside the industry. It's a century-long strategy. You take a crisis, use some public funding, you avoid pricing oversight, you expand into a big market. It's a narrative in real-time.
- 8.2 In my book, *Pharma: Greed, Lies, and the Poisoning of America*, which I'd spent five years essentially with Trisha researching and reporting, doing hundreds of interviews, tens of thousands of pages of documents, was published in March of 2020. The second last chapter in *Pharma* is titled *The Coming Pandemic*. The last paragraph in that chapter, talking to a number of doctors, says— I'm writing this and I say—"The prediction of a coming pandemic: unstoppable, because the pharmaceutical industry has put profits over its duty to develop drugs for the public good. It's no wild-eyed conspiracy theory heavy on drama and light on evidence. 'It's not a question of if,' warns Professor Bush."—Karen Bush. "'It's a question of when.'" Very dramatic.
- 8.3 But I wasn't talking about COVID. I was talking about a superbug pandemic that might take place because of antibiotic resistance. And these doctors who were talking to me were worried that the next pandemic would be a superbug, that they couldn't stop because companies have given up developing antibiotics because the market for them is so small: five to seven days; they'd rather do chronic drugs or orphan drugs.
- 8.4 As a matter of fact, COVID isn't mentioned in my book, *Pharma*, because it's published on March 10th—it happens to be Trisha's birthday—2020. The manuscript has been turned in to the publisher, Simon & Schuster, months in advance. It finally is closed off from any changes you can make a couple of months beforehand because they're already selling the book in terms of paperback sales down the road, for first serial rights and setting up the book tour.
- 8.5 The day after my book was published, the book tour started to get canceled: Bill Maher, things like that. We lost one show after another as people started to go on hiatus. But more than that, the World Health Organization came out with something. The World Health Organization, the day after *Pharma* was published, declared the coronavirus a worldwide pandemic. So not the best timing I've ever had.
- 8.6 A few days later, I was being interviewed by a newspaper called *The Intercept*, and I told that reporter, quote, "Pharmaceutical companies"—I knew something. I didn't know about COVID, the virus, what was taking place, but I knew what the response would be of the drug industry that I had studied, that we had talked about for the last seven hours, and now talking about in this eighth hour. I told this reporter without a caveat, quote, "Pharmaceutical companies view COVID-19 as a once-in-a-lifetime business opportunity." Oh, pharma companies hated that. I'd get pushback on that.

- 8.7** So then, I thought—you know I always like a challenge. So then I submitted op-eds to different newspapers, and The New York Times accepted it. I wrote an op-ed shortly after that. You should look it up once if you get a chance, called Big Pharma May Pose an Obstacle to Vaccine Development, talking about how in the past, they would want and demand significant profits for any vaccine.
- 8.8** Now, one of the things that happened in the very beginning of COVID, besides the fact that bookstores closed up a couple of weeks later across the country, and that was a very bad sign for a book that you're working on for a number of years and hoping to sell. People do buy online, but guess what? Book sales went up at the beginning of COVID, but they didn't go up for books, 800-page books about the pharmaceutical industry and exposés. They went up for—people wanted a break from all the angst and anxiety and fear of what was happening. They were reading novels; they were reading other things. It was hard to sell pharma as the idea of what they should read.
- 8.9** But one of the things that happened in this period, as it goes along, is I kept talking out, and I got a lot of pushback because in the early days, March, April, May, there was a lot of fear. There was a fear quotient. We forget about that a little bit. Afterwards, now we look back, and we have our own criticisms of what took place and the overreaching of the government, things like this. But in those early days of fear, when I'm talking about, "Hey, pharma is going to cash in," people would say to me, "Oh, just shut up. We aren't worried about pharma. We don't care if pharma makes a trillion dollars if they save the planet. We don't care how much money we spend. We just have to make sure that they can save us from this. Don't talk about that, that they shouldn't make a big profit from it. What's the matter with you?" So no one was focused on that or talking about that.
- 8.10** And then, in June of 2020, I got a new round of criticism because I was very vocal. I wish I had a bigger megaphone, one that could catch the administration's ear at the time, because the FDA set the rule that the vaccines down the road that we were then funding, that would be coming out at some point, they only had to prove to be 50% more effective than a placebo.
- 8.11** A lot of people weren't paying attention. They're all waiting for the vaccine. "Let's just get a vaccine out there." But the FDA set the bar 50% more effective than a placebo. Well, normally for infectious diseases, it's 70% at the very least. So 70% is what I expected. Fifty percent more than a placebo is like what you get for the annual flu shot. Sometimes the flu shot works; sometimes it doesn't. So they set a very low bar. And I thought that was a sign that pharma was going to come in and be able to develop vaccines and cash in again. But no one wanted to hear about the cashing in part, much to pharma's good graces.
- 8.12** Also, the fact that they had set the bar at 50 %—50 % better than a placebo meant you weren't going to eliminate a virus; you were going to control it, but it would be around, become part of our system. Maybe you'd need booster shots every year or two years or

that, like the flu vaccine. But that was early months. Nobody was yet worried about that. So that was my introduction to COVID crisis as catalyst.

- 8.13** We know this. So, 19th-century epidemics, major crises— patent medicines responded. If you had an outbreak of a disease in a particular city, you had a whole group of patent medicines that were being offered right away that would cure it if it happened again. Antibiotics—booming in World War II. Urgent needs, World War II puts the pressure on, and we end up getting penicillin. Tranquilizers prosper in these post-war anxieties that are developed, although pharma helped create the age of anxiety, they played off of it as well. Opioids surged amid chronic pain and economic despair in many of the communities in which they originally took a foothold. And there's no doubt about that.
- 8.14** COVID-19, though, was the single greatest crisis and opportunity in not just my lifetime, but in modern pharmaceutical years, because the entire industry was facing a potential crisis that had the world concerned. It presented this opportunity to leverage government funding and develop drugs.
- 8.15** What's interesting, as you know, is not every company was in a position to try to develop a vaccine. So you would think: why then is the pharmaceutical industry so interested about it? Because they realized the old line that a rising tide raises all ships. To the extent, this offered a once-in-a-lifetime opportunity—I used to say this to reporters and other reporters; some listened, most didn't—to rehabilitate their reputation.
- 8.16** Think about this: pharma had been thought of so well in the 1950s. They had come up with penicillin. They were saving lives. They had antibiotics. They were doing innovation. They were thought of as the good guys. And then over a period of time, we saw them push drugs too heavily, come out with things: they sparked the opioid prescription crisis. They're on our list of people that we no longer trust. And now, we're putting our faith in them. "You can help save us." And if they can do it cheaply—no one really cares about cost—if they can do it, then they will have rehabilitated themselves. Every drug company realizes this: this can do more for us in a short amount of time than a hundred PR companies can over a decade. So, there was an interest across the line on this.
- 8.17** The scientific achievement as a business model— COVID-19 vaccines, I say, represent a subsidized breakthrough of decades of research on mRNA platforms and scaling manufacturing. I'll tell you: you think penicillin manufacturing scaling was large? Nothing in comparison to this. There had been a focus on mRNA research by a number of companies for a while. When I say a scientific achievement in their research, I'm not talking about the quality of the vaccines or anything else, but in terms of those who are focused on mRNA, this was a potential goldmine because you had the interests of the government on this.

- 8.18** Government as risk taker, pharma as beneficiary—we had NIH, Operation Warp Speed, BARDA. I had hoped early on—I'm just one person. I finished a book on pharma, and suddenly I'm a know-it-all. I think I can tell the government what they should do. Obviously, I can't. What I was trying to tell the government to do here or anywhere else is you have the model. You know what to do.
- 8.19** It's the penicillin model: provide the money, give as much money as you need to get these companies to develop vaccines that you think you need, and make sure that nobody owns the patent. So at the very least, whatever they develop, you're going to expedite it, you're going to push it through. I understand that society is coming to a standstill, all the harms of COVID, when I say harms of COVID, I mean not just from the virus itself, but from the lockdowns and the kids not going to school and all the things that happen to us as a society. You're going to do all of that. Just make sure that they don't—they, meaning the drug companies— own the patent, that they're not going to profit on it on top of you providing the money.
- 8.20** That really doesn't become the course. So, public money: BARDA, NIH, Operation Warp Speed assume all of the financial risk, all the early R&D developments, clinical trials, subsidies, purchase guarantees. We're guaranteeing we're going to buy the vaccines if they're able to produce it. And of course, liability shields. You don't want to have pharmaceutical companies rushing to get a vaccine to stop a worldwide pandemic, and then somebody's going to sue them because they become paralyzed. So if there's a bad side effect that takes place, don't worry about it. You're not going to be able to get sued.
- 8.21** Intellectual property rights—it's unbelievable to me that after all these years, I understand the government's rushing for this, that they don't keep the intellectual property rights free. They let the pharmaceutical companies take them. It gives them long-term pricing power and global distribution profits that they develop from the public funding.
- 8.22** To me, this sounds like this arrangement echoed the pattern documented throughout the history of pharma. They're used to getting—what did we see with AZT? What did we see with Truvada on individual drugs? We saw the NIH and others go and do all the work and pay the money. Then, when it was done, the pharmaceutical companies are able to keep the patent. Here, nobody has the patent because there's no drug that's in existence. So you really were, like with penicillin, at a new opportunity to do it, but nobody's thinking of it.
- 8.23** Leveraging urgency into monopoly—I'll tell you: the drug makers were good at using the crisis to secure the purchase agreement. They don't want to do all that work unless they know it's going to sell. Indemnification, exclusive supply rights, and the protection of their own proprietary formulas.

- 8.24** Now, in the very beginning, there was a moment at which I knew not that it had been lost, but you saw it take place in Washington. In the Trump administration, the first Trump administration, you have a Republican President, and you have Democratic control of the House. Nancy Pelosi is the Speaker of the House. So you have to have a bipartisan agreement for this rush of the money out. Both parties are on board with a bipartisan agreement. Nobody wants to be known as the party who stopped discovery to save lives during a pandemic. So they're all saying yes.
- 8.25** The very first tranche of money that was given turns out to be about \$8 billion, of which \$3, \$4 billion ends up over in vaccine research. In the very first iteration of that emergency bill that was being offered, it was very interesting that they had the clause in that nobody would own essentially the intellectual property rights, no pharmaceutical company would. The idea of penicillin, somebody had drafted that correctly. It passed just a little over a week later, eight days later. They rushed through. And by the time the final bill gets through, that language is out.
- 8.26** The pharma lobby was in there making sure, "You don't want to slow us up. Oh, you don't want to give a hiccup to any of our development. We have all this—" What they used as the excuse— I now know, looking back—is they came back, and they essentially said to some of the legislators in Congress, "We've done all this mRNA research for years at our own cost"—supposedly—"and if you own the intellectual property rights and give it away for free, you will essentially, if we create a vaccine, be giving away our proprietary information about what we know about mRNA, and that would be unfair." The government said, "Well, maybe. Okay, you're right," and put it through at that.
- 8.27** Socializing risk, privatizing reward—so public funding drives research. We know that, right? Pharmaceutical companies have their hands out waiting for that type of research to come in sometimes. Private sector gains— the pharmaceutical companies earn the blockbuster profits, billion-dollar profits from drugs developed at public money.
- 8.28** The debate over fairness, to me, is equity and innovation. There's no question that it's not fair, but here's the factoid that I like so much. This is when I was writing Pharma. This is in 2020, so it's worse since then. Since the 1930s, the NIH has invested \$900 billion—we'll be getting up to a trillion dollars soon— into research that drug companies use to patent brand-name medications. The NIH's view would be, "We're helping public health. We're dealing with making people better." And there's something to that. That's taxpayer money.
- 8.29** We pay real estate taxes in every community we're in. Much of the real estate taxes are used for schools locally. You don't have to have children to benefit from schools. People graduate from the schools. They contribute to the community. There's a benefit to all of that. I understand the idea of a benefit from the NIH providing \$900 billion in funding over a period of time to help companies develop medications. But you would also think that those companies then can't be setting prices at \$23,000 a vial for something like Acthar gel.

- 8.30** The vaccine contract: secrecy, leverage, and guaranteed markets. Secret contracts—you know the COVID vaccine contracts that were originally done between the government? You had to sue to try to get them eventually. You had to apply. They weren't available. There was no transparency about it. Nobody said that's it because they didn't want to distract us under the details of what the contracts were like and the guaranteed purchases were, and when you did get them, they were heavily redacted.
- 8.31** Prioritize wealthy nations—no doubt about that. That, in part, is because the drug companies that are making the vaccines or selling them at a certain price, eventually are pressured to give away some of the vaccine to countries that could not otherwise afford it. But in the beginning, they're interested in selling their first batches available.
- 8.32** Guaranteed purchase agreements—they get this from governments, in which governments say, "We're essentially going to guarantee to a certain amount from you without knowing what the product looks like or whether it works." Indemnification against lawsuits we talked about before. Exclusive supply rights—these contracts gave drug makers exclusive rights to supply their COVID-19 formulas, and they would be able to keep them under wraps. Nobody else could see what they were. Protection of proprietary formulas—complete protection of the intellectual property, as I call it.
- 8.33** So what happens? Countries funding COVID-19 vaccine R&D. We could have an hour just talking about the efficiency and efficacy of the vaccines, an hour talking about the process of how they came to formulate it. What I've done, as I did in much of my research, is, it's a hackneyed phrase for a researcher, but it turns out to be used a lot because it's true: follow the money.
- 8.34** From the very beginning of COVID, I would say to people, "I'm not arguing with you." Before the vaccines were even to market, before they were invented. I said, "I'm not arguing with you about the vaccines that they're needed or whether or not. I'm arguing with you about the profit part of it. That's the part that I know drives the industry so much." People often couldn't see that because they thought that we were all one, joined together by this outside threat. It would be as if an alien came down from outer space and threatened the Earth. We'd be united as a people and not care about our differences. Now we were all one, including the pharmaceutical companies, because COVID had brought us together. I kept saying "not quite."
- 8.35** So the United States, Germany, and the European Union are the top providers for COVID-19. The US government alone provides some \$2 billion. Public and philanthropic funding dwarfed private investment. It was a world record in terms of a rush toward these vaccines. Everybody was throwing money at an early time: an initial \$10 billion appropriation to fund COVID-19 vaccine development. We were throwing money out. Ten billion dollars is unheard of inside the drug industry to suddenly have it come from public money and you're able to use it for your own research.

- 8.36** Eventually, \$22 billion is spent through Operation Warp Speed for major pharmaceutical companies to research, test, and produce COVID-19 vaccines. Research. It is an astonishing amount of money. I understand we're in a \$33 trillion debt, so maybe \$10 billion, \$22 billion isn't that much. But it was a lot of money for the drug industry at that time.
- 8.37** Remember, we know about this now. In the beginning, the government was also doing PPP loans. Remember, there was a rush to get money into people. The amount of fraud, we'll never know the amount of fraud. You're giving away free money, people are there to collect the money. If you're giving away free money, some people are going to collect two or three or four times because that's just their nature. Somebody's always gaming the system.
- 8.38** If you're giving \$22 billion to pharma companies, it's not that they're coming back to get it more and more and more, but it's fantastic because they're doing work that may help them on other drugs down the road, in addition to just the vaccine which is turning on here. So this massive government funding to me highlights the ability of the industry to take advantage of a health crisis at this moment, without the transparency that we would normally have with this amount of money in a public-private partnership.
- 8.39** Windfall gains—why am I not surprised that this turned into big-time windfall gains for companies like Moderna and Pfizer, cashing in on vaccine development? Almost always, the stock price responded— if you followed this—the day after somebody from one of the companies would say—and I have a list of these— "We're making real progress." Oh, boy, that was fantastic. "We're getting close. We think we'll be able to tame this." Those things would bump the stock prices up because people were anxious for it.
- 8.40** Moderna, of course, had never had a drug to market. Moderna was one of those companies focused on mRNA research, and drug after drug that they had tried to do with mRNA had never worked for them. So now they have a real opportunity. They might be able not just to get billions of dollars in government aid, they might be able to get their first drug to market, all paid for by the US government at a time of great crisis, at an expedited rating without having to go through all the clinical trials and without being worried about being sued down the road. That's pretty much as good as it could get from a Moderna executive.
- 8.41** Change in trading volumes—I follow this on Webull, one of the tradings. This chart shows the spike in trading volumes for vaccine makers like BioNTech, Moderna, Novavax—that was one—Pfizer during the pandemic. So the biggest jumps for Pfizer and BioNTech followed in November of 2020 when they had a joint announcement that their vaccine had been effective in 90 % of the trials that they tried it in. The keyword there is "effective."

- 8.42** It's like Alexander Fleming comes back from vacation in Scotland years ago, and he looks at his petri dishes, and there's some staph and microbes that he has there, and it looks like some of them have been eaten up by this green mold that he calls Penicillium. What does effective mean? Are you eating it at all? Are you turning it around? Have you destroyed the virus?
- 8.43** But that news caused the global stock market to surge, not just those vaccine makers, but everybody, so everybody with a 401k, everybody who's saving for retirement, nobody's complaining. It's the only good news in COVID. Everything else is bad. We're closed down; at least the stock market is going up. So nobody's complaining about this news causing these companies to go up.
- 8.44** BioNTech reached an all-time stock high, with the highest volume in August of 2021. Moderna got a 30% stock price [increase] in a single day in May of 2020, not long after we started to do the closedowns because it had positive results from a phase one clinical trial. The stock reached an all-time high a year later. It continues. They do a licensing deal with Sanofi. I can take you through these spikes to show you what the news is each time. I'm not saying that these companies manufactured the gains, but after a while, they knew that everything they said related to the vaccine was being examined under a microscope. Everything they said about the coming vaccine was getting front-page news and was being talked about. So they knew that it would move it.
- 8.45** Nobody, by the way, even those companies that eventually end up not developing a vaccine, like Johnson & Johnson, don't have an effective one. They have one, but it's not widely accepted. None of them ever gave you bad news. No one ever held a conference or issued a press release that said, "We're hitting a stumbling block. This is the most diabolical viral agent we've ever encountered, and we have no idea how we'll be able to master it." No. They only got good news time and time again from them, which affected the bottom line of the stocks.
- 8.46** Potential conflicts of interest and oversight—at every level, by pharmaceutical industry executives and their allies in the government. It makes sense. The administration says Operation Warp Speed is designed to make vaccines. So you can't exclude pharma. I don't mean to say that. You're not saying, "Okay, you work for a pharma company, we don't want you on." But they allowed the chief people who were running the companies and their top assistants to have access and control of Operation Warp Speed, which meant that it's only developed in terms of drug development.
- 8.47** Exemption from conflict of interest rules—the lawyer in me, I love to look at the nuances, the little things. Sometimes, I'll say to Trisha, "What about this?" She'll say, "Nobody cares." I'll say, "Yeah, but they should." This is one of those things I thought they should care about. Those executives were brought in as contractors, not subject, therefore, to federal conflict of interest regulations in place for employees. They're

specifically excluded for that. That's an important step for them. Nobody could bring a conflict of interest claim against them at any time.

- 8.48** The investments and earnings protected—the Operation Warp Speed advisors with connections and investments were able to keep their holdings and earnings, with only a requirement to assign some of their COVID-19 profits to the NIH after they died. Now, that's remarkable. Think about that. It's fantastic. We say to you, essentially, as a government, "Come in here. You don't have to divest yourself of your stock. Nothing like that. We don't want you not to make money if this is going along. We want that vaccine. And if you make a lot of money on the profits and everything else, be sure, after you die, to leave a legacy gift to the NIH." I hope I live long enough to see if those are fulfilled.
- 8.49** COVID-19 is an opportunity for the government to essentially restructure in many ways the profit-driven health care, instead of taxpayer money flowing to a small group of capitalists with minimal oversight and accountability. So what ends up happening here? The US government spending over, in the end, \$25.3 billion. We start with \$2 billion in the first tranche that comes out, and it keeps going up: \$25.3 billion in average price—this is what they're buying in terms of the vaccine, not what they're funding in terms of research—at \$20.69 a dose.
- 8.50** What I think was interesting is that in the end, it's about \$31 billion total that comes directly into the development, production, and purchasing of the mRNA COVID-19 vaccines. The National Institutes of Health helped offset part of the price for this big public launch. The \$100 billion that comes in sales revenues from Moderna and Pfizer is 20 times greater than the budget of the World Health Organization. These were big hit drugs. And one of the things that was fantastic—you'll understand this completely. We just finished the last hour talking about orphan drugs and exorbitant prices for these specialty drugs that are priced at \$500,000, \$1 million, \$3 million a drug.
- 8.51** Pfizer, there are no fools there. Moderna, they're not foolish. There was this anxiety about what will they price it at? They have the pricing control. And so they knew they would have to keep the price what appeared to be low: \$20 a dose, average price of \$20 a dose. Everyone's used to paying high drug price. You pay more for that in your copay when you go to fill in a drug. So people say, "Oh, they only want \$20, \$22, \$25. They raise the price a little bit afterwards. What a bargain. Thank goodness they aren't gouging us." They knew they had a market of potentially billions of people. They had the largest market possible. They were still making an enormous profit, but they had priced at a stage.
- 8.52** So I would have people that I talked to inside the reporting world and newspaper world and in television, reporters who would come back to me and say, "See, Posner, you were worried about nothing. They didn't price gouge." I would say, "No, they're not going to price it \$1,000 a shot. The planet will be burning down their headquarters. They're pricing at a level at which they've calculated how much they can make." I do

think that it's a very clever price, but people are looking at it in terms of just that particular dollar and thinking that it's a good deal. So, the government invests \$31.9 billion in research and procurement, a fair amount of money.

- 8.53** Pandemic profits—Pfizer and BioNTech, you might remember, they work together. Pfizer is the American arm. BioNTech is the little German company that has some mRNA research. They formed together a their COVID-19 sales: Johnson & Johnson is far behind. They're just in \$151 million, not much at all; the international sales are small. AstraZeneca, nothing at all in the US. They had some international sales. But the big money, Moderna, never had a hit before, never had a drug before in the market, suddenly has \$6 billion in sales from January first to June 30th of 2021. So you're talking a real turnaround. And Pfizer runs up over \$11 billion in sales.
- 8.54** These sales far outpaced. Pfizer and Moderna far outpaced the competition without any doubt. They became the economic giants of COVID-19. Their combined revenues soared into the hundreds of billions of dollars. And pharma had learned from previous crises that goodwill and strong scientific narratives help. They were always presenting themselves as the one who is saving the planet.
- 8.55** Let me tell you—easy for me to say, let me tell you, if you will—this is not cynicism. It's just a skepticism I have always about the good intent of the pharmaceutical industry. It is decades of refined messaging they continue to do, and here they were able to play it very, very well as we went along.
- 8.56** This chart shows the revenue from COVID-19 vaccines. It's another way of looking at it. Pfizer and Moderna in '21, way ahead of the curve. Now look what happens. It's very interesting as you go along here, because if you're looking from 2021 to 2025 on each of these, you'll see the numbers, the bars, really start to drop. So their annual sales through 2025 are much smaller. Their big sales are in that first big scare year, the full year of 2021. There's some expectation of sales that come up through booster shots. But here's the interesting thing. Now, what does the scientific community believe, and what do the analysts on Wall Street believe that the market value of DNA and RNA therapeutics will be over this very same time? Look at it from 2021 and follow it. Where is it in 2025? It's an all-time high because here's what happens.
- 8.57** So, COVID-19 sales dropped for the vaccine because COVID doesn't turn out to be a planet-ending virus. But the billions of dollars that we give to the pharmaceutical companies to develop the vaccine become useful for them inside the company to also develop other mRNA therapeutics and treatments. You're going to continue to see a list of those becoming public and getting FDA approval over the next few years. Nobody at Moderna, Pfizer, or anywhere else is going to break down how much of the research done on that came from the COVID funding that they were able to do. There's a benefit to that. I understand that it's almost impossible to figure out, but all you have to do is look at the figures what the expectations are versus what the COVID numbers are, and you can see how some of that plays in.

- 8.58** Vision research reports—mRNA therapeutics market size to attain \$45 billion by 2034. I will be around in 2034 unless something terrible happens. If I'm here in 2034, I hope I remember to look back at that because I would think the \$45 billion might be dwarfed. It might be larger than that, but we'll see. Long-term predictions are very hard make.
- 8.59** Profits soar. So, pandemic profits, the world's 20 biggest pharmaceutical companies saw their profits double during COVID-19: \$83 billion in profits in 2020, \$188 billion in 2022. No doubt the bulk of that is in the vaccine makers, but it's that rising tide. A number of pharmaceutical companies saw their all extra money from a whole host of respiratory drugs that they had. People were concerned every time they had the equivalent of a flu, or they were getting sick, they thought it was COVID. COVID test. Oh, my God. Remember, initially, you had to get a test, and then there were just kits. The kits were a moneymaker made in China for less than a dollar each. Then the government bought so many kits. You lost count of them at a certain point. So the profits were soaring.
- 8.60** Payouts to shareholders versus R&D spending—I always like this. They spent nearly as much on shareholder payouts, dividends, and executive compensation as they did on research and development during the pandemic, with some companies spending more on shareholders than on R&D. This brings us back to what we talked about the other day. I get it. You want to do R&D, you need to develop new drugs, you're here in the middle of COVID and you're doing vaccines. Fantastic. And you're still giving out dividends and shareholder buybacks, and you're raising the price of the stock, in some cases more than you're spending on R&D? It's just something that doesn't sit right with me, but then I'm not the drug czar.
- 8.61** Profit margins rival oil and gas, no doubt. Top 20 pharmaceutical companies had a profit margin of 19% during COVID. That rivals the profit margins of oil and gas. Although oil and gas varies, they usually have an outsized profit margins. One of the comparisons: to make 19 % return is a very healthy profit margin, and that's what it was during COVID. So while businesses suffered, many didn't survive, we're dealing with the retail, with the public, for the pharma industry, it was one of their best times across the board. COVID-19 vaccine maker profits, some of the highest: 54% profit margins for BioNTech, 51% for Moderna, and 28% for Pfizer. Those are margins they won't have for a while.
- 8.62** Calls for pandemic accord action—governments were asked by public interest groups and others and activists and even lone people like me to prioritize public health over the pharmaceutical industry profits in this pandemic accord. They never got around to that. They were doing everything but. So vaccines pump up pharma profit margins. We just said BioNTech: 54%, Moderna: 51%, Pfizer: 28%.
- 8.63** Here's the stock buybacks in some detail: Pfizer spent \$8.5 billion buying their stock back so they could make sure the stock price stayed strong. Executive bonuses: \$21 million. Now, you look at that, you know what? Moderna: \$3 billion in stock buybacks,

\$18 million in executive bonuses. J&J: stock buyback, \$7.5 billion, \$29 million. AstraZeneca and Merck.

- 8.64** The sad part of this is—I think it's sad, because I still work for a living to pay the bills. So I tend to be rather amazed, even though I know those are executive bonus figures. But we've gotten to a stage in this world where we look at executive bonus figures, and you say \$21 million, and you say, "That wasn't so bad. I've heard of somebody who got \$100 million. Gee, I heard somebody who knows cashing in on this, and my God, they walked away with nearly a billion dollars on a sale. Twenty-one million, those guys were really being very, very careful and not taking too much."
- 8.65** I guarantee you, during the development of the vaccine, you're receiving billions of dollars from the government to develop those. You would think that for a while, you've already been asked to donate excess profits after your death, maybe to the NIH, maybe you might say, "I'm not taking a bonus this year." That would be amazing. The public would think you're in the fight with them. But the public wasn't really angry about this. It was just business as usual. So they were able to do business as usual and still get away with it.
- 8.66** This chart is too much to see at a single glance, but I will just tell you that what it's essentially showing you here is an analysis of how the world's largest pharma companies prioritize these payouts to shareholders and executives over R&D and COVID-19. And what you end up doing is you look at the revenue for Pfizer, you go all the way down, AbbVie, BioNTech, Novo Nordisk—they make one of the big GLP-1 drugs—Moderna, Eli Lilly, and others. You see the revenue, you see the profit figures, and then you see the amount broken out for R&D. You can start to see that if Pfizer has \$223 million, there's \$30 million going in over here to R&D. Then, what the payouts are back on dividend stocks and executive compensation. It runs in most instances, the equivalent of the R&D, and in some instances, over the R&D.
- 8.67** So AbbVie, for instance, is spending nearly 50% more on its dividend share buybacks and executive compensation than it is on its R&D. This, to me, it shows you the profit margins on the side, is one of the things that you really have to—also, there's something else here, by the way. If you look near the bottom, Takeda is a Japanese company. Bristol-Myers Squibb: 3% profit margin. What's the matter with Bristol-Myers, you think? How are they a drug company? They call themselves a drug company, and they're only making a 3% profit margin? Somebody'd better get over the Sackler-inspired marketeers over there. But they make more than just drugs. These have become really conglomerates that make consumer products and everything else. Some of those areas lag behind. The ones focused just on drugs are the ones with the 30%, 35%, 25% profit margin. So again, the profit margins rival oil and gas, which averaged 17% in the middle of the pandemic in 2022.
- 8.68** The booster economy—crisis becomes continuity. This is one of the areas in which a one-time vaccine is profitable, but an ongoing series of shots or boosters is far more

profitable. So it's a recurring vaccine demand. Trish and I walk into our local Walgreens in Miami, where we live, Miami Beach, and we know the pharmacist. We say, "Hedra"—she's fantastic. "How are you? How's everything? How are your children and that?" "Did you get your COVID shot yet?" she says. It used to be just the flu shot they asked for, and they want to know. Then, at a certain age, you've got your pneumonia shot, you get your shingles shot, and they're still asking about the COVID shot at the big retail chain pharmacies. That's where you have to have it. If you're drug companies, you want Walgreens, CVS, Rite Aid, those types of companies to be able to be the ones with pharmacists saying, "By the way, did you get your COVID shot?"

- 8.69** Predictable annual revenue—if you can get a regular schedule for this, like the flu shot, you know how much money you make on that particular shot. It shifts it from a pandemic emergency to what I call an endemic management. That's what you try to do. Justification for premium pricing and new product lines for age and sub-risk groups, so there's still money being made on COVID, just not as much as there had been before.
- 8.70** Intellectual property values, what I call a vaccine knowledge lockdown— this is the percentage of COVID-19 doses covered by intellectual property protections. So 70% of the vaccines that J&J ended up getting out, they had their own IP protection on their own patent. They were able to price it at a higher point. The 75% with AstraZeneca, no partnership, no cross-licensing, nothing else, 85% with Moderna and 90% with Pfizer and BioNTech. Again, this is just a reality at this time of pharma. During the height of the pandemic, these companies are holding on to the IP very, very strongly.
- 8.71** I know you'd like to think that in COVID, they targeted everybody equally, men and women, and they did for the most part. They really didn't care if the jab was going into my arm or Trisha's arm. They liked that it was going into maybe both if that was the case. But in the COVID messaging for women, there is something similar in here. It's this framing of the vaccination as—I found this, maybe it was because I had done the research on the early days of targeting women, that when I saw the pharmaceutical companies issue these brochures, ads, or public service bulletins, and they would appeal to maternal caregiving, protect your family, "Hey, you mothers out there, protect your family." If you didn't, it was your responsibility. No one was saying it in another way.
- 8.72** Those campaigns emphasized anxiety, guilt, emotional burden, framing vaccinations as maternal care giving. Pharmaceutical companies targeted with that message. You protect your family, echoed the past techniques to me in a different way of saying to women, "Here's why you should be doing this." It did emphasize time and time again this anxiety and guilt, disproportionately, my view, targeting women for this burden that you had to have to protect the entire family, not just yourself.
- 8.73** Social media campaigns—without any doubt, we're in the age of social media. It was big. Everything went online targeting women. The companies' social media for COVID-19 research, I wish I had printed out everything that I came across. They're

really looking for women between the ages of 25 and 54 because that's the demographic, the pharmaceutical companies have determined, that demographic, 25 to 54 women, they make the decisions for the household. If a wife says, "Do something," the husband goes along with it. The rest of the family follows. If a sister says, "Do something," the brother will follow and other relatives will follow. I don't know if that's necessarily true, but that's how they've determined it, and that's one of the reasons that they were pushing for it.

- 8.74** Global inequity is a feature, not a bug. So the wealthy nations secure priority. I get it because this is about money. If it was really a vaccine, you think that it was going to save the planet from catastrophe, it would be distributed to everyone. Poor nations waited. The booster access was stratified by gross domestic product, not by medical need. They actually went down the line of figuring out how they were going to do the boosters after doing the initial vaccine.
- 8.75** Non-existent pricing transparency—that \$21, \$22 from Pfizer, how much did it actually cost them to make each dose of the vaccine? It's guesswork. We don't know. There have been any number of articles that talk about what they think it is at a few dollars, but they can't prove it because Pfizer doesn't break it out like that.
- 8.76** The permanent transformation of pharma—so mRNA platforms become commercial assets. I really think this is a core commercial asset for the future: vaccines, therapeutics, pharmaceutical industries, product pipeline. You're going to see more and more in the next few years, mRNA, and that's going to be used as a badge. The same way that on a food item, "organic" is something that you'd like to see: all natural. We think of those things as good. I don't necessarily think of mRNA stamped on a new drug therapeutic as a good thing. It might or might not be. I have to see what the drug is and what the treatment is for. But the pharmaceutical company thinks that people will like it because it's cutting-edge. It's science. And they're selling that apart.
- 8.77** Governments underwrite the industry risk without any doubt during these public health emergencies, shielding the companies from market uncertainties. The public has accepted these unprecedented levels of premium pricing, not just for COVID-19, but for biologics in general. We're entering that world of biotechnology, and I think that in the future, the profit margins you will see on those particular items will be bigger than anything we've seen in the past.
- 8.78** The control of intellectual property—never, never more important than it was on COVID. That's the ultimate test for the pharmaceutical company because their fear was a replay of what had happened with penicillin. They weren't in the negotiating position. They didn't have anything. They didn't have any power, and as a result, when the government said, "By the way, we need you as partners and we'll create the factories for you and we'll provide the money to develop penicillin and no one owns the price and it will go down to near nothing," they all said, "Great, we'll play along." Now they

had leverage. Now they're big pharma. And they were able to push back, and they were able to hold their ground, and this was a major test for them.

- 8.79** The normalization of annualized vaccine markets—I'll tell you: this isn't just COVID-19, but it's flu shots. They have higher profit margins. There's a point in my book on pharma in which I talk about the fact that there's a point at which, 20, 25 years ago, pharma companies had almost left the realm of vaccines because there was too much litigation, too many bad effects for them, and they had left it behind. Then they got the protection of not really being sued. So what COVID does is just an extension of what we'd already done for 20 years. And now companies have come in with much higher vaccine markets and higher margins in a much bigger number.
- 8.80** So to me, the COVID-19 pandemic shows how pharmaceutical innovation is tied to this idea of strategies and priorities. It really tells you more than anything what fear does. That initial process of fear for any industry creates an opportunity. When I said to the newspaper reporter a few days after the World Health Organization had declared a pandemic, that this was once in a lifetime opportunity for big pharma, I meant that. And that's not in a bad way. I don't mean that they created COVID. I'm not a conspiracy believer who thinks that they created it in order to profit from it. But once they saw it play out, they were ready to come in and make as much money as possible by it.
- 8.81** Smarter patient summary: so now I think in some ways, I think about this as we wrap it up because I'm just talking about the money end on COVID, and we could talk about more. But you're not just smarter or more informed, but you're system-aware, I think. It's not just about one drug. It's awareness of behavior during a crisis. That certainly happens during COVID. Recognition how emergencies reshape markets and sometimes do it permanently.
- 8.82** Question the urgency. I'll tell you, try to do that in the next pandemic. Say, "I don't know if it's as bad as they're all saying." People will run over you in their rush for the exit, but you try to say, "I'm not sure it's quite as bad as they are saying."
- 8.83** Separate innovation from pricing. Tough to do because you think it's new. It's an mRNA vaccine, so it must be an innovation. It's an innovation in terms of the marketplace. In terms of science, you don't always know how good that effect is going to be.
- 8.84** Understand the limits of regulation. We've talked about that time and time again. There's no doubt. Don't put all your faith in the FDA and the government. And even outside of the FDA, we see Republican administration, Democratic control the House, they both join hands in rushing billions of dollars to pharma. So at the right crisis, they forget about the limits of power, and they're happy to do it. Recognize marketing is a sales strategy. Time and time again, we've seen that. And assume financial incentive-shaped behavior, no doubt about it.
- 8.85** A final thought, or maybe a few final thoughts. Modern medicine saved lives. We've had these eight hours talking about greedy pharma executives, and they do develop

drugs at times that are really helpful and do save lives and do make lives better for people. In the book, I talk about some of those scientists working inside of labs, trying to come up with cures, and sometimes they're really great. They're not just all side effects of something that grow hair or whatever.

- 8.86** There are real drugs that make a difference, and we are lucky in many ways to be alive at a time when so many of the diseases and simple infections that had killed our ancestors 100 or 150 years ago are no longer there. We have other things to worry about: evening rush hour and the price of food and all these other matters that have us in an age of anxiety. But we don't have to worry about dying if we cut ourselves on a rosebush while we're pruning roses or something like that. Those types of things and infectious diseases, medicine has made a difference in our lives. Sometimes it's easy to lose track of that when I've given you eight hours of the greed part of it. So that's one of the things to remember.
- 8.87** It's also shaped by profits, patents, and pressure. So you knew I wasn't going to give them everything is a good thing. Don't forget that part as well. Being a smarter patient does not mean rejecting care. So don't go through all of this and think, "Oh, my God, they're just out to make money. They've probably just developed something now and they don't even care. We don't even know what the side effects are."
- 8.88** The doctor says, "By the way, you're really, really sick, and I think I should give you this pill or this drug," and you may ask a couple of questions about it. But don't say, "No, don't give it to me, because I'm sure it's just a money-making opportunity by those greedy people who are running the pharmaceutical industry." But do understand the system behind it because you get it now.
- 8.89** Stay engaged in your own treatment. I can tell you that I often think that I must be a pain to any doctor who suggests something to me because I'm always pushing back and asking why and what can be done as an alternative, and what about this? Be engaged in that. Be an informed consumer.
- 8.90** People spend more time going ahead—I have friends who do this—they spend more time buying a new notebook computer or a phone, which edition they need. They'll spend a couple of hours looking through different reviews and whatever else, but when it comes to health care, they just take whatever the doctor tells them to do. It's remarkable. There's no effort to become an informed consumer. They walk into a Best Buy and they know exactly what TV they want because they've checked out every possible feature. But here, they aren't informed consumers. This is an area in which it helps to be an informed consumer.
- 8.91** Skepticism. Be skeptical. It's not distrust. There's a fine line, but you can be skeptical. It's—I like to say—informed participation. Healthcare works best, from my view, when patients are not passive, but informed partners in it. That is how harm is reduced. It's also how trust is earned. There's no magic solution here. I wish I could tell you that I

had a list of five things you should do every time you're going to see the doctor or go in to fill in a drug or whatever else, and those will make sure that you never take a drug that has bad side effects, whatever. It doesn't exist. We're all prey to what the system has and what it offers. But if you approach it with a healthy bit of skepticism, just know that eventually you will be a customer, if you aren't already, if you live long enough, of the drug industry. And hopefully, that will be a good experience for the time that you have to use one of their products.

- 8.92** You say doctors are being paid millions in promotions and other benefits, and there's a revolving door between Big Pharma executives and leadership in the FDA. Would it be fair to view both doctors and the FDA as an arm or extension of Big Pharma? And should we be not distrustful, but equally skeptical of doctors' advice and approvals from the FDA?
- 8.93** It's hard to be skeptical of approvals from the FDA because initially we give them the benefit of the doubt because we don't know everything that took place behind the scenes in all the clinical trials. Now, you can go to Pubmed online. Pubmed is great. You can look at some of the work. It's published there eventually, and you can see what the sausage-making involved. But my initial reaction most of the time is to give the FDA the benefit that they've actually done and researched and looked at the clinical trials correctly. Then, I follow up to see if, in fact, they've done that correctly. In some drugs, it's fine, and other times, you're right, we place our trust in them unfairly.
- 8.94** As for doctors, there is a site, as a matter of fact, it was done—ProPublica had done a great public service a few years ago. They reported, together with The New York Times, on the amount of money that was being paid to doctors. Part of what I liked so much about that ProPublica result, and you will see it, is that they created a searchable database that is updated by reports from the drug companies and from physicians about how much money physicians get from the drug companies.
- 8.95** It turns out that you can go into that database and do a search for your doctor and find out. I did a search for the doctor I see, and I was ready to leave that doctor if they were a big beneficiary. They had taken, I think, \$384 or something. It turned out to be a couple of dinners. Okay, still it would be good if it was zero. You'd be surprised how many times doctors take nothing. But there are other doctors who have taken millions from that.
- 8.96** What was fascinating about that work, when it was done, it was a big story, is that you get, in journalism, to put your finger on the scale by how you write the headline. A lot of people only read the headline. Headline writers are a big thing. If I write an op-ed for some paper, they write the headline. I don't get to write the headline. So the headline is what people look at. Sometimes they look at the subtitle underneath the headline. They don't read the piece. They take that away as a thing. So, what I call the pharma-friendly outlets said that most doctors average just a few hundred dollars in payments

from pharma companies, which is true, but a handful cashed in big time. So millions go to doctors, but not all.

- 8.97** So the question is: is your doctor one of those? And then the doctors who were taking millions often say, "Well, that's for research. That's for this or that." It wasn't so clear. The great news is it's above the board. Some things aren't counted like lectures and necessary and things like this, but it's a great database to be able to find out.
- 8.98** I don't view doctors and Big Pharma as—they are allied. They move together in the same group. They're dependent on each other in the sense that pharma needs doctors to prescribe their products without any doubt. But the doctors are not getting the same benefit in reverse. Except for those who are getting money, it's not as though they get a benefit. The pharmaceutical companies can't work without doctors. They need them. But the doctors, on the other hand, could still see patients and prescribe without having any relationship with the pharma company. They just don't do it that way.
- 8.99** So, it depends on the doctor. I urge you to find the ProPublica database for how much money doctors are getting, put in your doctors 'til your heart's content and see what they are.